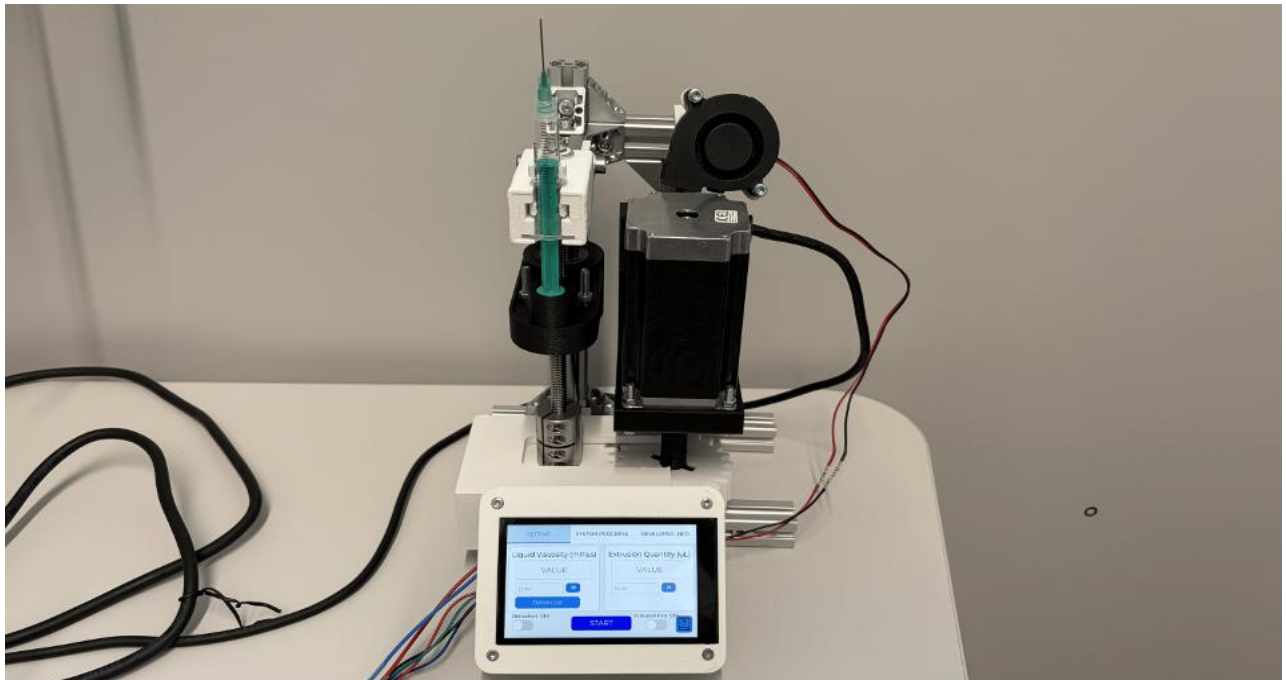


Viscosity-controlled syringe pump for precise filling of a urological implant with high-viscosity oils

Master Thesis in Computational Engineering

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1 Introduction and Objective

Stress urinary incontinence refers to the involuntary leakage of urine caused by a sudden increase in abdominal pressure (such as coughing, laughing, sneezing, or lifting heavy objects) [1] due to the pelvic floor muscles being unable to provide sufficient support. It is a common urological condition that can affect a patient's quality of life and necessitate long-term medical care. Epidemiological studies indicate that urinary incontinence is highly prevalent among the global population, with stress urinary incontinence representing an important subtype, particularly in women [2, 3]. Although urinary incontinence is particularly prevalent among women, when the focus of research shifts to implantable urinary control systems, relevant clinical studies and applications have primarily focused on severe male non-neurogenic stress urinary incontinence. Consequently, the high prevalence among women underscores the overall clinical significance of this condition, while research into male artificial urinary sphincter (AUS) serves as the primary engineering context for the AUS system. Clinical guidelines regard urinary incontinence as part of lower urinary tract symptoms and emphasize that it may cause distress and embarrassment for patients, as well as imposing a certain degree of financial burden [4]. AUS systems are currently one of the key treatment options for severe stress urinary incontinence and are particularly commonly used in male non-neurogenic cases. However, their long-term use remains associated with clinical and technical challenges, including revision surgery, device malfunction, infection, urethral erosion, and urethral atrophy. In addition, conventional AUS architectures require multiple implanted components and manual operation by the patient. These factors have prompted researchers to focus on simplified implant designs [5, 6, 7, 8]. However, the simplification of implant structures does not imply that their fabrication processes are correspondingly straightforward. Therefore, from an engineering perspective, these novel implants still require stable and reproducible manufacturing and validation processes. In particular, when device functionality depends on precise internal filling, control of the filling process becomes a key factor influencing device consistency and verifiability.

The Chair of Factory Automation and Production Systems (FAPS) previously proposed a new concept for an intraurethral AUS, aimed at addressing some of these limitations using an autonomous mechanical closure principle [9]. In the implant variant discussed in this thesis, the central occlusion component contains an internal cavity, which must be filled with high-viscosity oil in a predetermined and repeatable manner. Because of the small implant dimensions, the filling medium must be delivered via a small cannula, which presents a demanding syringe-based extrusion problem. Within the viscosity range considered in this thesis, which reaches up to 60,000 mPa·s, this high-viscosity oil generates significant flow resistance and can lead to pressure build-up, delayed flow initiation, residual pressure release, and deviations between the commanded plunger displacement and the actual dispensed volume. Previous research on syringe-based infusion and drug delivery devices has shown that, due to the syringe-based delivery method, the delivery process may be affected by start-up delay, compliance, flow-rate dependency, response time, and accessory-related effects [10, 11]. In the present application, these effects are particularly relevant because the filling medium used has a viscosity far higher than that of typical infusion solutions and must be extruded through a geometrically constrained cannula interface.

Therefore, the aim of this thesis is to develop a viscosity-controlled syringe pump system for the precise filling of the central occlusion component in urological implants. The system shall

translate application-specific input parameters, particularly the target filling volume and the viscosity of the filling medium, into reproducible syringe actuation commands. Based on these parameters and compensation factors derived from experiments, suitable motion parameters and extrusion durations are derived for the controlled extrusion of high-viscosity oil. The work covers the mechanical design of the pump system, the development of embedded control and user-interface software implemented on an Arduino GIGA Display Shield platform, and the experimental validation of the system under laboratory conditions. In addition to generating accurate syringe extrusion motion, the system supports user interaction, parameter handling, and validation-oriented process execution. These requirements are relevant in the development of medical devices, as traceability, repeatability, documentation, and validation-oriented process control are core elements of quality-management-oriented development [12].

1.1 Overview of Stress Urinary Incontinence (SUI)

Stress urinary incontinence refers to the involuntary leakage of urine during physical effort or exertion [1]. From a functional perspective, stress urinary incontinence (SUI) occurs when intra-abdominal pressure and bladder pressure temporarily rise, and urethral closure resistance is insufficient to counteract this pressure. This imbalance is commonly attributed to urethral sphincter insufficiency, weakened pelvic floor support, or a combination of both mechanisms.

Two functional mechanisms are commonly considered when describing SUI: reduced urethral support (often discussed in relation to urethral hypermobility) and insufficient urethral closure function (often described as intrinsic sphincter deficiency) [4]. In many patients, both of these mechanisms may contribute to the development of symptoms. The condition is particularly common in women, and its prevalence increases with age [3, 2]. Common risk factors in female patients include pregnancy, childbirth, obesity, and a history of pelvic surgery [4]. In men, SUI is less common and is often associated with sphincter impairment after prostate surgery [5, 8].

Clinically, SUI can range from occasional leakage during intense physical activity to frequent leakage during routine movements. Depending on the severity of symptoms, this may limit the patient's daily activities, exercise, and social participation [4]. Therefore, treatment options should be selected based on symptom severity, underlying functional impairment, and the expected benefit to the patient. Before considering surgical or device-based interventions, conservative treatments, such as pelvic floor muscle exercises and lifestyle modifications, are usually prioritized [4].

When conservative treatment does not provide adequate symptom control, surgical or device-based approaches may be considered. For female patients, common treatment options include sling procedures. On the other hand, artificial urinary sphincter systems are more commonly used in men with severe non-neurogenic SUI and in specific complex cases [4, 5, 8]. Although these treatment options offer significant clinical benefits, they may also be associated with surgical risks, device-related complications, and long-term maintenance requirements. These limitations have prompted researchers to explore alternative device concepts with the aim of improving ease of use, reducing invasiveness, and ensuring reliable functional performance. For such device concepts, reliable performance also depends on reproducible preparation processes, particularly when the device functionality relies on precise internal volume filling. Within this thesis, these clinical considerations are not addressed as medical research ques-

tions but are used to define the application context for a syringe-based high-viscosity filling process.

1.2 Limitations of Conventional Artificial Urinary Sphincters

The AUS system is the standard treatment option for severe stress urinary incontinence and is particularly suitable for men with non-neurogenic cases or post-prostatectomy cases. Despite their clinical value is significant, long-term studies and systematic literature reviews report relevant device-related limitations, including revision surgery, device malfunction, infection, urethral erosion, and urethral atrophy [5, 6, 7, 8]. From an engineering perspective, these complications are of critical importance, as long-term urinary control efficacy depends on the reliability of mechanical and hydraulic components, the integrity of pressure transmission, and the usability of the implanted control interface. Failure of the urethral cuff, control pump, pressure-regulating balloon, or connecting tubing can compromise pressure transmission and lead to recurrent incontinence or revision surgery [8, 7]. Infection, urethral erosion, and tissue-related complications further limit the device's longevity and increase the burden of follow-up care and potential revision surgery.

Another limitation of the conventional AUS system is the requirement for manual operation. Conventional AUS systems rely on a control pump, which the patient must press to initiate voiding. Figure 1 illustrates the basic structure of such a system, including the urethral cuff, control pump, pressure-regulating balloon, and connecting tubing. The need for manual operation introduces an additional usability requirement, since the patient must be able to locate and operate the implanted pump. This may prove challenging for elderly patients or those with reduced manual dexterity. Therefore, this factor should be considered when evaluating the suitability of conventional AUS systems [8].

The multi-component architecture of conventional AUS systems also increases the complexity of the implantation procedure. The cuff, control pump, pressure-regulating balloon, and connecting tubing must be positioned and connected across different anatomical regions, which increases the structural and procedural complexity of the system. Furthermore, conventional AUS have been predominantly applied in male patients, whereas their use in female patients is generally limited to recurrent or complicated SUI. Since the literature reports risks of complications, mechanical failure, and device removal, careful consideration is required before their use in female patients [4]. These aspects restrict broader clinical applicability and motivate the investigation of device concepts with reduced spatial requirements, simplified actuation, and reliable functional performance.

Overall, conventional AUS systems remain associated with limitations in terms of long-term reliability, usability, and clinical applicability. These limitations motivate the development of alternative device concepts that reduce system complexity and avoid manual actuation. For fluid-filled implant concepts, however, simplified device architecture also introduces an engineering requirement for a controlled and reproducible preparation process. If the functional behavior of such a device depends on a precisely defined internal oil volume, the filling process becomes a critical preparation step and requires accurate syringe-based extrusion control.

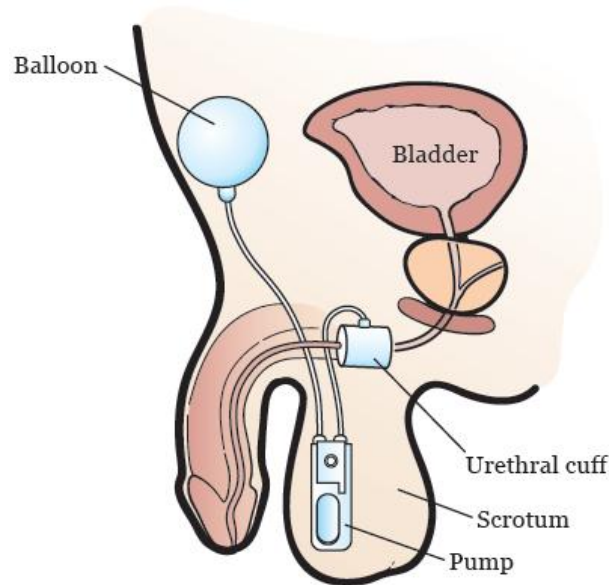


Figure 1: Schematic representation of a conventional artificial urinary sphincter system, consisting of a urethral cuff, a control pump, and a pressure-regulating balloon connected via tubing and implanted in different anatomical regions. Reproduced from Healthline [13].

1.3 Application Context: Mechanical Intraurethral Implant

The application context of the present thesis is a mechanical intraurethral AUS concept developed at the Chair of FAPS. The concept was proposed as an alternative to conventional AUS systems and aims to address limitations related to system complexity, manual handling, and sex-independent applicability [9]. In contrast to conventional patient-operated AUS architectures, the implant is based on an autonomous mechanical operating principle and does not require an external energy supply. Its function is controlled by pressure conditions within the lower urinary tract. Within this thesis, however, the implant concept is not investigated as a medical or implant-design research object. Instead, it defines the application-specific boundary conditions for a syringe-based high-viscosity filling process.

The basic closure mechanism consists of a pressurized closure balloon positioned inside a cylindrical intraurethral structure. The balloon acts as the occlusion element and blocks urine flow by pressing against the inner contour of the surrounding tube. It is connected through a throttle to a compensating reservoir, which enables pressure-dependent fluid exchange between the balloon and the reservoir. The interaction between these three components determines the opening and closing behavior of the implant [9]. Figure 2 illustrates the structure of the mechanical intraurethral AUS.

During normal conditions, the closure balloon remains sufficiently pressurized to seal the urethral passage, while short-term pressure peaks, such as those caused by coughing, are not intended to open the system. Opening occurs only when bladder pressure is increased over a longer period of time by active abdominal straining during micturition. Under this pressure state, fluid is displaced from the balloon through the throttle into the compensating reservoir. The resulting reduction in balloon volume decreases the sealing effect and allows urine flow. After the pressure condition changes, fluid can return from the compensating reservoir to the balloon, allowing the system to return to its closed state [9]. This pressure-dependent oper-

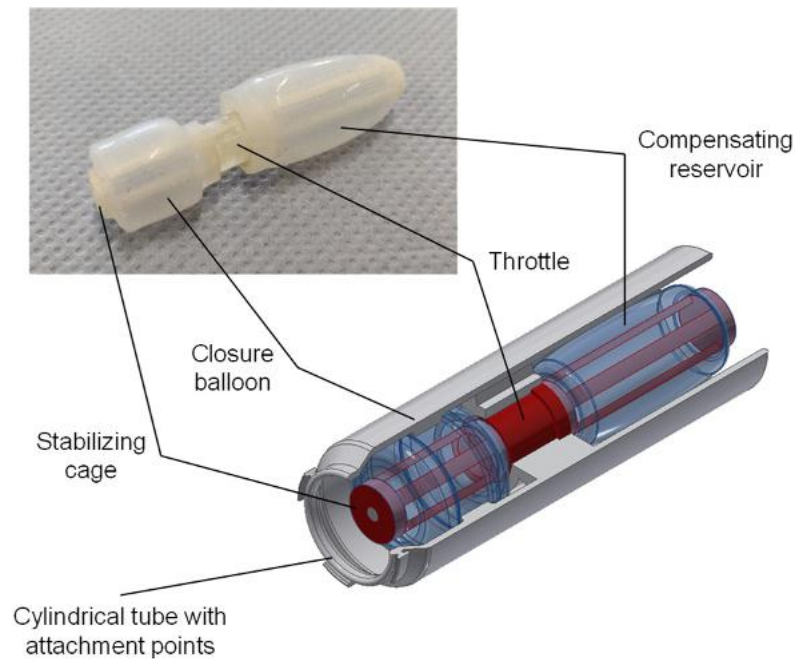


Figure 2: Prototype of the mechanical intraurethral AUS concept. The system consists of a closure balloon, a throttle, and a compensating reservoir for pressure-dependent regulation. Reproduced from Preis et al. [9].

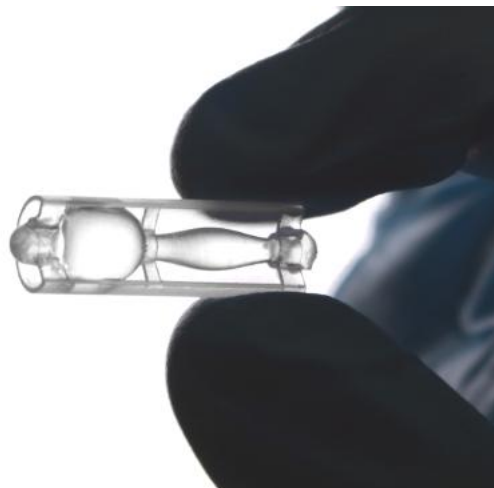


Figure 3: Cross-sectional illustration of the mechanical intraurethral AUS concept considered in this thesis. The closure balloon is positioned within a cylindrical intraurethral structure and connected through a throttle to a compensating reservoir for pressure-dependent regulation. Source: Chair of Factory Automation and Production Systems (FAPS), used with permission.

ating principle eliminates the need for manual actuation by the patient and therefore differs fundamentally from conventional artificial urinary sphincter systems with manually operated control pumps. In addition, the intraurethral arrangement reduces the number of spatially distributed components. This architecture is expected to reduce system complexity. However, its actual clinical applicability depends on the final device geometry, implantation procedure, and clinical evaluation.

For the present thesis, the implant is relevant primarily because its functional behavior depends on a defined filling state of the closure balloon. Before operation, the balloon must be filled with a defined liquid volume in order to establish the required internal pressure and sealing behavior. Deviations in filling volume or filling pressure can influence the opening and closing characteristics of the implant. The filling process is therefore not only an assembly step, but a functionally relevant preparation and validation process. The development and optimization of the implant itself are not the focus of this thesis. Instead, this work addresses the engineering task of developing a controlled syringe-based filling system for the implant's occlusion component. Because the implant dimensions are small, the filling medium must be delivered through a small cannula. In combination with high-viscosity oil, this creates a high-resistance extrusion process in which pressure build-up, delayed flow response, and deviations between commanded plunger displacement and actual dispensed volume become relevant. This requirement forms the basis for the viscosity-controlled syringe pump investigated in the following chapters.

1.4 Challenges in Filling Implant Components over a Wide Viscosity Range

Filling the occlusion component of the implant is a functionally relevant step in device preparation, as the filling state affects the internal pressure and mechanical response of the occlusion mechanism. In the previously proposed concept of a mechanical intraurethral AUS, the closure function is achieved via a fluid-filled closure balloon [9]. In the implant variant considered in this thesis, the corresponding occlusion component contains a defined internal oil volume. Deviations in the filled volume may alter the pressure state and affect the opening and closing characteristics of the implant. The filling process is therefore not only an assembly step, but also a preparation and validation relevant process that must be controlled with sufficient accuracy and repeatability.

A central challenge lies in the fact that the filling system must handle a wide viscosity range. In this study, water with a viscosity of approximately 1 mPa·s was used as a low-viscosity validation boundary, while the viscosity of the application-relevant oils reaches up to 60,000 mPa·s. This viscosity range leads to substantially differences process conditions during syringe-based extrusion. While low-viscosity media can usually be displaced with comparatively low pressure, high-viscosity media require a larger pressure build-up to initiate and maintain flow. This behavior is consistent with the expected hydraulic resistance generated by pressure-driven flow through a narrow channel. According to the Hagen–Poiseuille relation, the required pressure drop increases with viscosity, volumetric flow-rate, and channel length, and decreases strongly with the fourth power of the channel radius [14]. Although the actual filling process may deviate from an ideal rigid circular tube, this relation illustrates the strong sensitivity of the process to the cannula radius. In this application, this effect is particularly relevant because the small

implant dimensions require the filling medium to be delivered through a small cannula. Depending on the material properties and process conditions, this may lead to delayed extrusion, increased flow resistance, residual pressure release, and deviations between the commanded actuator motion and the actual volume delivered.

The filling result is not determined by viscosity alone, but rather depends on the interaction between multiple process and system parameters. Temperature variations can change the effective viscosity of silicone oils, and previous work has modeled silicone-oil viscosity as a function of temperature and shear rate under application-specific flow conditions [15]. The geometry of the syringe defines the relationship between plunger displacement and theoretical volume output, while the syringe material, plunger seal, and connector design influence the mechanical boundary conditions of the filling path. In this thesis, the filling platform is based on a standardized disposable B. Braun Injekt Luer Lock Solo syringe, which provides a Luer Lock connection and a fixed syringe format for the mechanical interface [16]. The conversion between actuator motion and theoretical volume output is derived from the syringe-specific plunger displacement relationship determined for the selected syringe geometry. In addition, plunger friction, sealing behavior, and the mechanical compliance of the syringe and the connected filling path can influence pressure build-up and flow response. These effects are particularly relevant under high-resistance conditions, where part of the actuator movement may first contribute to elastic deformation or pressure generation before a steady-state material flow is established.

A further difficulty is the transfer of a desired filling volume into appropriate actuator commands. In an ideal rigid system, the dispensed volume could be calculated directly from the syringe cross-section and the plunger displacement. However, in practice, friction, compliance, start-up behavior, and viscosity-dependent flow resistance, can cause deviations from this ideal relationship. Similar effects have been reported in the literature for syringe-based drug delivery systems, where start-up delay, compliance, flow-rate dependency, response time, and accessory-related effects can influence the delivered volume or flow behavior [10, 11]. Consequently, without compensation for specific materials and systems, cannot reliably provide consistent results across the entire operating range, when using a fixed actuator speed or purely time-controlled strategy.

These challenges show that a suitable filling system must do more than generate linear actuator motion. It must support viscosity-dependent parameter calculation, controlled syringe actuation, and repeatable execution of a defined filling process. This forms the engineering basis for the development of a viscosity-controlled syringe pump system, which is designed to reproducibly deliver high-viscosity oils.

1.5 Requirements for an Integrated Viscosity-Controlled Filling System

The filling challenges described in the previous section translate into specific requirements for the design of the syringe pump system. A suitable filling system must not rely solely on manual operation or fixed time-based dispensing parameters. Instead, it must convert application-specific input parameters, including the target filling volume, fluid viscosity, and syringe configuration, into reproducible actuator commands. In this application, the viscosity ranges from 1 mPa s for water to 60,000 mPa s for high-viscosity oils. Under these conditions, if viscosity-

related flow resistance, start-up behavior, and system compliance are not considered, the same actuator setting may lead to different filling results.

For the implant component considered in this thesis, the enclosed fluid volume and the resulting pressure state represent preparation-relevant boundary conditions that influence the mechanical response of the occlusion component. Thus, the filling process should be accurate, predictable, and reproducible. Deviation in the filled volume will influence the preparation of the occlusion component and increase the workload required for further testing and validation. It is relevant especially in small target volumes since even minor absolute deviations may represent a significant relative error. Therefore, the filling system must support defined process parameters and reproducible execution of filling operations under controlled laboratory conditions.

Additionally, the system must take into account mechanical restrictions of the syringe and the filling pathway. The syringe geometric, material properties, wall thickness, plunger sealing, and the design of connections all influence the required actuation force and the possible pressure that may arise during the extrusion process of the medium. It is especially important when extruding high-viscosity media, as the pressure necessary for extrusion can significantly increase. Since the current system cannot directly measure the fluid pressure, the control strategy must operate within empirically defined actuation limits, utilizing controlled motion parameters rather than closed-loop pressure regulation.

An integrated system architecture is required to combine actuation, parameter processing, user interaction, and process monitoring within a single platform. The control software must calculate the appropriate motion parameters based on the selected filling volume, viscosity, syringe configuration, and compensation factors determined through experimentation. At the same time, the user interface must provide a clear and reproducible workflow for entering process parameters, starting the filling sequence, and displaying relevant process information. Process monitoring, including commanded and actual motor states, current-related driver information, elapsed process time, and estimated remaining duration, can support the identification or later interpretation of abnormal process states and provide a basis for subsequent verification and qualification in a quality-management-oriented development environment [12].

The aim of this study is to develop a filling platform that integrates viscosity-dependent parameter calculation, controlled syringe actuation, a graphical user interface, and validation-oriented process monitoring. The system is designed to support reproducible filling operations under laboratory conditions and to provide a foundation for future process optimization, qualification-oriented testing, and early-stage medical device development workflows.

1.6 Research Question and Objectives

The main research question of this thesis is how to implement a viscosity-dependent syringe actuation on a desktop platform to enable accurate, reproducible, and configurable filling through a small cannula over a wide viscosity range. More specifically, this thesis investigates how to integrate viscosity-dependent parameter calculation, parameterizable syringe and process models, a graphical user interface, and process monitoring into a single filling platform. The aim is to provide a platform that supports controlled syringe actuation, defined filling sequences, and validation-oriented process execution under laboratory conditions.

To address this research question, the following research objectives have been established:

1. Design and implement a desktop syringe pump platform capable of controlled filling or extrusion through a small cannula.
2. Develop a viscosity-dependent parameterization and compensation strategy for adapting the extrusion process to media with viscosities between 1 mPa s and 60,000 mPa s.
3. Implement parameterizable syringe and process models that account for filling volume, fluid viscosity, syringe geometry, specified operating limits, and compensation factors.
4. Evaluate whether the filling accuracy whether the absolute volumetric deviation remains below 1% for a nominal filling volume of 1 mL under defined laboratory conditions.
5. Develop a graphical user interface with a simplified standard operating mode and an advanced mode with extended parameter access.
6. Develop monitoring functions capable of integrating motor-driver data and current-related driver information for process monitoring and diagnostic analysis.
7. Evaluate the filling process within a qualification-oriented framework, which includes Design Qualification (DQ), Installation Qualification (IQ), Operational Qualification (OQ) and Performance Qualification (PQ).

1.7 Structure of the thesis

this thesis is divided into seven chapters. Chapter 1 introduces the clinical background, the technical motivation, and the application context of the mechanical intraurethral implant. It also explains why a viscosity-controlled syringe-based filling system is required for the present application. The chapter then defines the research question and the main objectives of this thesis.

Chapter 2 provides the theoretical and technical background needed for the system development. The chapter discusses syringe-based dispensing, viscosity-dependent extrusion behavior, the relationship between syringe geometry and delivered volume, and the quality-management-related validation.

Chapter 3 derives the requirements for the syringe pump platform, which are based on the filling task defined in the previous section. The chapter identifies the mechanical, actuation-related, user-interface, control, monitoring, and validation requirements that must be fulfilled by a viscosity-controlled filling system for the present application.

Chapter 4 describes how these requirements are implemented in the syringe pump platform. The chapter begins with the overall system architecture and then presents the mechanical design, including syringe fixation, the standardized cannula connection, the plunger contact interface, actuator mounting, and linear guidance. The drive and actuation subsystem is then described with respect to the stepper motor integration, gear transmission, the gear-to-lead screw interface, and rotary-to-linear motion conversion. The remaining sections describe the user interface and guidance concept, the control and compensation method, and the experimental validation method based on Design Qualification, Installation Qualification, Operational Qualification, and Performance Qualification.

Chapter 5 presents the results and evaluation of the developed system. The chapter reports the Design Qualification and Installation Qualification results, including the evaluation of the mechanical design, user-interface design, and installation state. It then presents the Operational Qualification results, including motion execution, process monitoring, repeatability, baseline deviation, and viscosity-dependent compensation parameter selection. Finally, the Performance Qualification results evaluate the compensated extrusion performance and the post-extrusion passive release and waiting-time behavior.

Chapter 6 discusses the developed syringe pump system in relation to the research question and the derived requirements. The discussion evaluates the achieved filling accuracy, repeatability, viscosity-dependent compensation behavior, mechanical and control limitations, and the extent to which the system can support reproducible high-viscosity filling through a small cannula under laboratory conditions.

Chapter 7 summarizes the main findings of this thesis, evaluates the developed syringe pump system in relation to the defined objectives, and outlines remaining limitations and possible directions for future work. Particular attention is given to further optimization of the filling process.

2 Foundations and State of the Art

This chapter provides the theoretical and technical background for developing a viscosity-controlled syringe pump for filling through a small cannula. The focus is on the physical and system-related aspects that influence high-viscosity extrusion over a wide viscosity range. These include viscosity-dependent flow resistance, pressure build-up in the cannula, force transmission to the syringe plunger, mechanical limitations of the syringe system, and dynamic constraints of actuator motion.

The chapter begins with the principles of syringe-based filling and the analytical fluid-mechanical model used to describe pressure-driven flow through a cannula. Based on this model, the relationship between flow rate, viscosity, cannula geometry, and required pressure is formulated. The resulting pressure is then related to the required plunger force, allowing mechanical limitations such as plunger stability, allowable force, and acceleration-related dynamic loads to be considered. Subsequently, existing syringe pump systems and common extrusion control strategies are discussed with respect to their suitability for high-viscosity and viscosity-dependent filling processes. The role of embedded systems and graphical user interfaces in laboratory-scale devices is then addressed, followed by an overview of validation and quality-management-related considerations relevant to medical-device-oriented laboratory development. The chapter concludes by identifying the technological gaps that motivate the integrated filling system developed in this thesis.

2.1 Syringe-Based Filling of High-Viscosity Media

Syringe-based filling is used when small and defined liquid volumes must be delivered in a controlled manner. The basic principle relies on the axial movement of the syringe plunger, which forces the fluid from the syringe barrel through an outlet such as a cannula or filling needle. For low-viscosity fluids and ideally rigid components, this relationship can often be approximated from the syringe cross-section and the plunger displacement. However, for viscous media the delivered volume and the required extrusion force are also affected by the syringe geometry, plunger motion, outlet geometry, fluid viscosity, plunger friction, and downstream pressure conditions. Previous studies on syringe-based delivery and injectability report that viscosity, syringe and needle geometry, plunger friction, back pressure, and mechanical testing conditions can influence both the required extrusion force and the resulting delivery behavior [17, 18, 19]. Figure 4 illustrates the basic syringe-based extrusion principle considered in this thesis.

Under ideal geometric conditions, the theoretical dispensed volume can be related to the plunger displacement by

$$V = A_s \cdot s = \pi R_s^2 \cdot s \quad (2.1)$$

where V is the displaced volume, A_s is the internal cross-sectional area of the syringe barrel, R_s is the inner radius of the syringe barrel, and s is the axial plunger displacement. This equation provides a geometric relation between actuator motion and theoretical volume output. In practice, however, it remains an approximation. The actual dispensed volume can differ from

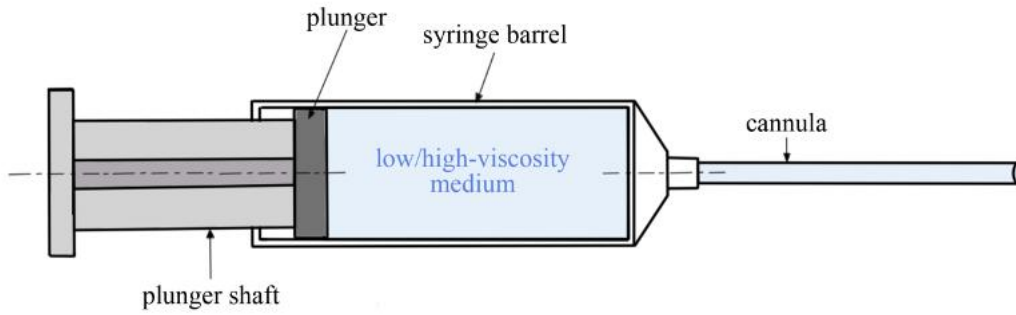


Figure 4: Schematic representation of syringe-based extrusion of low- and high-viscosity media. The axial movement of the plunger displaces the fluid inside the syringe barrel and forces it through the connected cannula. The figure was created by the author.

the calculated value because not all actuator motion is converted directly into fluid displacement.

For a target volume and an assumed constant volumetric flow rate, the required extrusion time can be estimated as

$$t = \frac{V}{Q} \quad (2.2)$$

where t is the extrusion time and Q is the volumetric flow rate. For an ideal rigid syringe system, the flow rate can be related directly to the plunger velocity. In practice, this relation is limited by the mechanical limits of the syringe and actuator. For this reason, injectability studies usually consider not only the delivered volume, but also the force–displacement or force–time curves of the syringe–plunger system. These curves indicate the force required to extrude a material through a defined needle or cannula geometry [20, 21].

In the present work, the filling media range from 1 mPa s for water up to 60,000 mPa s for highly viscous oils. This range produces substantially different extrusion conditions. Low-viscosity fluids can be displaced with relatively low pressure, whereas highly viscous media require much higher pressure to initiate and maintain flow through the cannula. If viscosity-dependent flow resistance is not considered, the same actuator command may therefore produce different filling behavior.

The pressure demand of cannula-based extrusion can be estimated using an idealized pressure–flow relation for laminar flow through a cylindrical channel. The following section formulates this pressure–flow relation and links viscosity, cannula geometry, flow rate, and pressure demand. For the present application, this relationship is particularly relevant because the small implant dimensions require the filling medium to be delivered through a small cannula, making the outlet geometry a dominant factor for the required pressure and plunger force.

The ideal Hagen–Poiseuille relationship is useful for estimating the basic pressure–flow behavior, but real filling processes may deviate from this simplified model. First, the actual flow path may include connectors, transitions, or local restrictions that are not represented by an ideal cylindrical tube. Second, the effective viscosity of high-viscosity silicone oils may vary with temperature and, depending on the material and shear conditions, with local shear rate. Previous work on silicone-oil flow in viscous-damper applications has shown that temperature and shear-rate-dependent viscosity models can be relevant under application-specific flow condi-

tions [15]. In the present thesis, viscosity is treated as an input parameter for process calculation, while shear and temperature-dependent effects are considered as possible sources of deviation between ideal model behavior and practical filling performance.

In addition to viscosity-dependent pressure losses, high-viscosity filling processes can exhibit transient start-up effects. After the actuator begins to move, part of the motion may initially contribute to pressure build-up, elastic deformation of the syringe or connected filling path, and the overcoming of static friction before stable material flow is established. These transient and compliance-related effects are consistent with the non-ideal delivery behavior reported for syringe-based delivery systems [10, 11]. Under high-resistance conditions, this can create a delay between commanded plunger motion and actual fluid extrusion. The actual dispensed volume may therefore differ from the theoretical displacement volume.

Syringe-based filling of high-viscosity media therefore cannot be treated as a displacement-only process. The filling result depends on syringe geometry, cannula geometry, fluid viscosity, pressure build-up, friction, and transient start-up behavior. A fixed time-based or purely manual dispensing approach is therefore unlikely to provide reproducible filling over the full viscosity range considered in this thesis. The following sections use these dependencies to derive the fluid-mechanical and force-transmission models needed for viscosity-dependent actuator parameter selection.

2.2 Fluid Mechanical Model of Cannula-Based Extrusion

Since plunger displacement alone does not define the outlet behavior under high-resistance conditions, this section introduces the pressure–flow relation used to estimate the pressure required for cannula-based filling. In the simplified model used in this thesis, the narrow cannula is treated as the dominant hydraulic resistance of the filling path. The implant itself is not modeled in this section; its relevance lies in the small geometric constraints that require the filling medium to pass through a small cannula. As a result, the cannula radius, cannula length, volumetric flow rate, and fluid viscosity become the main parameters for estimating the pressure required to sustain flow. Figure 5 shows the simplified syringe–cannula flow model used in this thesis.

For an incompressible Newtonian fluid under steady, laminar, and fully developed flow conditions in a rigid cylindrical channel, the relationship between pressure drop and volumetric flow rate can be described by the Hagen–Poiseuille equation [14]:

$$Q = \frac{\pi r_c^4 \Delta p_c}{8\mu L_c} \quad (2.3)$$

where Q is the volumetric flow rate, r_c is the inner radius of the cannula, Δp_c is the pressure drop across the cannula, μ is the dynamic viscosity of the filling medium, and L_c is the effective flow length of the cannula. In this thesis, Equation 2.3 is used as a first-order analytical model for estimating the pressure demand of cannula-based extrusion.

For process planning, Equation 2.3 can be rearranged to express the pressure drop required for a selected flow rate:

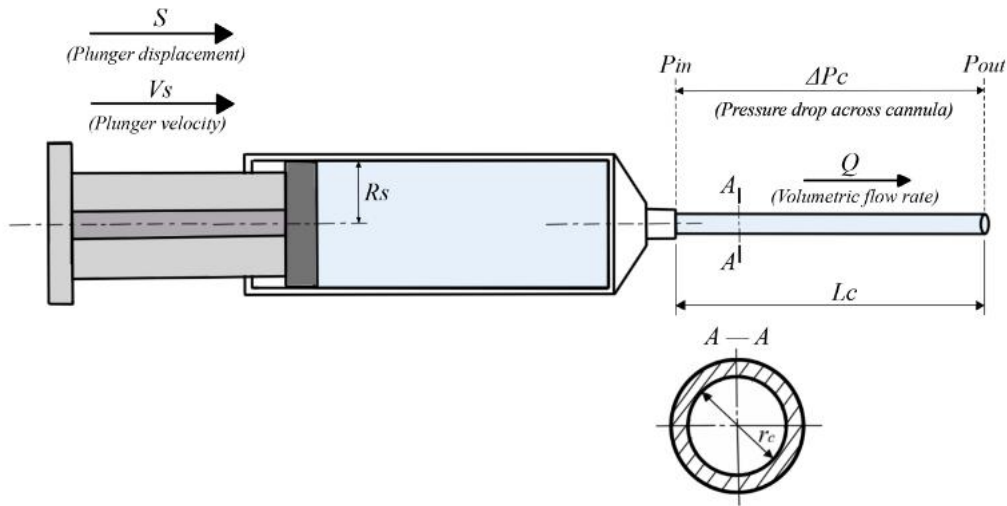


Figure 5: Simplified syringe–cannula flow model used for the fluid-mechanical description of the filling process. The plunger displacement generates a volumetric flow rate Q , while the cannula geometry, defined by the inner radius r_c and length L_c , determines the pressure drop Δp_c across the cannula. The figure was created by the author.

$$\Delta p_c = \frac{8\mu L_c Q}{\pi r_c^4} \quad (2.4)$$

In the present filling task, the equation is used primarily to identify the parameters that increase the pressure load on the syringe system. At a fixed cannula radius, higher viscosity, higher volumetric flow rate, and a longer effective cannula length all increase the required pressure. By contrast, the influence of the cannula radius is nonlinear because the radius appears to the fourth power in the denominator. A small reduction in the inner radius can therefore lead to a large increase in pressure demand. Similar dependencies between viscosity, needle or cannula geometry, flow rate, and required injection force have been described in syringe and injection-device studies [17, 18].

The sensitivity to cannula radius is relevant for the present system because the filling path is constrained by the small implant geometry. Figure 6 illustrates the calculated pressure drop as a function of the cannula inner radius for selected viscosity levels used in this work. The calculation illustrates the strong sensitivity of the pressure demand to the cannula radius under otherwise constant process conditions.

To separate the effect of viscosity from the geometric influence of the cannula, the cannula geometry and flow rate can be kept constant in the analytical calculation. Figure 7 shows the calculated pressure drop for the 21G cannula used in this work over the viscosity range considered in this thesis. With constant cannula radius, cannula length, and flow rate, Equation 2.4 predicts a linear increase in pressure drop with dynamic viscosity.

Together, Figures 6 and 7 show that a fixed target volume can require substantially different pressure levels depending on cannula geometry, extrusion duration, and fluid viscosity. These dependencies define the operating window in which the syringe pump must generate actuator motion without exceeding the mechanical limits of the syringe and plunger interface.

The volumetric flow rate can be related to the target filling volume and extrusion duration by

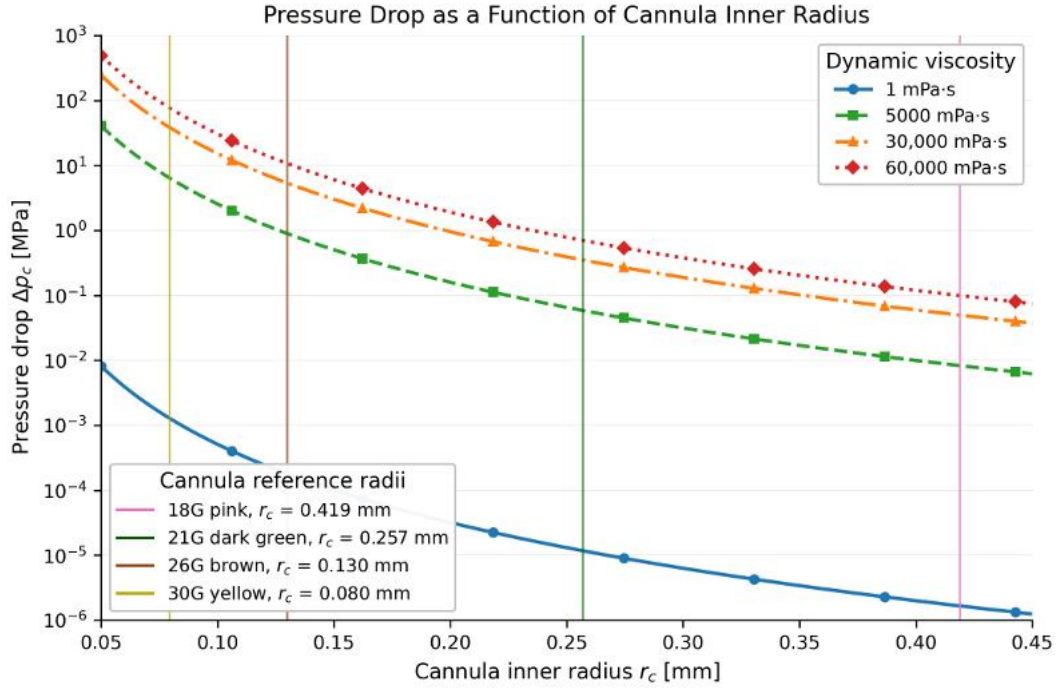


Figure 6: Calculated pressure drop as a function of the cannula inner radius for selected viscosity values. The vertical reference lines indicate the inner radii of the standard cannulas considered in this work. The strong increase in pressure drop at smaller radii results from the fourth-power dependence of the Hagen–Poiseuille equation. The figure was calculated and created by the author.

$$Q = \frac{V}{t} \quad (2.5)$$

where V is the filling volume and t is the extrusion duration. Substituting Equation 2.5 into Equation 2.4 gives

$$\Delta p_c = \frac{8\mu L_c V}{\pi r_c^4 t} \quad (2.6)$$

For a given filling volume, extrusion duration affects not only process time but also the pressure load of the filling process. Reducing the extrusion time increases the required volumetric flow rate and, according to Equation 2.6, the pressure drop across the cannula. This is particularly important for highly viscous oils, where short extrusion times can lead to high plunger forces and may bring the syringe, the plunger contact interface, or the actuator system close to their mechanical operating limits.

The flow rate can also be written in terms of the plunger velocity. With the syringe barrel cross-section $A_s = \pi R_s^2$, the flow rate generated by the plunger motion is

$$Q = A_s v_s = \pi R_s^2 v_s \quad (2.7)$$

where R_s is the inner radius of the syringe barrel and v_s is the axial plunger velocity. Combining

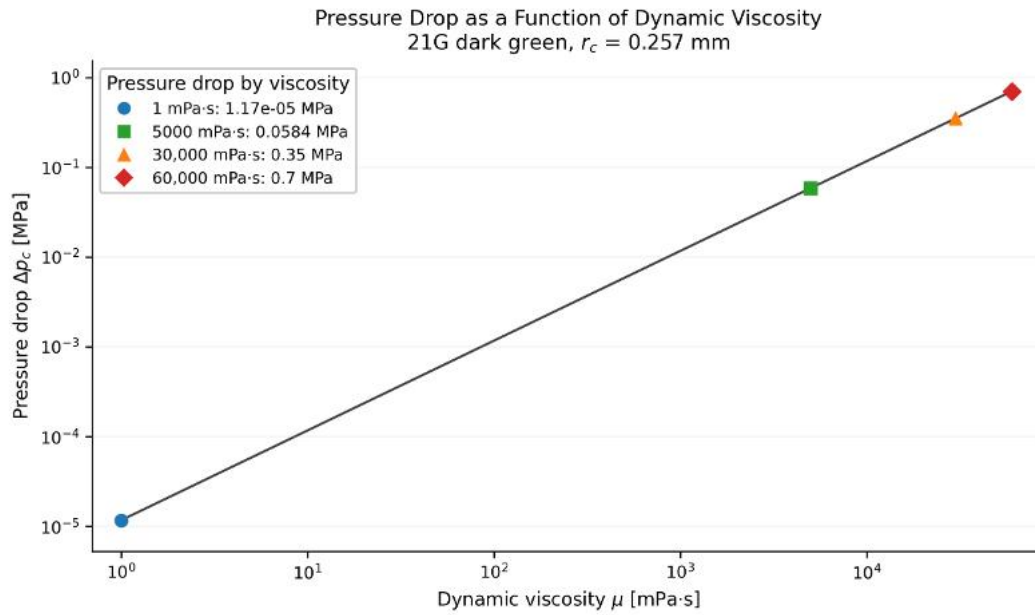


Figure 7: Calculated pressure drop as a function of dynamic viscosity for the 21G cannula used in this work with an inner radius of $r_c = 0.257$ mm. The cannula length and volumetric flow rate are kept constant, while the viscosity is varied from water-like reference conditions to high-viscosity oil. The figure was calculated and created by the author.

Equation 2.7 with Equation 2.4 gives

$$\Delta p_c = \frac{8\mu L_c R_s^2 v_s}{r_c^4} \quad (2.8)$$

For a fixed syringe and cannula geometry, this equation shows that the pressure drop increases with plunger velocity and viscosity. The selected plunger velocity therefore acts as both a motion parameter and a process parameter, because it directly affects the pressure load of the filling path. In the present system, it must be chosen with respect to the viscosity of the filling medium and the cannula geometry.

The analytical model is used to estimate the magnitude and sensitivity of the pressure demand, rather than to predict the exact transient pressure during filling. It assumes a rigid circular channel, constant viscosity, steady laminar flow, and negligible entrance, connector, and compliance effects. In practice, deviations may occur because the actual filling path includes connectors and transitions, the syringe and plunger seal can deform under load, and the filling medium may show temperature- or shear-rate-dependent viscosity behavior. In addition, start-up delay, compliance-related behavior, response-time limitations, and accessory-dependent effects are known limitations of syringe-based delivery systems [10, 11]. The model is therefore treated as a design-oriented approximation for identifying how viscosity, cannula geometry, filling volume, and extrusion duration shift the operating conditions of the syringe actuator.

The calculated pressure values are therefore interpreted as theoretical estimates for process design, not as direct measurements or exact predictions of the real filling pressure. The model identifies the dominant dependencies between viscosity, flow rate, cannula geometry, and pressure demand. These pressure-related estimates are then translated into plunger force,

actuator capability, and viscosity-dependent motion parameters in the following sections.

2.3 Force Transmission and Mechanical Limits of Syringe Systems

During cannula-based extrusion, the pressure required to move the filling medium is converted into an axial load on the syringe plunger. In the developed platform, this load is not carried by a rigid industrial dosing cartridge, but by a disposable B. Braun 2 mL syringe and its associated mechanical interfaces. The plunger, syringe barrel, Luer Lock outlet, and plunger contact interface therefore define practical load limits for the filling process. Excessive loading may lead to plunger deformation, leakage, misalignment, or interface failure. In high-viscosity filling through a small cannula, the required extrusion force can increase substantially. Consequently, the extrusion speed used in the control strategy cannot be selected only to minimize filling time. It must also remain within a mechanically reasonable operating range for the disposable syringe, the plunger contact interface, and the connected cannula path. Similar dependencies have been reported in syringe and injection-device studies, where viscosity, syringe and needle geometry, plunger friction, and back pressure influence the required extrusion or injection force [17, 18, 19].

2.3.1 Pressure-Induced Plunger Force

The pressure difference required for cannula-based extrusion acts on the effective internal cross-sectional area of the syringe barrel. As illustrated in Figure 8, this pressure contribution is converted into an axial force component that must be generated by the actuator and transmitted through the plunger contact interface.

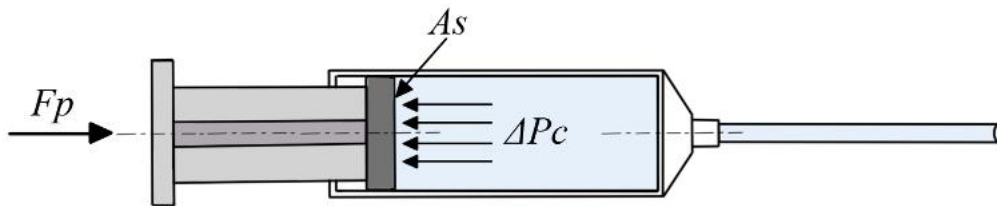


Figure 8: Pressure-to-force transmission on the syringe plunger. The pressure difference required for cannula-based extrusion acts on the effective syringe cross-sectional area A_s and results in the pressure-induced axial force component F_p . The figure was created by the author.

The estimated pressure drop can be converted into an equivalent plunger force by using the internal cross-sectional area of the syringe barrel:

$$F_p = F_{p, \text{byd}} = \Delta p_c A_s = \Delta p_c \pi R_s^2 \quad (2.9)$$

where $F_{p,byd}$ is the pressure-related component of the plunger force, Δp_c is the estimated cannula pressure drop, A_s is the internal cross-sectional area of the syringe barrel, and R_s is the inner radius of the syringe barrel.

Substituting the pressure drop from Equation 2.4 gives

$$F_{p,byd} = \frac{8\mu L_c Q R_s^2}{r_c^4} \quad (2.10)$$

where μ is the dynamic viscosity, L_c is the effective cannula length, Q is the volumetric flow rate, and r_c is the inner radius of the cannula. The fourth-power dependence on the cannula radius further indicates that the small cannula required by the application is not merely a geometric interface, but a dominant factor in the mechanical loading of the syringe actuation system.

The force calculated in Equation 2.10 represents only the pressure-related force component. In a real syringe system, the actuator must also overcome additional force components, including plunger friction, seal deformation, guide friction, and acceleration-related dynamic loads. Therefore, the total actuation force can be interpreted conceptually as

$$F_{act} \approx F_{p,byd} + F_{fric} + F_{dyv} \quad (2.11)$$

where F_{fric} represents friction-related contributions and F_{dyv} represents transient dynamic loads. Because the developed system does not include a force sensor, Equation 2.11 is not used as an identified quantitative force model. Instead, the equation is used to show which load components may become limiting when the extrusion speed is increased.

For the present design, Equation 2.10 is mainly used to relate the selected flow rate to the pressure-related load on the syringe plunger. For a fixed filling volume, a higher flow rate reduces the extrusion duration, but it also increases the pressure-induced force that the actuator must generate and transfer through the plunger contact interface. This trade-off is especially important in the present application. Since the small implant dimensions require a small cannula, the fourth-power dependence on the cannula radius can strongly limit the feasible flow-rate range when high-viscosity oils are used.

2.3.2 Buckling Limit of the Syringe Plunger

The disposable syringe plunger also introduces an axial stability limit. During high-viscosity extrusion, the actuator pushes the plunger in the axial direction, while the resistance of the filling medium generates a reaction load through the syringe system. If this load becomes too high, the plunger may bend laterally instead of transferring the actuator motion reliably into pressure build-up and material flow. The critical load for an ideal slender member can be estimated with Euler's buckling [22]:

$$F_{crit} = \frac{\pi^2 EI}{(KL_p)^2} \quad (2.12)$$

where F_{crit} is the critical buckling force, E is the Young's modulus of the plunger material, I is the second moment of area of the plunger cross-section about the weakest bending axis,

L_p is the unsupported plunger length, and K is the effective length factor determined by the boundary conditions.

The effective length factor K represents the idealized end constraints of the compressed plunger. Since the actual contact between the plunger and the actuator interface cannot be treated as a rigid moment-transmitting clamp, different boundary assumptions lead to different design estimates of the critical load. Figure 9 illustrates the typical boundary conditions used for the design-oriented comparison in this thesis.

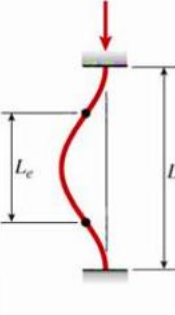
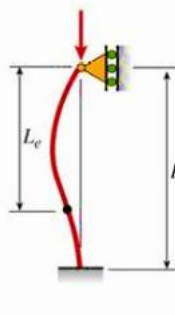
(a) Pinned-pinned column	(b) Fixed-free column	(c) Fixed-fixed column	(d) Fixed-pinned column
$P_{cr} = \frac{\pi^2 EI}{L^2}$	$P_{cr} = \frac{\pi^2 EI}{4L^2}$	$P_{cr} = \frac{4\pi^2 EI}{L^2}$	$P_{cr} = \frac{2.046 \pi^2 EI}{L^2}$
			
$L_e = L$	$L_e = 2L$	$L_e = 0.5L$	$L_e = 0.699L$
$K = 1$	$K = 2$	$K = 0.5$	$K = 0.699$

Figure 9: Buckling boundary conditions and corresponding effective buckling lengths for different column end constraints. The effective length factor K depends on whether the column ends are pinned, fixed, free, or partially constrained. Reproduced from [23].

In the developed syringe pump setup, the syringe barrel is fixed near the outlet side, while the plunger is pushed from the opposite side by a flat plunger contact interface. This interface transfers axial load but does not rigidly clamp the plunger or reliably transmit a bending moment. A fixed–fixed assumption would therefore overestimate the plunger stability. For the design calculation, the pinned–pinned case with $K = 1.0$ is used as the reference assumption. The resulting value is interpreted as a conservative design estimate relative to a fixed–fixed model, not as an experimentally verified failure load.

The unsupported plunger length L_p is also relevant because the critical buckling force decreases with the square of the effective length. In practice, the unsupported length changes with the plunger position during the filling sequence. For a conservative design estimate, L_p should therefore be selected as the maximum unsupported length occurring within the relevant operating range. The second moment of area I should also correspond to the weakest bending direction of the actual plunger geometry, especially when the plunger has a ribbed or non-circular cross-section. Because the disposable plunger geometry and contact conditions are simplified in the Euler model, the calculated force is used only as an approximate design limit.

To avoid operation close to the estimated structural limit, a safety factor is introduced. The allowable plunger force is defined as

$$F_{\text{allow}} = \frac{F_{\text{crit}}}{S} \quad (2.13)$$

where S is the safety factor. The process parameters should therefore be selected so that the expected pressure-related and friction-related loads remain below the allowable plunger force. In the present system, this limit is related not only to plunger buckling but also to possible barrel deformation, plunger-seal leakage, Luer Lock interface loading, and failure of the connected filling path.

The buckling calculation is not used as a complete structural verification of the disposable syringe, because the actual plunger geometry, material data, and contact conditions are represented only in simplified form. Instead, it provides a mechanical boundary condition for process planning. Together with the pressure-induced force model, it supports the selection of viscosity-dependent motion parameters that remain within a mechanically reasonable operating range for the syringe, plunger contact interface, and actuator system.

2.3.3 Maximum Allowable Flow Rate

By combining the pressure-induced plunger force with the allowable force, a maximum allowable volumetric flow rate can be estimated. Setting $F_p \leq F_{\text{allow}}$ and rearranging Equation 2.10 gives

$$Q_{\text{max}} = \frac{F_{\text{allow}} r_c^4}{8\mu L_c R_s^2} \quad (2.14)$$

Equation 2.14 is used in this thesis for selecting a flow-rate range that does not intentionally exceed the estimated plunger load limit. For the selected syringe and cannula geometry, the allowable flow rate decreases with increasing viscosity and cannula length, while it increases strongly with the inner radius of the cannula. A larger syringe barrel radius also reduces the allowable flow rate, because the same pressure drop acts on a larger plunger area and therefore produces a higher axial load.

Only the pinned–pinned, fixed–pinned, and fixed–fixed cases are included in the flow-rate comparison, since the fixed–free case does not represent the support condition of the developed syringe pump. Figure 10 shows the maximum allowable volumetric flow rate as a function of dynamic viscosity for different assumed plunger boundary conditions. The curves should not be interpreted as selectable operating modes of the developed pump. They are included to show the sensitivity of the calculated flow-rate limit to the assumed plunger support condition.

As shown in Figure 10, the maximum allowable flow rate decreases strongly with increasing viscosity. The assumed boundary condition affects the allowable plunger force and therefore the feasible flow rate. The fixed–fixed case gives the highest theoretical flow-rate limit, while the pinned–pinned case is the most conservative of the three displayed cases. Since the actual plunger support in the developed setup cannot be regarded as fully fixed, the pinned–pinned condition is used as the reference case for estimating a mechanically conservative operating range.

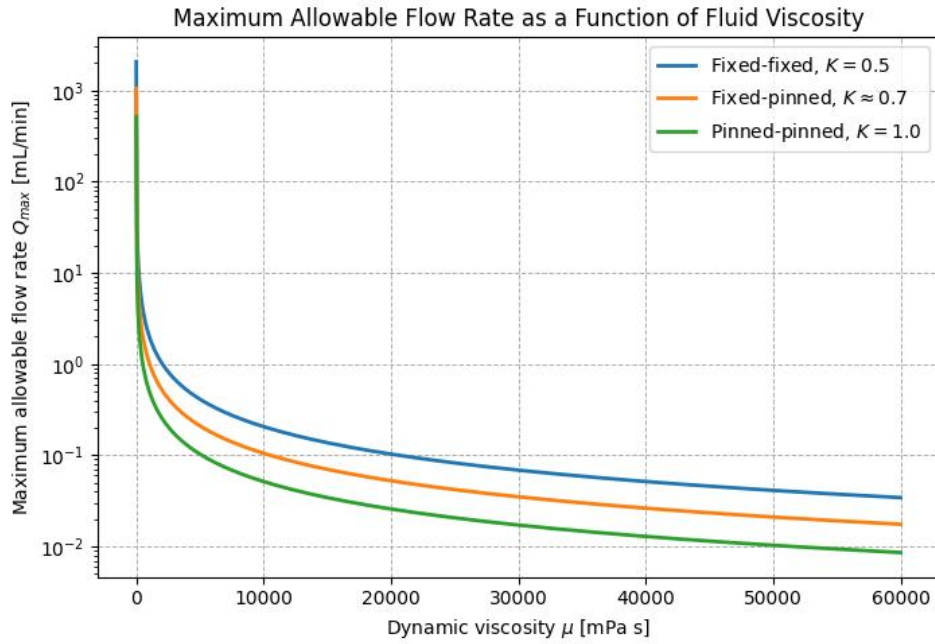


Figure 10: Influence of the assumed plunger boundary condition on the maximum allowable volumetric flow rate as a function of dynamic viscosity. The figure was calculated and created by the author.

For a defined target filling volume, the minimum theoretical extrusion time can be estimated as

$$t_{\text{theor}} = \frac{V}{Q_{\text{max}}} \quad (2.15)$$

where V is the target filling volume. This relationship provides a theoretical lower bound for the extrusion duration under the selected fluidic and mechanical assumptions. In practical operation, the actual extrusion time may be longer because of plunger friction, compliance, start-up effects, and additional safety margins.

2.3.4 Dynamic Load Limitation

In addition to quasi-static extrusion forces, actuator acceleration can introduce dynamic loads into the syringe system. In a complete force balance, the available margin for acceleration would depend on the pressure-induced force, friction-related force components, and the allowable mechanical load. Since these quantities are not directly measured in the present setup, the acceleration limit is not calculated from a complete residual-force balance. Instead, the inertial load is used as a separate design indicator according to Newton's second law:

$$F_{\text{inertia}} = m_{\text{eff}} a \quad (2.16)$$

where m_{eff} is the effective moving mass of the actuator-plunger system and a is the axial acceleration of the plunger.

If the allowable force is used only as an upper reference for the separately considered inertial

contribution, a reference acceleration can be estimated as

$$a_{\text{ref}} = \frac{F_{\text{allow}}}{m_{\text{eff}}} \quad (2.17)$$

This value is not interpreted as the maximum physically admissible acceleration during extrusion, because part of the allowable force is already consumed by pressure-induced and friction-related loads. Instead, it is used as a design indicator for avoiding unnecessarily abrupt start-up, stopping, or velocity transitions. The acceleration limit is therefore treated as a conservative motion-planning constraint rather than as a complete structural force balance.

For the developed syringe pump, the equations in this section are used to constrain the selection of motion parameters before experimental compensation is applied. The actuator profile must deliver the target volume through the small cannula, but the selected flow rate and acceleration must remain compatible with the disposable syringe, the plunger contact interface, and the connected filling path. The control problem is therefore treated as a mechanically constrained extrusion task rather than a purely kinematic positioning task.

2.4 Existing Syringe Pump Systems and Dispensing Concepts

Syringe pump systems are established tools for delivering defined liquid volumes in laboratory, biomedical, and manufacturing environments. Their basic operating principle is to move the syringe plunger linearly with a motor-driven actuator. If the syringe geometry is known, the commanded plunger velocity and travel distance can be converted into a theoretical flow rate and displaced volume. This displacement-based principle makes syringe pumps suitable for infusion, dosing, dispensing, and experimental fluid-handling tasks. In practice, however, the delivered output is not determined by actuator motion alone. It is also affected by syringe mechanics, outlet-path resistance, and the transient behavior of the syringe–fluid system [24, 25].

For low-viscosity fluids and moderate outlet resistance, the displaced volume inside the syringe can often be treated as a reasonable approximation of the delivered volume. This assumption becomes less reliable when high-viscosity media are extruded through a small cannula. Under these conditions, pressure build-up, plunger friction, syringe compliance, cannula flow resistance, and delayed material release can all cause deviations between commanded plunger displacement and actual delivered volume. The nominal positioning accuracy of a syringe pump therefore does not necessarily translate directly into volume accuracy at the cannula outlet.

Existing approaches can be grouped into three categories: conventional syringe pump systems, high-viscosity dispensing and industrial dispensing systems, and manual or semi-automated filling approaches. Conventional syringe pump systems provide syringe-based displacement control, but they usually do not treat viscosity-dependent effects, pressure-induced deformation, or delayed extrusion as primary process variables. High-viscosity dispensing and industrial dispensing systems address viscous-material metering more directly, but many of them are based on dispensing, potting, progressive-cavity, valve-based, or piston-metering principles. They do not usually use a disposable syringe as both the fluid reservoir and the

mechanical actuation interface. Manual and semi-automated filling approaches are flexible during early laboratory work, but they depend strongly on operator handling and provide limited process traceability.

These approaches are not considered direct replacements for the platform developed in this thesis. They are used instead to define the technical gap between general syringe-based displacement control and application-specific high-viscosity implant filling. For the filling task considered in this work, the required platform must link the target filling volume, fluid viscosity, syringe geometry, cannula resistance, mechanical constraints, and experimentally determined compensation factors to an executable and verifiable actuation strategy. Section 3.2 will compare these existing approaches using representative commercial examples and figures.

2.5 Control Strategies for Extrusion Processes

In the present filling task, extrusion control addresses the mismatch between commanded plunger motion and material output at the cannula outlet. The controller commands the syringe plunger, but the actual extrusion behavior depends on the coupled interaction of plunger displacement, fluid viscosity, cannula resistance, plunger friction, syringe compliance, and transient pressure build-up. For this reason, the control problem in high-viscosity syringe filling cannot be reduced to a purely kinematic positioning task. It must also define how the plunger motion is planned, limited, and observed during the filling sequence.

In syringe-based dispensing, open-loop displacement control is a common and direct control strategy. Its basic principle is to convert a target delivery volume into an axial plunger displacement based on the known syringe barrel geometry. The actuator then executes this displacement without continuously correcting the plunger motion from real-time process feedback, such as measured flow rate, pressure, or dispensed volume. This approach is simple, transparent, and cost-effective. It can be sufficient for many standard syringe dispensing tasks, especially when the flow resistance is low, the outlet path is not strongly restricted, and the actual fluid output follows the commanded plunger motion with acceptable accuracy.

The filling task considered in this thesis goes beyond these typical operating conditions. When high-viscosity oil is extruded through a small cannula, the flow resistance increases substantially. Pressure build-up, plunger friction, syringe compliance, outlet resistance, back pressure, and delayed extrusion can all contribute to deviations between commanded plunger displacement and actual delivered volume. These effects become more relevant when the target filling volume is small, because even a short delay or a small elastic deformation may represent a noticeable fraction of the target volume. Studies on injection force and injectability also show that fluid viscosity, syringe geometry, needle or cannula dimensions, plunger friction, and back pressure influence the force required for syringe-based delivery [17, 18, 19]. Open-loop displacement control is therefore used as the starting point rather than as a complete control description. In the developed system, it is extended with viscosity-dependent actuation parameters and compensation to reduce the conversion error between plunger displacement and actual output under high-viscosity, small-cannula conditions.

Because direct flow and pressure feedback are not available in the present system, model-based feedforward planning is used as an extension of open-loop displacement control. This planning step defines suitable actuation parameters before the filling sequence starts, using the target filling volume, syringe geometry, cannula geometry, and fluid viscosity as input

information. The fluid-mechanical and force-transmission models introduced in the previous sections provide estimates of pressure demand, plunger force, and feasible flow-rate range. These estimates help to avoid using the same fixed plunger velocity for all viscosity conditions. Related work on high-viscosity robotic fluid deposition has also shown that model-based compensation can be useful when material response and system dynamics create deviations between actuator commands and actual output [26]. In contrast to systems with vision-based flow sensing or real-time output correction, the present thesis uses a pre-parameterized feed-forward strategy. The model is used to constrain and adjust syringe actuation before extrusion, making open-loop displacement control more suitable for high-viscosity oil delivery through a small cannula.

For high-viscosity syringe filling, the shape of the actuator motion profile is also relevant. Abrupt velocity changes may introduce unnecessary dynamic loading, promote pressure peaks, or increase deformation at the syringe plunger and connected filling path. The motion command should therefore include defined acceleration and deceleration phases. The target velocity should not be reached by an instantaneous step change. In the developed system, this motion planning is used to reduce abrupt mechanical loading and to keep the extrusion process within a mechanically reasonable operating window. The purpose of the motion profile is not to implement a fully optimized dynamic controller, but to provide a controlled and reproducible actuation sequence that is compatible with the disposable syringe, the plunger contact interface, and the small cannula.

The control strategy implemented in this work is described as model-based open-loop extrusion control. In this strategy, the extrusion command is not defined directly by a fixed velocity or a fixed dispensing time. Instead, it is calculated before the start of extrusion based on the selected filling volume, fluid viscosity, syringe geometry, cannula geometry, and experimentally determined compensation factors. The geometric volume relation is used to convert the target filling volume into a theoretical plunger displacement, while the fluid-mechanical and force-transmission models are used to estimate the pressure demand, plunger force, and acceptable actuation range. Since no real-time flow or pressure feedback is used in this work, the extrusion process is not corrected during execution based on the actual output state. Systematic deviations between the theoretical model and the practical filling result are therefore addressed mainly through predefined compensation factors. The resulting strategy combines geometric volume calculation, viscosity-dependent process planning, mechanical load limitation, and experimental compensation for high-viscosity filling through a small cannula.

2.6 Embedded Systems and User Interfaces for Laboratory Devices

In a viscosity-controlled syringe pump, the embedded system not only drives the actuator but also converts the user-defined filling task into a reproducible process sequence. Compared with a computer-based test bench, a compact embedded platform integrates actuator control, motor-driver communication, current monitoring, parameter handling, and user interaction directly in the device. This integration is important for the present filling task because the target volume, viscosity, syringe geometry, cannula geometry, compensation factors, and motion limits must be processed consistently before the syringe plunger is moved. The embedded controller therefore links the selected process parameters to the physical extrusion process.

For reproducible laboratory operation, the user input must be clearly linked to the actuator behavior. If motor speed or extrusion duration is adjusted manually without a defined relation to displacement, velocity, and compensation parameters, the filling result may vary between runs. The graphical user interface therefore guides the user through the required inputs, shows the corresponding physical units, limits parameter ranges, and displays calculated process values before the filling sequence is started. In this way, the interface supports visibility of the system state, error prevention, recognition rather than recall, and clear user feedback, which are central usability principles described by Nielsen [27].

A user interface for experimental syringe filling should distinguish between routine operation and advanced configuration. In routine operation, the user should only interact with the main process variables, such as the target filling volume and fluid viscosity. This reduces the risk of unintentionally changing to internal calculation parameters and supports repeatable operation. In contrast, the advanced expert mode provides access to syringe dimensions, cannula geometry, motor settings, mechanical limits, and compensation factors. This separation is useful during process development because the system must support both standardized filling sequences and flexible parameter adjustment.

Process visualization is also required because high-viscosity extrusion can involve long filling times, delayed flow initiation, and passive post-extrusion behavior. During operation, the user needs information about the current process state, remaining duration, active motion phase, and relevant actuator status. In the syringe pump developed in this work, this information can include motor-driver data from the TMC5160-TA, current-related electrical measurements from the INA228, commanded or driver-reported actuator states, planned motion phases, and process duration. These values improve process transparency and support later interpretation of experimental results.

The embedded software must handle motor control, motor-driver communication, display updates, and user interaction at the same time. Motor control and motor-driver communication are time-critical tasks because they directly affect plunger motion. In contrast, display updates and graphical user-interface rendering should not take priority over actuator control during active extrusion. If graphical updates or interaction routines occupy too much processing time, motor-command updates may be delayed. This can cause the actual motion to deviate from the planned velocity profile. The software structure should therefore separate actuator control from non-critical display tasks, so that the motor motion can be executed reliably. This separation is especially important during ramped motion. The motor should not be forced into abrupt velocity changes.

Parameter validation and data consistency provide an additional layer of process reliability. When high-viscosity media are extruded through a small cannula, the pressure demand and plunger load can increase substantially. The embedded system should therefore check before extrusion whether the selected parameters remain within the defined operating range. This includes the target volume, plunger velocity, acceleration, syringe geometry, cannula dimensions, and the mechanical load estimated by the model. In addition, the input parameters and internal calculation results used for each filling sequence should remain identifiable, including extrusion time, plunger displacement, velocity limit, and compensation factor. For the viscosity-controlled syringe pump developed in this thesis, the embedded system and graphical user interface form the central interface between model-based parameter calculation, mechanically constrained actuation, process visualization, and reproducible laboratory operation.

2.7 Validation and Quality Management in Medical Device Development

Validation and quality-management principles are used in this thesis as a structured basis for evaluating the developed syringe pump. The system is not treated as a certified medical device and is not intended to represent a manufacturing system for commercial production. Here, validation refers to the engineering evaluation of a laboratory prototype. It examines whether the platform fulfills the defined functional and process-related requirements, including controlled syringe-based filling of implant components and extrusion of high-viscosity media through a small cannula. The focus is not regulatory approval. Instead, the validation provides a traceable basis for assessing motion accuracy, filling accuracy, repeatability, parameter handling, user interaction, and process documentation under defined laboratory conditions.

For the experimental work in this thesis, verification and validation must be distinguished. Verification addresses whether the system has been implemented according to the defined technical requirements. This includes actuator displacement, motor direction, parameter calculation, graphical user-interface functions, and the display of driver or sensor data. In contrast, validation addresses whether the integrated system is suitable for its intended application. In the present work, this means evaluating whether the developed syringe pump can repeatedly and controllably deliver defined volumes of high-viscosity oil through a small cannula under defined laboratory conditions. This distinction follows a quality-management-oriented view of medical-device development, where requirements, design outputs, test procedures, and documented results should remain traceable throughout the development process [12].

Before the prototype can be validated, the system requirements must be defined clearly. In this work, these requirements include small-volume liquid delivery, compatibility with high-viscosity media, limitation of mechanical load, repeatable actuator motion, correct parameter handling, and understandable operation through the graphical user interface. Each requirement should be linked to a suitable test method or evaluation criterion. This allows the result to be assessed objectively, rather than assuming that the requirement is fulfilled because the prototype completes a single filling operation. For the syringe pump developed in this thesis, this requires consistent documentation of the target volume, viscosity, syringe geometry, cannula geometry, motion settings, compensation factor, and measured filling result. This traceability is especially important for high-viscosity filling, because the final filling error can result from several coupled factors, including model assumptions, plunger friction, syringe compliance, temperature-dependent viscosity, leakage, and residual droplets during measurement.

Although this thesis does not establish a complete regulatory risk-management file, risk-based thinking is still useful for prototype design and validation. The aim is not to perform medical-device risk certification, but to identify engineering failure modes that may affect filling accuracy, system integrity, or user operation. Relevant examples include excessive pressure demand, plunger deformation, leakage at the syringe or Luer Lock interface, incorrect parameter input, unintended motor direction, unstable motion execution, and misinterpretation of process information by the user. These considerations support the definition of parameter limits, safety margins, software checks, and test conditions. They also explain why the filling process is treated in this work as a mechanically constrained extrusion task, rather than as a simple plunger displacement command.

Validation of filling accuracy and repeatability requires a consistent and reproducible mea-

surement procedure. A single accurate filling result is not sufficient to demonstrate process reliability. Repeated trials are therefore performed with defined parameter settings. The measured filling volume is then compared with the target volume. Since this work involves small liquid volumes, the weighing procedure and volume conversion are sensitive to the measurement conditions. Taring, balance resolution, density assumptions, environmental conditions, residual droplets, and the order of operation can all influence the calculated filling result. For high-viscosity media, the actual output volume may also be affected by pressure build-up, elastic relaxation, plunger friction, temperature-dependent viscosity, and delayed residual release. Filling error and repeatability must therefore be interpreted with regard to both the measurement procedure and the non-ideal behavior of the syringe–fluid system.

Software behavior, user interaction, and process visualization are also part of the validation scope, because they directly affect how the filling sequence is parameterized and executed. The graphical user interface must process input values correctly, display consistent units, apply parameter limits, update calculated process values, and execute the selected motion mode as expected. At the same time, the functional boundary of process monitoring must be stated clearly. In the current system, motor-driver data from the TMC5160-TA and current-related measurements from the INA228 are mainly used to make the process state more transparent and to support documentation of the operating state. These signals are not used for automatic fault detection, closed-loop pressure control, closed-loop flow control, or online correction of the extrusion command. For the viscosity-controlled syringe pump developed in this thesis, validation therefore goes beyond the final filling result. It also covers requirements, parameter consistency, mechanically constrained actuator motion, measurement repeatability, and documentation of the process state. This broader validation scope supports evaluation of high-viscosity filling at laboratory scale.

2.8 Summary of Technological Gaps

The preceding sections have shown that precise syringe-based filling through a small cannula is not determined by actuator displacement alone, particularly when high-viscosity media are used. It is affected by the combined influence of fluid viscosity, cannula geometry, syringe mechanics, plunger friction, actuator motion, parameter handling, and measurement conditions. Conventional syringe pumps provide a useful basis for controlled displacement and volume-based dispensing. However, their standard operating logic is usually based on syringe size, target volume, flow rate, or dispensing duration. For the filling task investigated in this thesis, this is insufficient. Viscosity directly affects the pressure demand, feasible plunger velocity, extrusion duration, and mechanical load in the disposable syringe system. High-viscosity filling therefore requires more than accurate displacement control. It requires a process-specific control strategy that links fluid properties, geometry, mechanical limits, and compensation parameters.

A central technical gap is therefore the limited connection between fluid-mechanical process modeling and actuator motion planning. For example, the Hagen–Poiseuille model describes how the pressure drop depends on viscosity, flow rate, cannula length, and cannula radius. However, this information is not necessarily converted into feasible motion commands in general-purpose syringe pumps [14, 18, 17]. In the application considered in this thesis, the small cannula is both a connection interface and a major factor in pressure demand and

plunger force. This indicates that the control strategy should link the target volume, viscosity, syringe geometry, cannula geometry, and mechanical force limits before extrusion starts. Without this integration, the user may select a plunger velocity that is either mechanically unsuitable for high-viscosity oils or unnecessarily slow for lower-viscosity media.

Another limitation concerns the handling of non-ideal and transient behavior. High-viscosity extrusion may involve start-up delay, compliance of the syringe and flow path, residual pressure release, plunger friction, leakage, temperature-dependent viscosity, and deviations between the theoretical plunger displacement and the measured delivered volume. These effects are especially relevant for small target volumes. Even a small absolute deviation can then produce a noticeable relative error. Existing syringe-pump systems can provide accurate motorized plunger motion, but they usually do not combine viscosity-dependent parameter calculation, syringe-specific load constraints, experimentally determined compensation factors, and an application-specific filling workflow in a compact laboratory platform. For the application considered in this thesis, displacement accuracy is therefore only one part of the problem. The non-ideal behavior of high-viscosity filling must also be considered during parameter planning and compensation under mechanically constrained conditions.

Based on these gaps, the technical aim of this thesis is not to build a complete closed-loop extrusion controller. The developed system does not include direct inline flow or pressure measurement. Instead, this work develops a model-based open-loop filling platform. The platform retains open-loop actuation as the basic control principle while adding viscosity-dependent process planning, mechanically constrained actuator motion, calibration-based compensation, structured user interaction, real-time process visualization, and validation documentation. These considerations define the development target for the viscosity-controlled syringe pump described in the following chapters: reproducible filling of high-viscosity media through a small cannula under defined laboratory conditions, with traceable process parameters, operating limits, and validation results.

3 Need for Action

The preceding chapters showed that the investigated filling task cannot be described sufficiently by ideal plunger displacement alone. Under high-viscosity conditions, the delivered volume depends on the combined behavior of the filling medium, the narrow cannula, the disposable syringe, the actuator, and compliance-related relaxation effects.

This chapter uses this gap to derive the practical development needs of the syringe pump platform. It does not present the final design solution. Instead, it defines the application-specific constraints that make a dedicated viscosity-controlled syringe pump necessary. The following sections examine the limitations of existing syringe-based filling approaches and translate them into requirements for motion-parameter adaptation, force transmission, process transparency, and user guidance.

3.1 Application-Specific Filling Requirements

The investigated filling process is defined by small target volumes, highly viscous oils, a small implant-related filling interface, and the use of a standardized disposable syringe as the fluid container and actuator interface. The application requires reproducible syringe-based delivery of high-viscosity oil through a small cannula under laboratory conditions. Due to the small dimensions of the implant component, a narrow cannula is required as the transfer interface between the syringe and the filling region. In this thesis, the implant geometry defines the application need and the boundary conditions for the filling process, while the engineering focus remains on controlled syringe–cannula extrusion of the high-viscosity medium. The internal filling behavior of the implant component itself is not investigated in detail.

A fixed boundary condition of this thesis is the use of a B. Braun Injekt Luer Lock Solo 2 mL disposable syringe, as shown in Fig. 11. For the present process, the relevant syringe features are the barrel, barrel flange, plunger, plunger flange, seal, and Luer Lock outlet. These features define the available filling volume, the displacement-to-volume relation, the actuator contact region, and the standardized screw connection for the cannula. The Luer Lock outlet provides a leak-tight mechanical connection between the syringe and the cannula when properly assembled. Unlike reinforced high-pressure dosing systems, the investigated setup must operate with a small disposable plastic syringe, narrow cannula geometries, and oils with viscosities up to 60,000 mPa·s. These conditions increase the sensitivity of the process to pressure build-up, mechanical deformation, plunger friction, and delayed material release.

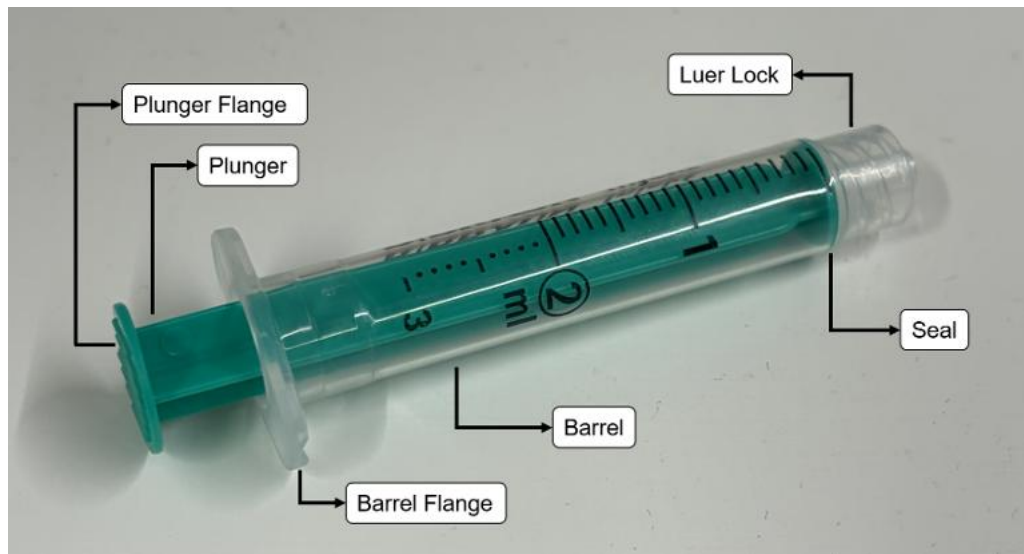


Figure 11: B. Braun Injekt Luer Lock Solo 2 mL disposable syringe used as the application-specific fluid container and mechanical interface in the investigated filling process. The figure was created by the author.

The main engineering challenge results from the fact that commanded plunger displacement and delivered volume are not necessarily identical under high-viscosity conditions. During extrusion, the cannula generates a viscosity- and geometry-dependent flow resistance. The resulting counterpressure can lead to elastic deformation of the syringe barrel, plunger, fixture, and actuator structure. After the actuator motion has stopped, stored elastic energy may still drive a delayed outflow from the cannula. In addition, misalignment, plunger friction, and local leakage may further affect the effective delivered volume. The filling result is therefore governed by the coupled behavior of the fluid, syringe, cannula, and mechanical drive system.

Accordingly, the scope of this thesis is limited to the controlled syringe–cannula extrusion process before the fluid enters or fills the implant component. The validation focuses on the relationship between commanded actuator motion and delivered outlet volume under defined syringe, cannula, viscosity, and laboratory boundary conditions. Effects inside the implant component, such as internal fluid distribution, implant deformation, or pressure-dependent behavior of a closed implant cavity, are outside the main experimental scope of this work.

For the validation of the developed syringe pump platform, a target filling volume of 1 mL and a maximum permissible deviation of 1% are defined as the primary accuracy criterion. This criterion is used as an engineering validation target for the laboratory system developed in this thesis. It does not represent a general product requirement for all urological implant variants unless such a requirement is specified separately by the corresponding product or manufacturing specification.

Based on these boundary conditions, the syringe pump platform must fulfill the following application-specific requirements:

1. Reproducible extrusion of small liquid volumes using the specified 2 mL disposable syringe
2. Accurate filling of a 1 mL target volume with a deviation not exceeding 1%

3. Processing capability for high-viscosity oils up to 60,000 mPa·s
4. Stable force transmission from the motor-driven actuator to the syringe plunger
5. Mechanical support of the syringe to reduce deformation, tilting, and misalignment
6. Consideration of compliance-related effects during and after extrusion
7. Viscosity-dependent adjustment of motion parameters for different filling conditions
8. Transparent process operation through visualization of relevant input and output parameters

Overall, the developed system must move the plunger in a controlled way, provide sufficient pushing force, adapt the motion parameters to the selected viscosity, and make the applied process settings visible to the user. These points also define the criteria against which existing filling approaches are assessed in the next section.

3.2 Limitations of Existing Filling Approaches

Existing filling approaches provide an established technical basis for controlled liquid dispensing, but they are not fully aligned with the application-specific requirements of the investigated implant filling process. The purpose of this section is not to argue that existing systems are generally unsuitable for precise liquid handling. Instead, the limitations are analyzed with respect to the specific combination of a standardized disposable syringe, narrow cannula-based extrusion, high-viscosity oils, and a validation-oriented laboratory workflow. Three categories are considered: conventional laboratory syringe pumps, high-viscosity dispensing systems, and manual or semi-automated filling approaches.

3.2.1 Conventional syringe pump systems

Conventional syringe pump systems are widely used for controlled metering tasks in laboratory, research, and clinical settings. Representative commercial examples include laboratory syringe pumps, such as the New Era NE-1000, and clinical syringe infusion systems, such as the B. Braun Perfusor Space shown in Fig. 12. These systems use the same basic operating principle. User inputs are converted into plunger motion commands based on the syringe geometry, target volume, and flow-rate settings. For many low-load applications, this approach is sufficient, because the volume displaced inside the syringe can usually be treated as a reasonable approximation of the volume delivered at the outlet.

However, for the high-viscosity filling process investigated in this work, this assumption is not sufficient. The present application depends not only on the geometric relation between syringe diameter and plunger displacement, but also on cannula flow resistance, oil viscosity, disposable syringe compliance, and the load-dependent behavior of the drive structure. Under these conditions, nominal actuator accuracy does not necessarily lead to volume accuracy at the cannula outlet. Part of the commanded plunger displacement may first be absorbed by elastic deformation of the syringe barrel, the plunger interface, the fixation structure, or the actuator structure before the fluid is actually extruded.



Figure 12: Commercial B. Braun Perfusor Space syringe pump as an example of a standard medical syringe-based infusion and dispensing system. Reproduced from B. Braun [28].

Conventional syringe pump systems are also limited by their general-purpose operating logic. Most of them execute predefined flow-rate or volume commands for different syringes and fluids. They are usually not designed to account for application-specific effects such as delayed outflow after the actuator stops, pressure-induced syringe deformation, or viscosity-dependent changes in extrusion time. When high-viscosity media are extruded through a narrow cannula, suitable process parameters may therefore still have to be determined empirically.

For this thesis, the relevant limitation is not the basic positioning capability of conventional syringe pumps. It is the mismatch between general displacement-based operation and the coupled behavior of the syringe–cannula–fluid system during implant filling. The required system must link the selected filling volume and viscosity to specific actuator settings, including extrusion time, motion profile, and the displayed process state.

3.2.2 High-Viscosity Dispensing and Industrial Dispensing Systems

High-viscosity dispensing and industrial dispensing systems are mainly used when viscous materials must be metered, dispensed, or potted in a controlled manner. Representative examples include the preeflow eco-PEN XS 180 shown in Fig. 13 and the Atlas Copco piston dispensing or potting system shown in Fig. 14. Such systems are commonly used for materials such as adhesives, sealants, resins, silicones, greases, and thermally conductive pastes. They are technically relevant to this thesis because they show that precise metering of high-viscosity materials is already a recognized engineering task when viscosity and flow resistance become limiting factors.

However, the design logic of these industrial dispensing systems differs from the platform required in this thesis. They are often based on progressive-cavity dispensing, piston metering, potting, or valve-based dispensing principles. They do not usually use a standardized disposable syringe as both the fluid reservoir and the mechanical actuation interface. These systems can address the metering of viscous materials, but their mechanical architecture, material supply concept, process integration, and control objectives do not directly match the syringe-based filling task investigated in this work.

For this thesis, the main issue is not whether a system can dispense high-viscosity material in general. The filling process must be reproducible under the boundary conditions of a B. Braun Injekt Luer Lock Solo 2 mL disposable syringe, a small cannula, and a mechanically limited actuation structure. In this configuration, syringe barrel compliance, plunger friction, the Luer



Figure 13: Preeflow eco-PEN XS 180 high-viscosity precision dispensing system as an example of an industrial viscous-material dispensing approach. Reproduced from ViscoTec/preeflow [29].

Lock interface, cannula flow resistance, and residual pressure release can all affect the relation between commanded plunger displacement and actual delivered volume. Industrial high-viscosity dispensing systems can serve as reference cases for viscous-material metering, but they cannot directly replace the viscosity-controlled syringe-based filling platform required in this work.

The technical gap addressed in this thesis lies between conventional syringe pump systems and industrial high-viscosity dispensing systems. Conventional syringe pumps provide syringe-based displacement control, but they usually do not include viscosity-dependent compensation for high-viscosity filling through a small cannula. Industrial dispensing systems are designed for viscous-material metering, but they usually do not match the boundary conditions of a disposable syringe-based implant filling workflow. The required platform must convert filling volume, fluid viscosity, cannula geometry, mechanical load constraints, and experimentally determined compensation factors into an executable and verifiable actuation strategy for a specific actuator–syringe–cannula configuration.

3.2.3 Manual and semi-automated filling

Manual and semi-automated filling approaches are the most direct ways to transfer fluid into a component. In the context of this thesis, manual filling can be represented by direct operator actuation of the B. Braun Injekt Luer Lock Solo 2 mL disposable syringe. This approach can be attractive during early laboratory development, because it requires little equipment, allows direct operator intervention, and can be adapted quickly to experimental setups that are not yet fixed. For low-viscosity fluids, larger filling volumes, or non-critical preparation steps, manual syringe actuation may be sufficient.

However, manual actuation introduces clear operator dependence for the high-viscosity implant filling process investigated in this work. The applied force, plunger feed rate, stopping position, and post-extrusion waiting time can vary between operators. When the oil viscosity



Figure 14: Atlas Copco piston dispensing system as an example of industrial metering technology for viscous materials. Reproduced from Atlas Copco [30].

is high and the cannula flow resistance is significant, these variations affect pressure build-up, syringe compliance, elastic relaxation, and delayed outflow. As a result, the actual delivered volume may differ even when the same target volume is used.

Semi-automated filling approaches can reduce part of this operator variability by using dispensing equipment. For example, the Nordson EFD UltimusPlus shown in Fig. 15 can be used as a representative semi-automated dispensing system. Such systems can make fluid delivery more consistent than purely manual operation by controlling dispensing pressure, time, or dispensing cycles. However, if the motion parameters are not calculated and executed based on the filling volume, viscosity, syringe geometry, and cannula geometry, the process still provides only limited application-specific reproducibility. These approaches also often provide limited process-state information, such as expected extrusion time, velocity setting, compensation factor, or the influence of viscosity on the motion settings. This makes parameter tracing and interpretation of validation results more difficult.



Figure 15: Nordson EFD UltimusPlus dispensing system as an example of a semi-automated dispensing approach for controlled fluid delivery. Reproduced from Nordson EFD [31].

For a validation-oriented laboratory workflow, the main limitation of manual and semi-automated approaches is not their ability to complete a single filling operation. The limitation is

that they still depend strongly on operator experience, visual judgment, or simplified preset dispensing parameters. High-viscosity implant filling requires defined input parameters, repeatable actuator motion, documented process settings, and sufficient process transparency for assessing filling accuracy and repeatability. Based on these requirements, manual and semi-automated filling approaches do not provide the process stability and traceability required for this work.

3.2.4 Derived need for a dedicated viscosity-controlled platform

The comparison above shows that a dedicated system is not needed because syringe pumps or high-viscosity dispensing equipment are unavailable. Conventional syringe pump systems already provide accurate syringe-based displacement control. High-viscosity dispensing and industrial dispensing systems can provide higher force, pressure capability, or specialized dispensing principles for viscous materials. Manual and semi-automated methods can also be useful during early experimental work. However, these approaches do not match the specific boundary conditions of this thesis. The required system is not simply a more powerful commercial device, but a dedicated filling platform for a specific actuator–syringe–cannula configuration. This platform must combine viscosity-dependent parameter calculation, mechanically constrained actuation, experimental compensation, and process traceability. The main characteristics and limitations of the existing approaches are summarized in Table 1.

The required system must retain the advantage of controlled actuator motion while adding application-specific parameter adaptation. Rather than executing plunger displacement at a fixed speed or for a fixed duration, it should derive suitable motion parameters from the user-defined filling volume and fluid viscosity. The system must also transfer the actuation force reliably to the syringe plunger. In addition, the parameter planning and compensation strategy must account for possible deviations between commanded displacement and actual delivered volume during high-viscosity extrusion. To support a traceable laboratory workflow, the system should display the relevant input parameters, calculated output parameters, and process-state information. Examples include the expected extrusion time, plunger displacement, velocity setting, and compensation factor.

These limitations motivate the dedicated viscosity-controlled syringe pump platform developed in this work. The following sections further discuss the gap between ideal displacement control and practical high-viscosity extrusion. Based on this gap, the requirements for motion parameter adaptation, process monitoring, and user guidance are derived.

3.3 Gap Between Ideal Displacement Control and Practical High-Viscosity Extrusion

Ideal displacement control is based on a direct geometric relation between plunger travel and displaced fluid volume. In an ideal syringe system, the delivered volume (V) is determined by the syringe cross-sectional area (A_s) and the commanded plunger displacement (s), according to Equation 2.1. This relation forms the basis of many displacement-controlled syringe pump systems. If the syringe body is rigid, the fluid is incompressible, the outlet resistance is low,

Table 1: Simplified comparison of existing filling approaches relevant to the present filling task

Approach	Typical use and main limitation
Conventional syringe pumps	Typical use: Programmable syringe-based dosing based on syringe geometry, target volume, flow rate, or dispensing duration. Main Limitation: ■ Limited consideration of viscosity-dependent flow resistance. ■ Limited compensation of disposable syringe compliance and plunger friction. ■ Insufficient consideration of cannula-induced backpressure and delayed outflow.
High-viscosity dispensing and industrial dispensing systems	Typical use: Metering, dispensing, potting, or dosing of viscous materials such as adhesives, silicones, greases, sealants, resins, or thermally conductive pastes. Main Limitation: ■ Usually based on industrial dispensing, piston metering, valve-based, or progressive-cavity principles. ■ Not primarily designed around a disposable 2 mL syringe as both fluid reservoir and actuation interface. ■ Does not directly match the small-cannula implant-filling workflow investigated in this thesis.
Manual and semi-automated filling	Typical use: Flexible low-cost filling during early experiments, preparation steps, or simple laboratory transfer tasks. Main Limitation: ■ Strong operator dependence. ■ Limited repeatability of feed rate, stopping point, and waiting time. ■ Limited traceability of process parameters and process-state information during validation.

and no leakage or delayed outflow occurs, the commanded displacement can be assumed to correspond closely to the delivered volume.

This ideal relation becomes insufficient when highly viscous oil is extruded through a narrow cannula. The cannula introduces a flow resistance that depends on both the fluid viscosity and the cannula geometry. As viscosity increases, a larger pressure difference is required to achieve the same volumetric flow rate. This pressure acts on the fluid column and is transmitted to the syringe barrel, the plunger interface, the cannula connection, and the mechanical support structure. Consequently, part of the commanded actuator displacement may first be used to build up pressure and deform the mechanical system before the fluid is discharged at the cannula outlet.

The resulting counterpressure decouples commanded actuator motion from effective outlet volume. In a purely geometric model, every motor step is assumed to generate a proportional volume displacement. Under high-viscosity extrusion conditions, however, the system behaves as a coupled actuator–syringe–fluid system. The syringe barrel may expand slightly, the plunger interface may deform, the fixture may deflect, and friction may delay the transfer of actuator motion to the fluid. These effects are small in low-load dosing applications, but they become relevant when a highly viscous oil is forced through a narrow cannula using a disposable plastic syringe.

Preliminary filling tests confirmed that this decoupling can produce a process-relevant volumetric deviation. For a target filling volume of 1 mL under high-viscosity conditions, the difference between the commanded displacement volume and the delivered volume was observed to reach approximately 30% before process-specific compensation was applied. This value should not be interpreted as the deformation of a single component. Rather, it represents a global process deviation resulting from the combined effects of syringe compliance, plunger interface deformation, fixture and drive stiffness, cannula-induced backpressure, friction, and delayed material release. The observation confirms that the investigated process cannot be described sufficiently by ideal syringe geometry alone.

The gap between commanded displacement and delivered volume also continues after the actuator motion has stopped. If elastic energy has been stored in the syringe, plunger, fluid column, or mechanical structure during extrusion, the system may continue to release material from the cannula outlet. This delayed outflow is not represented in a simple displacement model, because the model assumes that the process ends when the actuator reaches its target position. In practice, the final delivered volume may depend on the pressure state at the end of the motion, the relaxation behavior of the syringe system, and the waiting time before the filling result is evaluated.

The practical filling result must be understood as a time-dependent extrusion process rather than a pure positioning task. A slower feed motion can reduce some dynamic effects, but it may lead to long process times and does not necessarily eliminate compliance-related deviations. Conversely, increasing the actuator force can overcome flow resistance, but it does not automatically ensure that the delivered volume matches the commanded displacement. The relevant challenge is therefore broader than precise plunger motion; it is to control the extrusion process under load.

The observed gap establishes the need for viscosity-dependent motion parameters and compensation. For the investigated application, the syringe pump must account for viscosity-dependent extrusion resistance, cannula-related pressure build-up, mechanical compliance,

and delayed outflow. The motion parameters cannot be selected only from the target volume and syringe geometry. They must be adapted to the practical extrusion behavior of the complete system. This observation leads directly to the viscosity-dependent motion parameters and compensation strategy discussed in the following section.

3.4 Need for Viscosity-Dependent Motion Parameters and Compensation

The limitations of ideal displacement control show that the motion of the syringe plunger must be adapted to the actual extrusion conditions. In a purely geometric dosing model, the target volume can be converted directly into a required plunger displacement. For the investigated filling process, this conversion is only the starting point. The final filling result also depends on the viscosity of the oil, the flow resistance of the cannula, the mechanical compliance of the syringe system, and the time-dependent release of stored elastic energy. Therefore, the pump cannot rely on a single predefined motion strategy for all filling conditions.

Viscosity is a primary parameter for defining the required motion behavior of the system. An increase in viscosity raises the flow resistance during extrusion and increases the pressure required to move the oil through the cannula. If the same plunger velocity is applied to oils with different viscosities, the resulting pressure build-up, deformation of the syringe system, and delayed outflow may differ substantially. A motion profile that is suitable for a low-viscosity oil may therefore lead to excessive loading, long relaxation times, or inaccurate filling when applied to a highly viscous oil.

The relevant motion parameters include more than the final plunger position. For high-viscosity extrusion, several parameters influence the delivered volume and must be considered as process-relevant variables:

- **Feed velocity:** The feed velocity determines how quickly pressure is generated in the fluid column. A higher velocity may shorten the extrusion time, but it can also increase pressure build-up and compliance-related deformation.
- **Acceleration and deceleration behavior:** The acceleration and deceleration phases influence the dynamic loading of the syringe, plunger interface, fixture, and actuator structure. Sudden changes in motion may increase transient deformation or unstable material release.
- **Stopping strategy:** The stopping behavior affects the pressure state at the end of extrusion. A sudden stop may leave stored elastic energy in the syringe system, whereas a controlled stop can reduce rapid pressure changes.
- **Waiting time after extrusion:** The waiting time determines how much delayed material release is included in the final filling result. This parameter is particularly relevant when residual pressure continues to drive outflow after the actuator has stopped.
- **Volume compensation factor:** A compensation factor can be used to correct systematic deviations between commanded displacement volume and delivered volume, provided that the deviation is repeatable under defined process conditions.

A fixed low-speed strategy is not sufficient as a general solution for the investigated process. Reducing the feed velocity can lower pressure build-up and may improve the stability of extrusion, but it also increases the total process duration. For highly viscous oils, this may result in

impractically long filling times and an inefficient laboratory workflow. In addition, low-speed motion alone does not eliminate compliance-related deviations or delayed outflow. The process requires a motion strategy that balances filling accuracy, repeatability, mechanical loading, and process time.

A compensation concept is required because viscosity-dependent effects cannot be fully avoided by mechanical design or conservative motion settings alone. Even with stable actuation and sufficient pushing force, part of the commanded displacement may be absorbed by elastic deformation or released only after the actuator has stopped. As a result, the relationship between commanded displacement volume and delivered volume must be adjusted based on the observed process behavior. In this context, compensation does not replace precise actuator control. It extends displacement control by accounting for systematic deviations that occur under specific viscosity and loading conditions.

The compensation strategy must be based on repeatable process behavior rather than uncontrolled trial-and-error adjustment. If the deviation between commanded and delivered volume is random, compensation cannot reliably improve the filling result. If, however, the deviation is systematic and reproducible under defined conditions, it can be used to derive correction factors for the motion command. This distinction is important for the present thesis, because the objective is not only to build a pump with sufficient force, but to establish a controllable filling process whose deviations can be identified, reduced, and documented.

The required control concept must link user-defined input parameters to executable motion parameters. The input parameters include at least the target filling volume and the viscosity of the oil. Based on these values, the system must determine suitable actuator settings, estimate and display the expected extrusion duration, and apply compensation where necessary. This creates a direct connection between the user interface, the parameter calculation, and the motor control. The syringe pump platform must therefore function as a parameter-driven process platform that connects user input, compensation logic, and actuator execution for high-viscosity filling.

The need for viscosity-dependent motion parameters and compensation follows directly from the gap between ideal displacement control and practical high-viscosity extrusion. A suitable system must adapt the motion command to the fluid and process conditions, reduce the influence of compliance-related deviations, and support a reproducible validation workflow. These requirements lead to the need for process monitoring and user guidance, as discussed in the following section.

3.5 Need for Integrated Process Monitoring and User Guidance

The need for viscosity-dependent motion parameters also creates a need for transparent process monitoring and user guidance. If the extrusion behavior changes with viscosity, filling volume, cannula resistance, and compensation settings, the operator must be able to see which process parameters are currently applied. The system should execute motor commands while also presenting the relevant input and output information in a clear and traceable way. This is especially important for a validation-oriented laboratory workflow. Reproducibility depends on mechanical design as well as consistent parameter selection and correct operation.

In this thesis, process monitoring mainly means displaying relevant process information. It does not mean that all internal physical quantities are measured directly in closed loop. The developed system does not measure variables, such as pressure, local syringe deformation, or instantaneous outlet flow. Instead, monitoring must give the operator access to the parameters that define the commanded process state. These include the selected filling volume, entered viscosity, calculated extrusion duration, and progress of the filling procedure. This information helps the user check the selected settings before and during operation.

User guidance is needed because the filling process includes several operator steps that can affect the final result. The operator must select the target filling volume, enter the correct viscosity value, prepare the syringe and cannula, start the extrusion procedure, and follow the required waiting or release phase after extrusion. If these steps are not performed consistently, the final filling volume may vary even when the mechanical system works correctly. The user interface should provide more than a panel for starting the motor. It should structure the workflow and present the required inputs, confirmation steps, and process-state information clearly to the operator. This reduces operator dependence and supports repeatability and traceability in the high-viscosity filling process.

Displaying the expected extrusion duration is especially important for high-viscosity filling. Long extrusion times may occur when conservative motion parameters are used. In contrast, faster motion strategies may require compensation and careful handling of the stopping and waiting phases. If the expected process duration is not shown clearly, the operator may stop the process too early, evaluate the filling result at an inconsistent time, or interpret delayed outflow as a malfunction. The calculated duration helps the operator plan the process and evaluate the filling result at a consistent time.

Integrated process information also supports traceability during laboratory validation. A filling result can only be interpreted correctly if the relevant parameter settings are known. These settings include the target volume, viscosity input, selected or calculated motion parameters, and compensation-related values. Even if the system does not provide a complete digital record of all parameters, the user interface should make the applied settings visible and unambiguous. This reduces the risk of undocumented parameter changes and supports a more systematic comparison between filling trials.

User guidance also requires a systematic approach to interface design. In this thesis, Jakob Nielsen's 10 usability heuristics [27] are used as a reference framework for structuring the user interface and evaluating whether the operator is sufficiently informed during the filling process. In particular, the principles of system status visibility, consistency, error prevention, and recognition rather than recall are relevant for the developed syringe pump, because the user must enter process parameters, verify calculated values, follow the extrusion sequence, and observe the post-extrusion waiting or release phase.

For the investigated syringe pump platform, process monitoring and user guidance define more than interface features. They determine how the operator selects parameters, follows the extrusion sequence, and interprets the filling result, and they are part of the process control concept. The interface must connect user inputs, parameter calculation, motor control, and process state visualization into a coherent workflow. This requirement follows directly from the need to operate a high-viscosity filling process with defined parameters, reproducible actuation, and transparent evaluation conditions. The resulting system requirements are summarized and formalized in the following section.

3.6 Consolidated Requirements and Chapter Summary

The preceding sections derived the need for a dedicated viscosity-controlled syringe pump platform from the application-specific filling requirements, the limitations of existing syringe-based approaches, and the gap between ideal displacement control and practical high-viscosity extrusion. The investigated process is not defined by plunger displacement alone, but by the combined influence of fluid viscosity, cannula-induced flow resistance, disposable syringe compliance, delayed material release, and operator-dependent process execution. This final section consolidates these findings into the main requirements for the developed platform and summarizes the role of this chapter as the transition from the technological background to the subsequent system design.

The derived requirements are summarized at the level of main requirement areas. The purpose is to define the functional scope that must be addressed by the developed laboratory system. The requirements focus on the aspects directly relevant to the investigated implant filling process: syringe compatibility, mechanical stability, viscosity-dependent motion control, compensation of repeatable deviations, process transparency, and validation-oriented operation.

Table 2: Consolidated requirements for the viscosity-controlled syringe pump platform

Requirement area	Derived requirement
Syringe compatibility and positioning	■ Ensure compatibility with the specified B. Braun Injekt Luer Lock Solo 2 mL disposable syringe ■ Support repeatable axial positioning of the syringe and plunger
Mechanical stability and force transmission	■ Transmit sufficient pushing force for high-viscosity extrusion ■ Reduce avoidable tilting, misalignment, and compliance-related deformation
Viscosity-dependent motion control	■ Convert the selected filling volume and oil viscosity into suitable motion parameters ■ Include feed velocity, acceleration and deceleration behavior, stopping behavior, and expected extrusion duration
Compensation of repeatable deviations	■ Allow correction of systematic deviations between commanded displacement volume and delivered volume ■ Apply correction under defined and repeatable process conditions
Process transparency and user guidance	■ Display relevant input and output parameters ■ Indicate the current process state. ■ Guide the operator through parameter input, preparation, and extrusion
Validation-oriented operation	■ Support reproducible filling trials with defined parameters ■ Enable consistent operating steps and sufficient traceability for laboratory validation

The requirements summarized in Table 2 show that the development task is not limited to constructing a mechanical actuator for syringe displacement. A purely mechanical solution would not be sufficient, because the filling result is affected by viscosity-dependent extrusion resistance, pressure-induced compliance, and delayed outflow after actuator stop. The required platform must combine mechanical support, controlled force transmission, parameter-driven motion generation, and process-oriented compensation.

The consolidated requirements also define the role of the user interface within the overall system concept. The interface is more than an operating panel for starting and stopping the motor. It connects user-defined process inputs, calculated motion parameters, motor control, and process state visualization. This is necessary because the operator must enter the filling volume and viscosity, verify calculated values, follow the extrusion sequence, and observe the post-extrusion waiting or release phase under consistent conditions.

In summary, this chapter shows that the investigated filling task cannot be addressed suffi-

ciently by conventional displacement-controlled syringe operation alone. The need for action results from the mismatch between ideal syringe dosing assumptions and the practical behavior of high-viscosity extrusion through a narrow cannula using a small disposable plastic syringe. The resulting development task is broader than the construction of a syringe actuator. The platform must hold the disposable syringe repeatably, transmit the actuator force without avoidable misalignment, adapt the motion command to viscosity-dependent resistance, correct repeatable volume deviations, and make the applied process parameters visible to the operator. These consolidated requirements guide the concept development, system design, implementation, and validation presented in the following chapters.

4 Methods

Chapter 3 derived the need for a dedicated viscosity-controlled syringe pump platform from the application-specific filling requirements and the limitations of conventional syringe-based approaches. Based on these requirements, this chapter describes the methodological approach used to develop the syringe pump system. The focus is on translating the identified requirements into a structured engineering method covering the mechanical design, actuator and drive configuration, viscosity-dependent parameter calculation, compensation strategy, user interface design, and experimental validation procedure. The chapter does not present the final performance results. Instead, it defines the design logic and methodological framework by which the developed system was conceived, implemented, and prepared for validation.

4.1 System Architecture of the Syringe Pump Platform

The development of the syringe pump platform followed a requirement-driven engineering approach. The consolidated requirements derived in Chapter 3 were used as the starting point for structuring the mechanical design, drive and actuation concept, user interface, control and compensation logic, and validation procedure. The purpose of this approach was to avoid treating the system as a stand-alone actuator and to develop it instead as an integrated process platform for high-viscosity implant filling.

The overall platform was decomposed into five interacting subsystems to separate the main engineering functions. This decomposition structures the following sections of this chapter. Each subsystem addresses a specific part of the filling process, while the final filling behavior results from the combined interaction of mechanical support, motion generation, user interaction, control logic, and validation. The subsystem structure used in this thesis is shown schematically in Fig. 16.

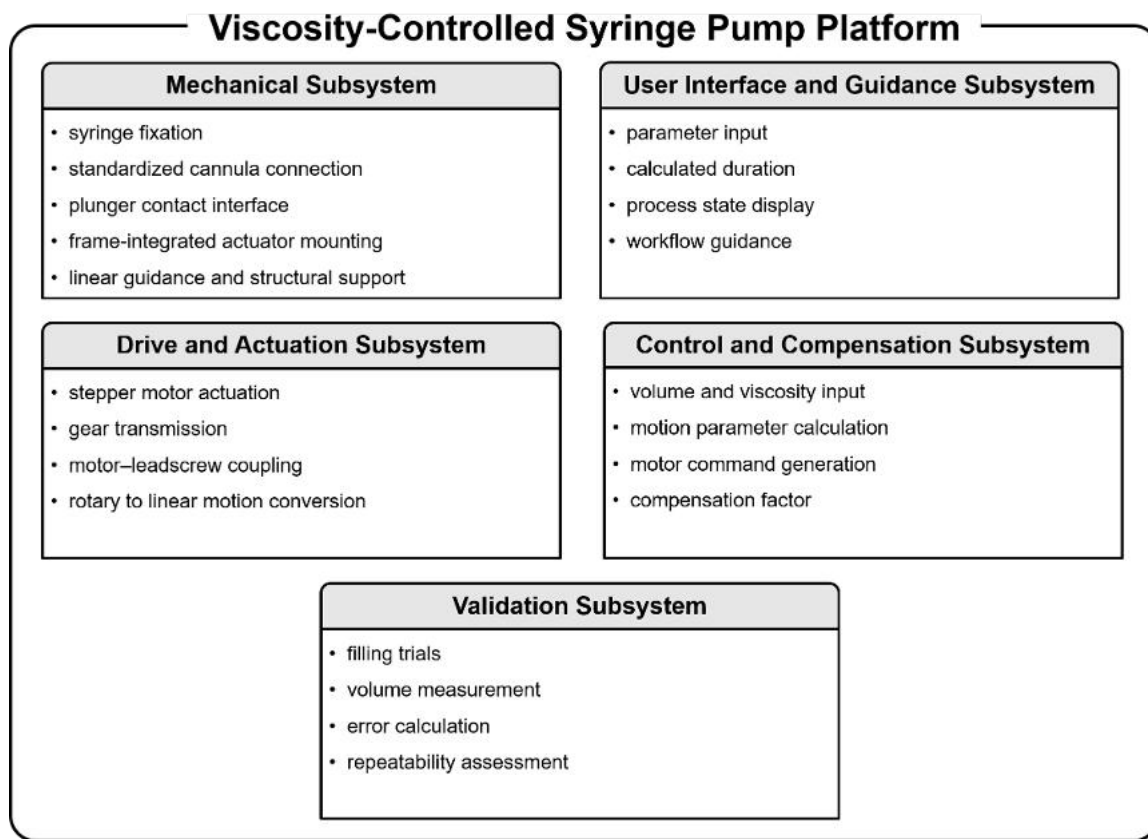


Figure 16: Functional subsystem structure of the viscosity-controlled syringe pump platform used to organize the development method in Chapter 4. The figure was created by the author.

The mechanical subsystem defines the structural interfaces of the syringe pump platform. It includes the syringe fixation, standardized cannula connection, plunger contact interface, frame-integrated actuator mounting, and linear guidance with structural support. Its methodological role is to provide repeatable syringe positioning, a defined plunger contact condition, stable actuator support, and a mechanically consistent boundary for the syringe–cannula assembly. The detailed design of the syringe support, frame, actuator mounting, guide structure, and plunger contact element is described in Section 4.2.

The drive and actuation subsystem defines how the commanded motion is transmitted and converted into plunger displacement. It includes the stepper motor actuation, gear transmission, motor–leadscrew coupling, and rotary-to-linear motion conversion. Its role is to provide reproducible low-speed motion, sufficient displacement resolution, and controlled mechanical actuation under high-viscosity loading conditions. The corresponding motor selection, transmission concept, coupling design, driver configuration, and motion-control implementation are described in Section 4.3.

The control and compensation subsystem defines how user-defined process inputs are converted into executable actuator commands. The relevant inputs include the target filling volume and the viscosity of the oil. Based on these values, the system calculates or selects motion parameters, generates motor commands, and applies compensation factors where required. This subsystem forms the connection between the process definition, the compensation logic, and the physical execution of the filling process. Its method is described in Section 4.4.

The user interface and guidance subsystem provides the operator-facing part of the process control concept. It enables parameter input, displays the calculated extrusion duration and process state, and guides the user through the filling workflow. Its role extends beyond starting and stopping the motor and includes supporting consistent preparation, extrusion, and post-extrusion waiting or release steps. The design and qualitative evaluation of the interface are described in Section 4.5.

The validation subsystem defines how the developed platform is evaluated under laboratory conditions. It includes the filling trials, volume measurement, error calculation, repeatability assessment, and documentation of relevant process parameters. Its role is to verify whether the developed platform can support the defined filling task under reproducible and traceable conditions. The validation method is described in Section 4.6.

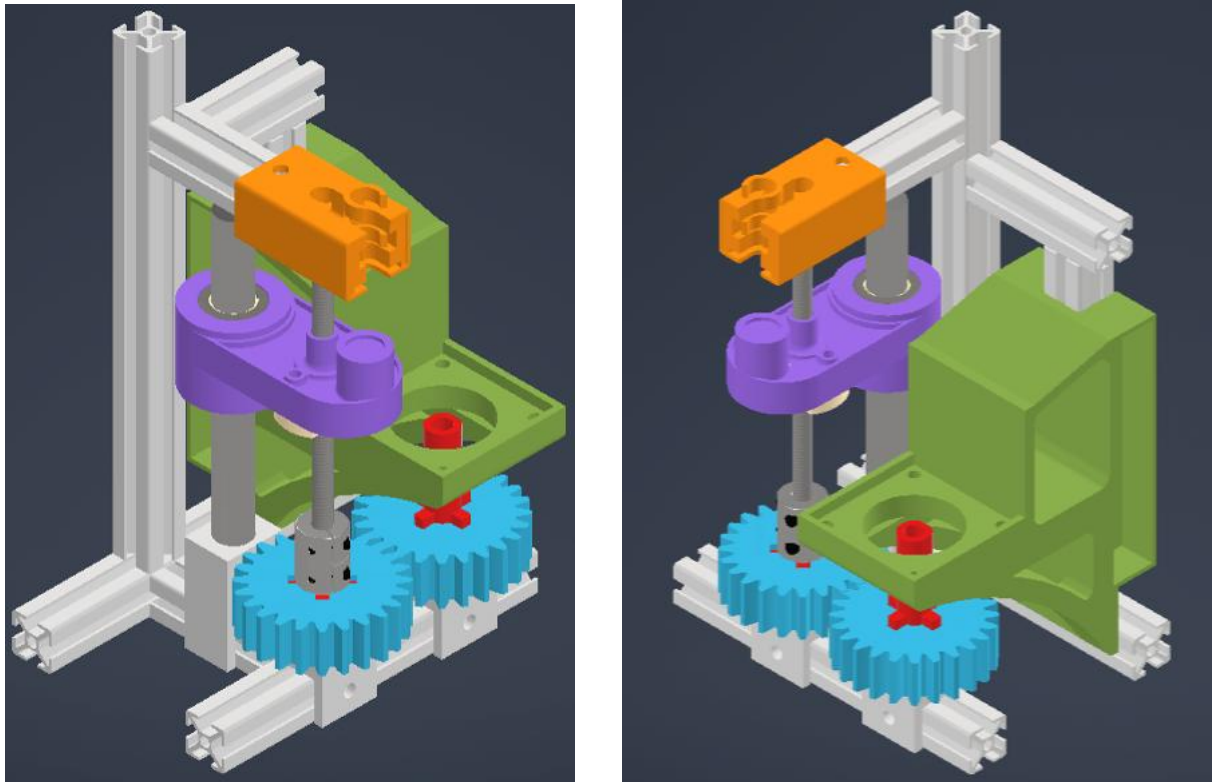
This subsystem-based approach defines the structure of Chapter 4. The following sections describe how the requirements from Chapter 3 are translated into concrete design and method decisions. The chapter focuses on the development logic and methodological implementation of the platform, while the final performance evaluation is presented in the subsequent validation and results chapters.

4.2 Mechanical Design Method

4.2.1 Design Objectives and CAD-Based Mechanical Concept

The mechanical subsystem provides a stable and repeatable structural basis for the syringe-based filling process. Its design logic was derived from the need to position the disposable syringe reproducibly, support the actuator within the frame, transmit the actuator force along the syringe axis, and guide the moving element without introducing unnecessary mechanical over-constraint. In this context, the mechanical subsystem does not describe how motor motion is generated or controlled. Instead, it defines the structural interfaces required for reliable high-viscosity extrusion.

The CAD model defines the main functional areas of the mechanical subsystem. These areas include the syringe fixation, the standardized cannula connection, the plunger contact interface, the frame-integrated actuator mounting, and the linear guidance. The CAD-based design process allowed the relative position of the syringe, plunger contact element, actuator mounting, and guide structure to be evaluated before manufacturing. This step was necessary because alignment errors can affect the force path between the actuator and the syringe plunger.



a) CAD overview from the left 45° perspective.

b) CAD overview from the right 45° perspective.

Figure 17: CAD overview of the mechanical subsystem from two 45° perspectives, with colors used to distinguish different components of the system. The figure was created by the author.

Figure 17 provides a CAD-based overview of the mechanical subsystem and locates the main functional design areas rather than every individual component. The orange component represents the syringe fixation, while the purple component forms the plunger contact element used to transmit the pushing motion to the syringe plunger. The dark grey components provide guidance and support for the purple moving element during linear motion. The actuator is mounted in the green frame section. The red transmission element, together with the blue gear, transfers the rotational motion generated by the actuator toward the syringe side, where it is converted into plunger displacement. The remaining white components are used as structural supports or constraints to define the assembly position and limit the degrees of freedom of the connected parts. The colors are used only for functional visualization in the CAD model and do not represent the final physical colors of the prototype. The following subsections focus on the design rationale of the mechanical subsystem and describe how the main structural interfaces were defined.

4.2.2 Syringe Fixation and Standardized Cannula Connection

The syringe fixation provides repeatable positioning of the specified disposable syringe during repeated filling trials. A central boundary condition of the mechanical design is the use of the B. Braun Injekt Luer Lock Solo 2 mL disposable syringe. Since the syringe is replaced during laboratory operation, the fixation concept had to support both reproducible positioning and practical handling. The syringe fixation constrains the syringe body in a defined axial

position while allowing the syringe to be inserted and removed without disassembling the pump structure.

Figure 18 shows the complete CAD view of the syringe fixation, including the overall geometry of the fixation element and its integration into the mechanical subsystem. It provides an overview of the snap-fit fixation, the semi-enclosing syringe support, the leadscrew slot, and the mounting holes for the aluminum profile.

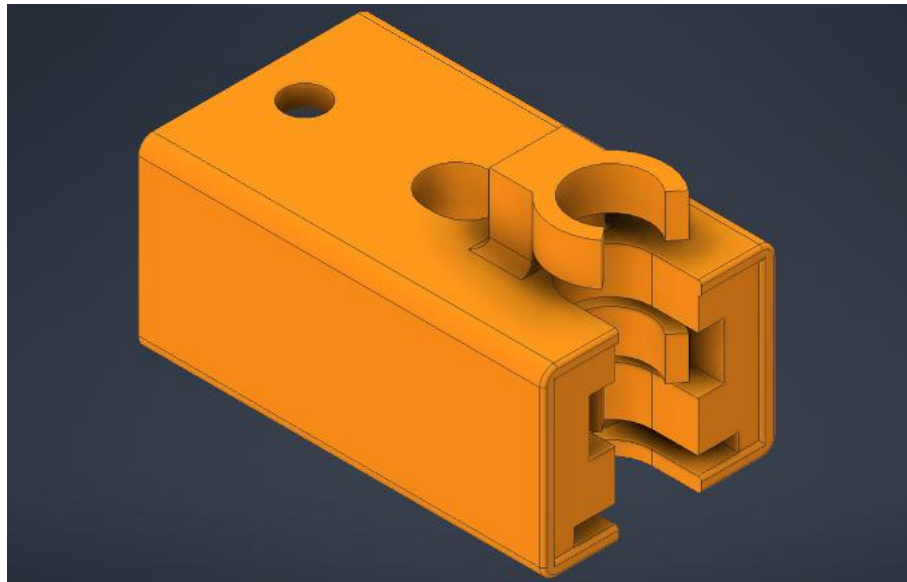


Figure 18: Complete CAD view of the syringe fixation. The figure was created by the author.

The syringe fixation was implemented as a quick-release interface to support an efficient laboratory workflow. The use of a disposable syringe requires repeated syringe exchange during preparation, testing, and validation. A snap-fit fixation was selected to allow rapid insertion and removal of the syringe. The quick-release function was treated as part of the mechanical repeatability concept rather than as a purely ergonomic feature, because syringe exchange should not introduce avoidable variation in the position of the syringe body relative to the plunger contact interface.

The fixation concept must reduce avoidable tilting and misalignment during extrusion. Under high-viscosity loading conditions, the axial force applied to the plunger can generate bending moments if the syringe axis, the plunger contact surface, and the actuator motion axis are not sufficiently aligned. Such misalignment may increase local deformation, friction, and variation in the effective delivered volume. The syringe fixation defines a repeatable mechanical boundary for the syringe body and supports a consistent force path during extrusion.

Figure 19 provides an annotated section view of the syringe fixation and highlights the snap-fit fixation, semi-enclosing syringe support, and barrel flange contour. It also shows how the leadscrew slot and the mounting holes for the aluminum profile are integrated into the fixation element.

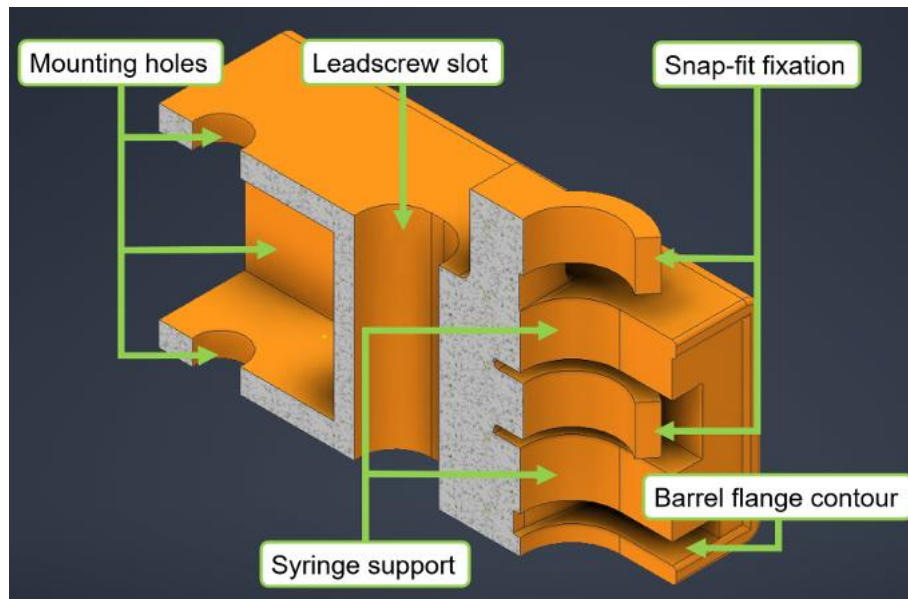


Figure 19: Annotated section view of the syringe fixation showing the snap-fit fixation, semi-enclosing syringe support, barrel flange contour, leadscrew slot, and mounting holes for the aluminum profile. The figure was created by the author.

The semi-enclosing syringe support and the barrel flange contour were used to improve syringe stability without creating an over-constrained fixation. The semi-enclosing geometry supports the syringe body laterally, while the barrel flange contour provides a form-fit boundary against displacement. This is relevant because lateral force components during high-viscosity extrusion may otherwise shift or tilt the syringe and thereby change the force path between the actuator and the plunger. By using the shape of the syringe barrel flange as a locating feature, the syringe fixation contributes to a more stable syringe position during loading.

The snap-fit fixation and alignment concept uses multiple locating points to increase stability while maintaining practical handling. A fully constrained fixation could increase assembly sensitivity or make syringe insertion and removal difficult, especially because disposable syringes can have small geometric tolerances. The selected fixation concept uses local contact and form-fit features to define the syringe position without turning the fixation into a rigid over-constrained clamp. This supports repeatability while still allowing efficient syringe exchange during laboratory trials.

The rounded edge of the barrel flange contour was considered in the syringe removal direction. The syringe can be rotated out of the snap-fit fixation instead of being pulled out directly against the retaining geometry. This reduces unnecessary local loading of the printed fixation features during syringe removal and supports the durability of the fixation during repeated laboratory use. The design balances secure retention during extrusion with practical and repeatable syringe removal.

The leadscrew slot and the mounting holes for the aluminum profile were integrated as mechanical integration features of the syringe fixation. The slot provides the required space for the leadscrew path, while the mounting holes define how the fixation element is attached to the aluminum profile. These features do not define the syringe position directly, but they are necessary to integrate the syringe fixation into the overall mechanical structure of the pump platform.

The standardized Luer Lock connection of the selected syringe defines the cannula attachment. In the developed system, the cannula is not fixed by a separate frame-mounted clamp, but is attached directly to the threaded outlet of the syringe. This design choice keeps the syringe–cannula connection consistent with the disposable syringe interface and reduces the need for an additional cannula fixation mechanism. At the same time, the standardized Luer Lock connection only defines the mechanical attachment of the cannula. It does not reduce the hydraulic resistance generated by the cannula during high-viscosity extrusion.

Overall, the syringe fixation provides a repeatable and practical interface between the disposable syringe and the mechanical platform. The snap-fit fixation supports rapid syringe exchange, the semi-enclosing syringe support and barrel flange contour reduce avoidable tilting and misalignment, and the standardized Luer Lock connection defines the cannula attachment without requiring an additional clamp. These design decisions support both the laboratory workflow and the mechanical repeatability required for the subsequent filling trials.

4.2.3 Plunger Contact Interface and Force Transmission Path

The plunger contact interface transfers the actuator force to the syringe plunger in a defined and repeatable manner. In the investigated filling process, the disposable syringe functions as both a fluid container and part of the mechanical force transmission path between the actuator and the high-viscosity oil. The contact condition between the plunger contact interface and the syringe plunger influences how the actuator load is introduced into the syringe system. The plunger contact interface was therefore treated as a functional mechanical interface rather than a simple pushing surface.

The main design objective of the plunger contact interface is axial force transmission. During extrusion, the actuator force should act as closely as possible along the syringe axis in order to reduce lateral force components. If the contact point is offset from the syringe axis, the syringe plunger may tilt inside the syringe barrel and generate additional friction. Such tilting can change the effective resistance of the syringe system and may increase variation between filling trials. The plunger contact interface was positioned relative to the syringe fixation so that the pushing direction remains aligned with the syringe axis.

Figure 20 provides a geometry overview of the plunger contact interface and identifies the main functional regions, including the plunger contact surface, stainless steel guide rod, sliding bearing region, leadscrew slot, screw slot, and interface for the trapezoidal flange nut. This overview is intended to explain the functional layout of the component rather than to document all manufacturing details.

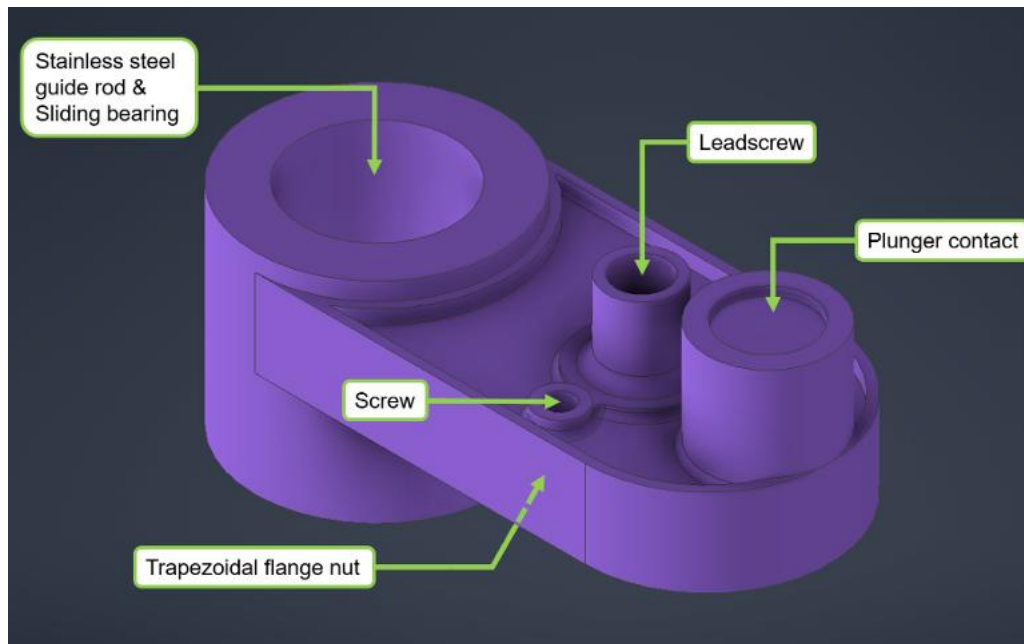


Figure 20: Geometry overview of the plunger contact interface with the main functional regions. The figure was created by the author.

The contact geometry within the plunger contact interface provides a stable pushing condition without rigidly coupling the syringe plunger to the actuator. The plunger contact surface was designed slightly larger than the actual contact area of the syringe plunger. This provides a small clearance that allows limited relative movement during operation and avoids an over-constrained contact condition. This design consideration is particularly relevant for FDM-printed parts, because their dimensional accuracy and alignment quality are lower than those of conventionally machined metal components. If the plunger contact interface were too tightly constrained, small misalignments between the syringe axis, the moving element, and the actuator motion axis could introduce bending of the syringe plunger and increase the risk of friction, jamming, or failure during extrusion. The developed system applies compressive force to the syringe plunger during extrusion, but the syringe plunger is not mechanically locked to the moving element. This distinction is important because the actuator should transmit the required pushing force during filling while still allowing pressure-related relaxation effects after the commanded motion has stopped. The plunger contact interface supports controlled pushing without creating a permanent mechanical constraint between the actuator and the syringe plunger.

The plunger contact interface assembly reduces the influence of dimensional inaccuracies from FDM-printed parts. Since 3D printed components may show small deviations in hole diameter, surface quality, and alignment, direct sliding contact between the printed part and the stainless steel guide rod could increase friction and reduce motion stability. A sliding bearing was used between the plunger contact interface and the stainless steel guide rod. The sliding bearing provides a defined sliding interface and reduces the dependency of the motion quality on the printed surface itself.

Figure 21 shows the assembly of the plunger contact interface with the syringe plunger, leadscrew, trapezoidal flange nut, sliding bearing, and stainless steel guide rod. The assembly view illustrates how the sliding bearing supports guided motion along the stainless steel guide rod

and how the trapezoidal flange nut is mounted below the leadscrew slot.

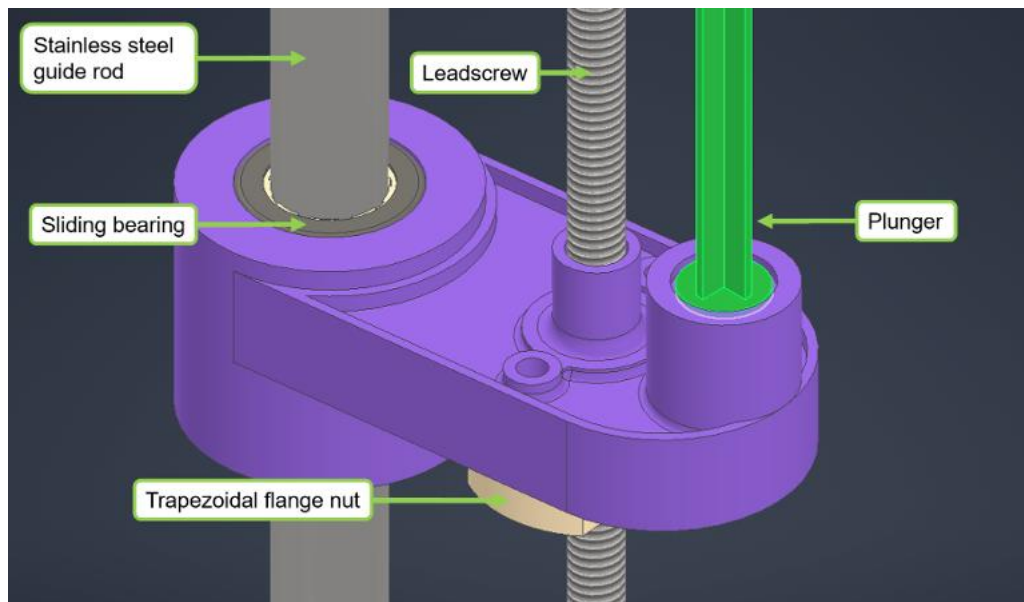


Figure 21: Assembly view of the plunger contact interface with the syringe plunger, leadscrew, stainless steel guide rod, sliding bearing, and trapezoidal flange nut. The figure was created by the author.

The trapezoidal flange nut was integrated below the leadscrew slot to connect the plunger contact interface to the leadscrew-driven motion. Its function is to convert the rotational motion of the leadscrew into linear displacement of the plunger contact interface. In this subsection, the trapezoidal flange nut is considered as part of the mechanical integration of the plunger contact interface. The detailed drive concept and motion conversion are discussed in the drive and actuation subsystem.

The force transmission path was considered from the actuator mounting through the transmission elements, the plunger contact interface, and finally into the syringe plunger. Figure 22 shows this intended force transmission path in the assembly view. The figure illustrates how the actuation input is transferred through the moving element to the plunger contact interface and then into the syringe plunger, while the syringe fixation provides the corresponding mechanical boundary for the syringe body. This path must remain sufficiently supported to avoid deformation of unsupported elements under high-viscosity loading conditions. If the force transmission path is not well defined, part of the actuator displacement may be absorbed by local deformation before it contributes to fluid extrusion.

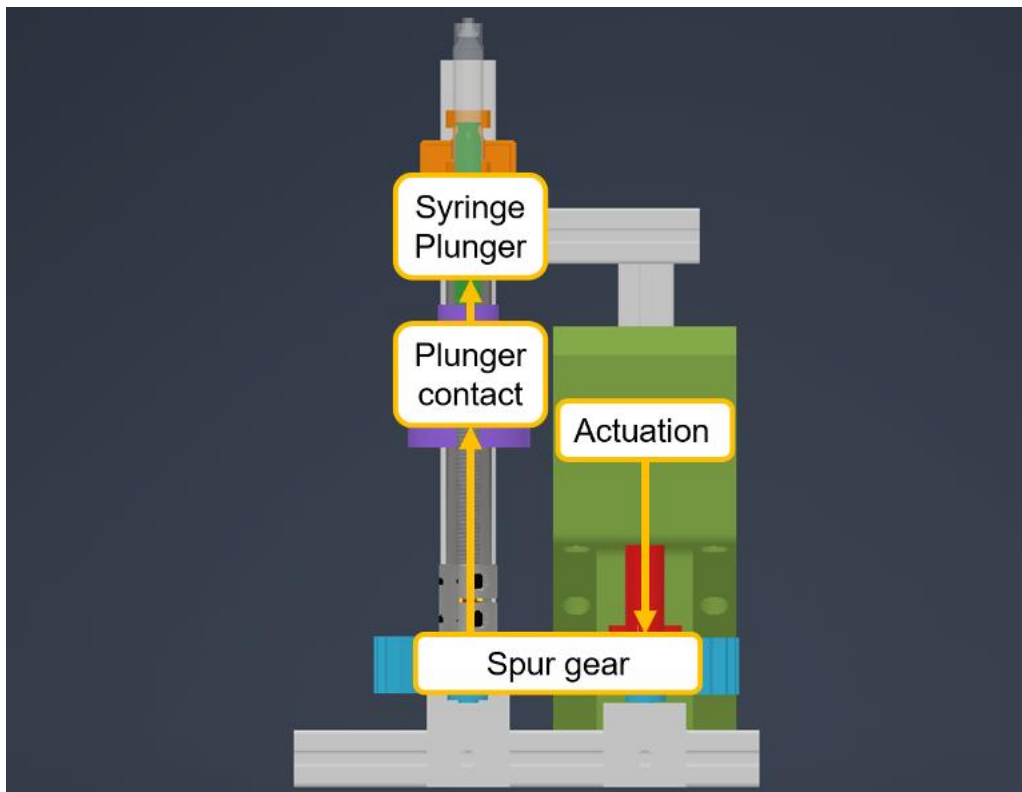


Figure 22: Assembly view of the plunger contact interface showing the actuation input via the spur gear and the intended force transmission path toward the syringe plunger. The figure was created by the author.

Overall, the plunger contact interface defines how actuator force enters the disposable syringe system. Its design supports axial force transmission, reduces avoidable syringe plunger tilting, uses a sliding bearing to improve guidance stability, and integrates the trapezoidal flange nut below the leadscrew slot for leadscrew-driven linear motion. Together with the syringe fixation, it forms the mechanical basis for controlled force application in the subsequent drive and actuation subsystem.

4.2.4 Frame-Integrated Actuator Mounting and Linear Guidance

The frame-integrated actuator mounting and linear guidance define the structural support of the moving system. While the syringe fixation and the plunger contact interface define how the syringe is positioned and loaded, the frame-integrated actuator mounting defines how the actuator-side components are supported within the mechanical subsystem. Its purpose is to maintain a stable relative position between the actuator, the transmission elements, the lead-screw, the stainless steel guide rod, the plunger contact interface, and the syringe fixation.

The frame-integrated actuator mounting supports the actuator in a defined position relative to the syringe axis. This support is necessary because the actuator and transmission elements generate both the intended driving motion and reaction forces that are transferred back into the frame during high-viscosity extrusion. If the actuator is insufficiently supported, these reaction forces may cause deflection or angular misalignment. Such deformation can influence the alignment between the leadscrew, the plunger contact interface, and the syringe plunger. Fig-

Figure 23 shows the frame-integrated actuator mounting in a partial assembly view. The left view provides the CAD overview of the actuator mounting region within the mechanical subsystem, while the right view focuses on the actuator mounting structure itself. Together, the views show the relative position of the actuator mounting and surrounding frame elements.

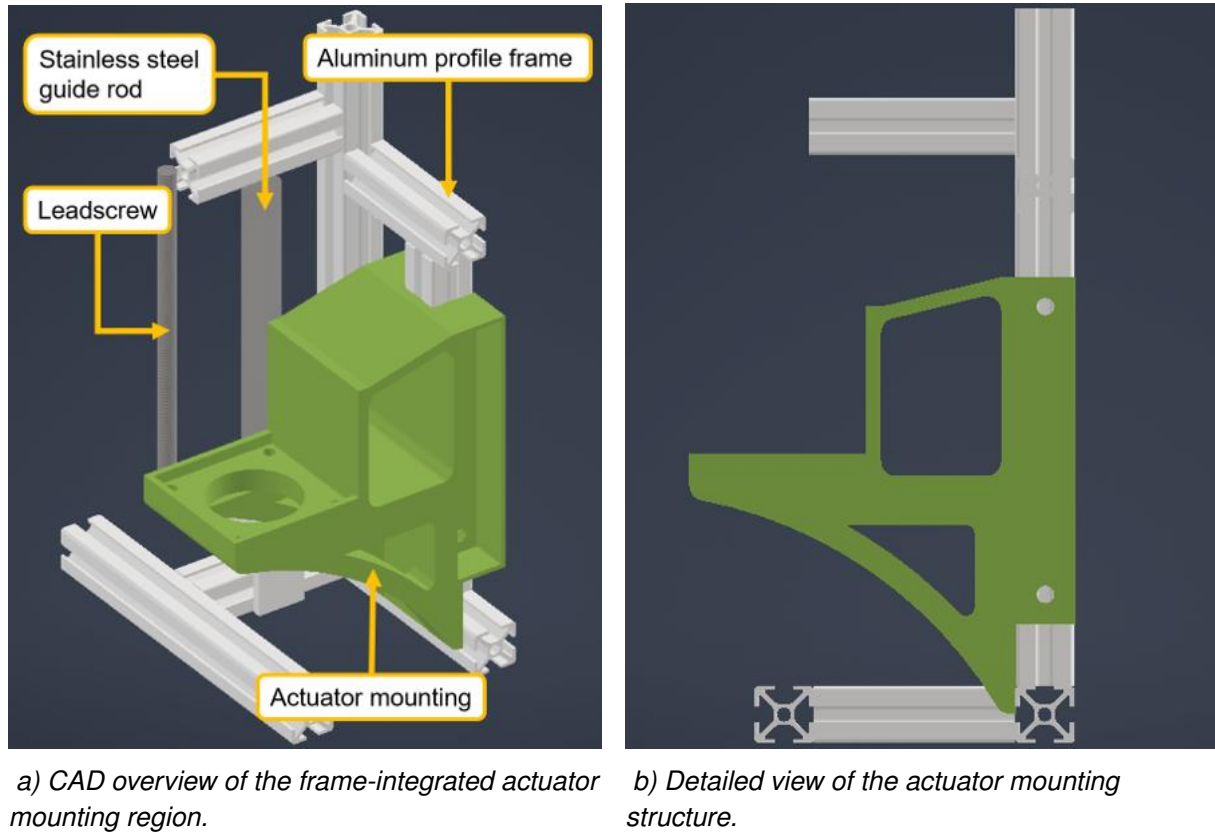


Figure 23: Partial assembly views of the frame-integrated actuator mounting showing the relative position of the actuator mounting and aluminum profile frame elements. The figure was created by the author.

The frame-integrated actuator mounting was treated as part of the load-bearing structure rather than as a simple motor attachment. In the developed design, the actuator is positioned above the spur gear and is not supported directly by the main frame surface underneath the transmission element. This arrangement creates a partially overhanging load-bearing region. To increase the support stability of this region, the actuator mounting uses a triangular support geometry. The inclined support member redirects part of the actuator load toward the frame and increases the stiffness of the actuator support platform. Near the aluminum profile side of the frame, mounting holes were integrated to define the fastening position of the support element. This design helps maintain the relative alignment between the actuator-side components and the syringe-side interfaces during extrusion.

The linear guidance consists of the stainless steel guide rod and the leadscrew. The leadscrew provides the axial driving motion of the plunger contact interface, while the stainless steel guide rod supports the plunger contact interface against rotation and lateral displacement. This arrangement separates the driving function from the guiding function and avoids introducing an unnecessarily complex guidance structure. The design supports controlled motion along

the extrusion axis while remaining compatible with the tolerance limitations of the FDM-printed frame.

Overall, the frame-integrated actuator mounting and linear guidance provide the structural basis for controlled plunger motion. The frame-integrated actuator mounting supports the actuator-side components and reduces avoidable deflection, while the linear guidance stabilizes the plunger contact interface along the extrusion axis. Together with the syringe fixation and the plunger contact interface, these design elements complete the mechanical subsystem and provide the mechanical basis for the drive and actuation method described in the following section.

4.3 Drive and Actuation Method

4.3.1 Drive Concept and Actuation Chain Overview

The drive and actuation method defines how the commanded motor motion is physically transmitted to the syringe plunger. In contrast to the mechanical subsystem, which defines the structural interfaces and support conditions, the present section focuses on the actuation chain itself. The drive concept was divided into three functional stages: stepper motor actuation, gear transmission, and the gear-to-lead screw interface. This structure separates torque generation, torque transmission, and motion conversion within the developed syringe pump platform.

Figure 24 provides an annotated overview of the drive and actuation chain and identifies the main transmission regions, including the motor end, motor-side connector, spur gear stage, leadscrew-side connector, coupling end, and leadscrew end. In the figure, red indicates the connectors, while blue indicates the spur gear.

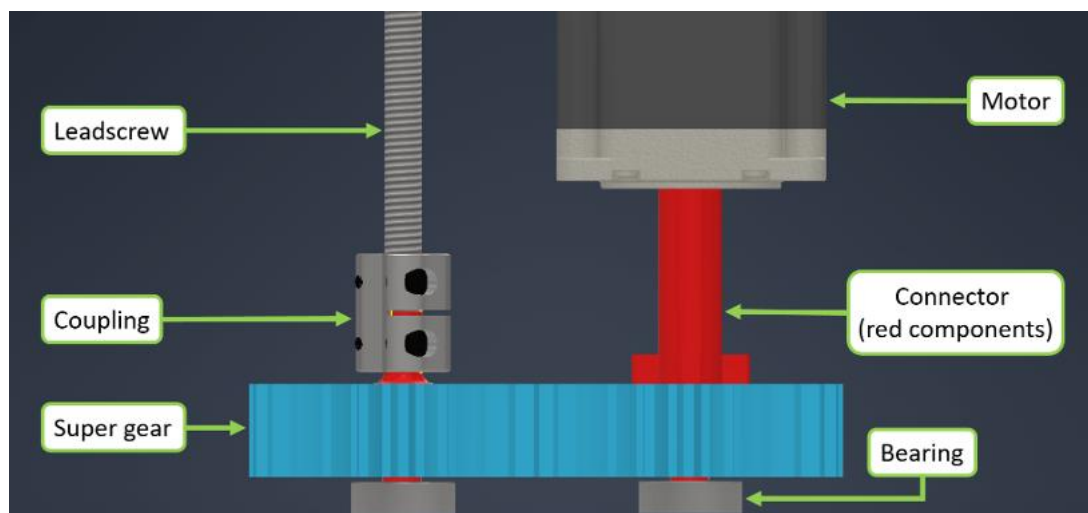


Figure 24: The overview of the drive and actuation chain showing the motor end, motor-side connector, spur gear, leadscrew-side connector, coupling end, and leadscrew end. The figure was created by the author.

The actuation chain was designed to transmit torque through modular interfaces rather than through a single continuous shaft. This modularity was necessary because the actuator, gear

transmission, and leadscrew are arranged as separate elements within the available installation space. At the same time, the interfaces between these elements had to remain sufficiently robust under high-viscosity loading conditions. The drive method relies on custom-designed connectors to define the torque-transmission path between the motor, the spur gear stage, and the leadscrew.

4.3.2 Stepper Motor Actuation and Driver Integration

Stepper motor actuation forms the basis of the drive concept. The developed system uses a NEMA 23 stepper motor because the investigated filling process requires sufficient torque at low speed rather than high-speed positioning. In the present application, the main objective of the actuator is to generate controlled and repeatable plunger displacement under high-viscosity loading conditions. The NEMA 23 motor provides the actuation source for the subsequent transmission stages.

The motor driver was implemented on a custom-designed circuit board integrating a TMC5160-TA. This driver was selected because it supports controlled stepper motor operation and provides an internal ramp generator for motion execution. In the developed system, the TMC5160-TA forms the low-level actuation interface between the embedded control system and the motor. The embedded control system is based on an Arduino Giga R1 WiFi with a Giga Display Shield. The higher-level generation and adaptation of motion commands are addressed later in the control and compensation method.

Figure 25 shows the custom-designed motor driver board with the integrated TMC5160-TA and documents the physical implementation of the drive electronics that execute the motor actuation commands.

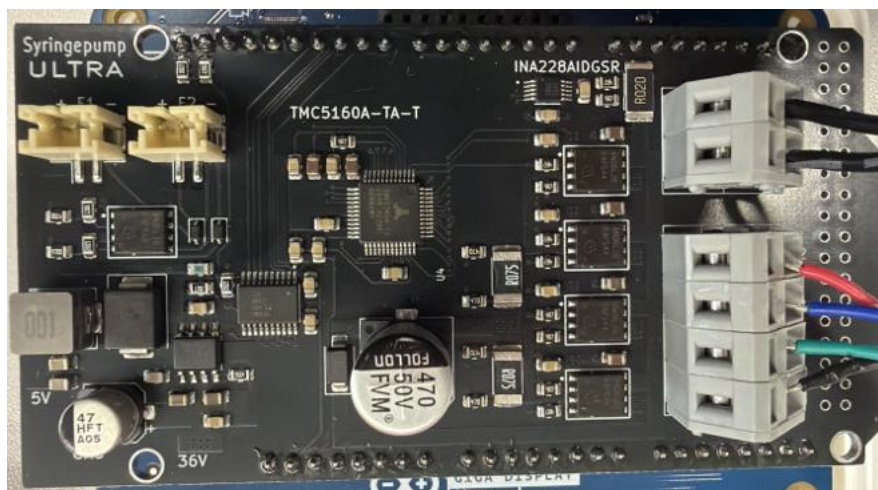


Figure 25: Custom-designed motor driver board integrating the TMC5160-TA for stepper motor actuation. The figure was created by the author.

Figure 26 shows the controller and display hardware used for the user interface and embedded control implementation. The left image shows the Arduino Giga R1 WiFi, which provides the main microcontroller platform for process control and communication with the motor drive. The right image shows the Giga Display Shield, which provides the touch-based graphical interface for parameter input, process monitoring, and user guidance.



a) Arduino Giga R1 WiFi used as the main controller platform.



b) Giga Display Shield used as the graphical user interface hardware.

Figure 26: Controller and display hardware used for embedded process control and user interaction. The figure was created by the author.

4.3.3 Motor-Side Connector and Gear Transmission

A custom-designed motor-side connector transfers torque from the stepper motor to the spur gear stage. A direct friction-based connection was not sufficient for the present application. Experimental observations showed that a conventional coupling approach alone did not transmit the motor torque to the gear reliably and could slip under load. The motor-side connector was introduced as a dedicated transmission element between the motor shaft and the spur gear stage.

Figure 27 compares the motor-side connector with the motor shaft. The left view presents the custom-designed connector, while the right view presents the motor shaft geometry. One end of the connector was designed with the same cross-sectional shape as the motor shaft. This matching geometry ensures that torque transmission relies on geometric correspondence between the shaft and the connector interface rather than friction alone.

The motor-side connector transmits the motor torque to the spur gear stage through a four-fin interface. The fins are inserted into corresponding slots in the gear and create a defined connection between the connector and the gear. Compared with a purely friction-based interface, this design introduces a more robust torque-transmission path. The axial thickness of the fins was increased to improve the mechanical strength of the connector during repeated operation under load.

Figure 28 shows the CAD-based view of the motor-side connector and the associated gear interface, highlighting the four-fin geometry and the relative position between the connector and the gear. The motor is shown with high transparency to illustrate the assembled relative position of the motor, connector, and gear.

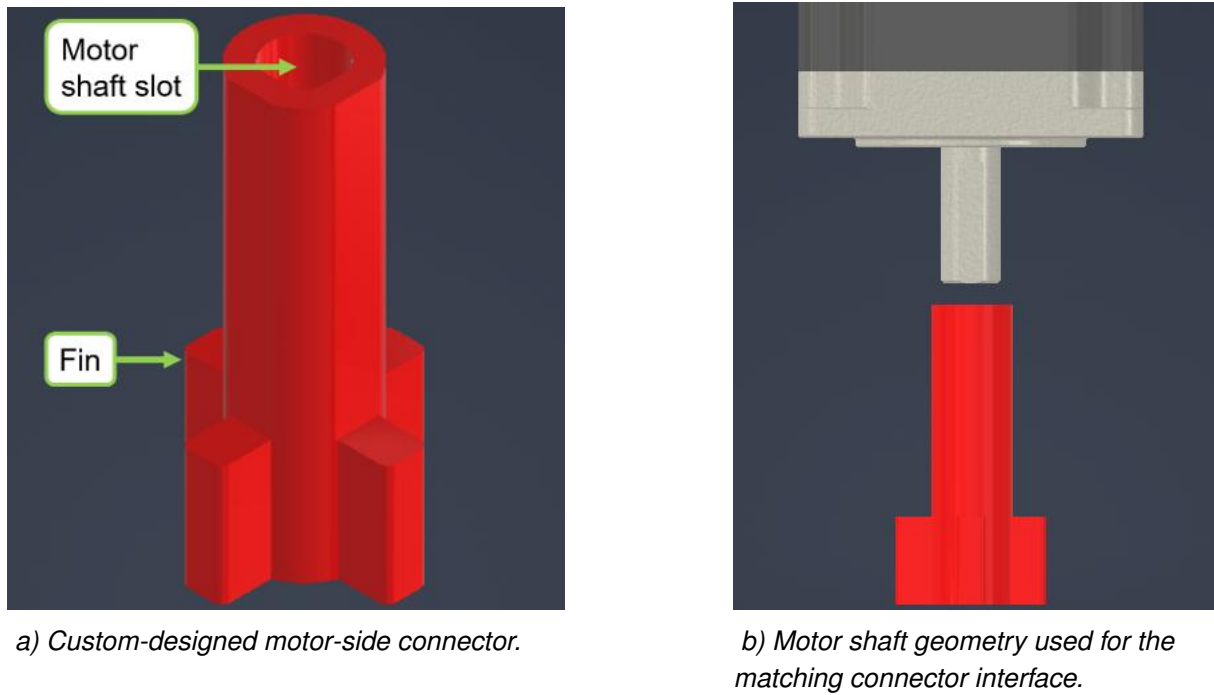


Figure 27: Geometry of the motor-side connector and schematic assembly with the motor. The figure was created by the author.

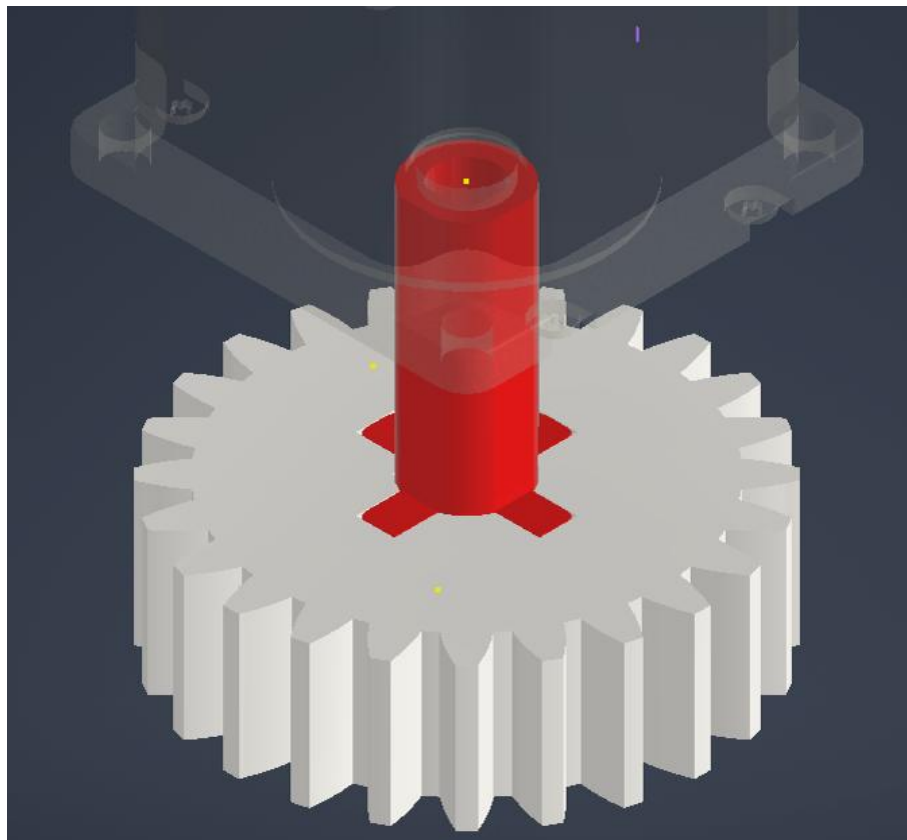


Figure 28: CAD-based view of the motor-side connector and gear interface, with the motor shown transparently to illustrate the assembled relative position of the motor, connector, and gear. The figure was created by the author.

The gear transmission itself is based on a 1:1 spur gear stage. In the developed design, the gear stage is not intended to provide speed reduction or torque multiplication through a transmission ratio. Instead, its function is to redirect and transfer the motor torque within the available installation geometry. The spur gear was designed with a larger pressure angle and a reinforced tooth body compared with a conventional general-purpose gear. This design choice improves the robustness of the gear under the higher loading conditions associated with high-viscosity extrusion.

The connector and the gear were intentionally designed as separate components. This modularity allows the motor-side connector and the gear to be treated as independent FDM-printed parts. If one of the two parts is damaged during operation, the failed part can be reprinted and replaced without requiring the entire transmission stage to be remanufactured. This approach improves the maintainability of the actuation chain during iterative laboratory development.

4.3.4 Gear-to-Leadscrew Interface and Rotary-to-Linear Motion Conversion

A second custom-designed connector, referred to here as the leadscrew-side connector, transfers torque from the spur gear stage to the leadscrew. At its gear-side end, this connector uses the same fin-based interface principle as the motor-side connector. This preserves the modular separation between the connector and the gear and allows both parts to remain independently replaceable. The gear-side interface maintains the same basic torque-transmission logic in the second stage of the actuation chain.

At its opposite end, the leadscrew-side connector is connected to the leadscrew through a coupling that is tightened by screws. Figure 29 shows the CAD view of the leadscrew-side connector on the left and the coupling on the right. This view illustrates the interface between the custom-designed connector and the coupling-based leadscrew connection.

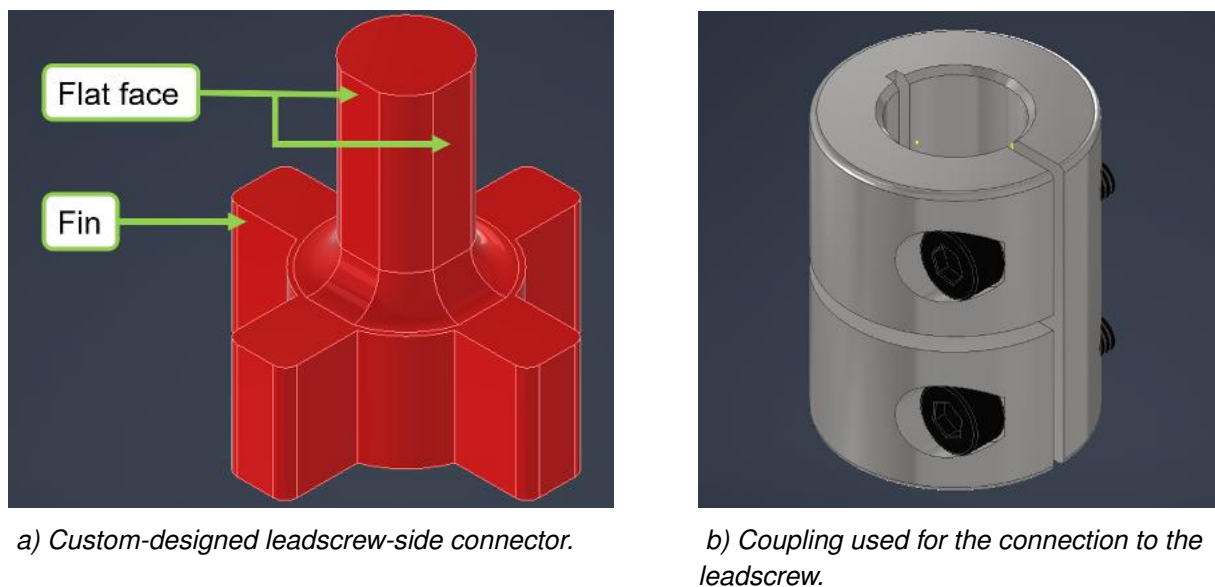


Figure 29: Gear-to-leadscrew interface with the leadscrew-side connector and the coupling. The figure was created by the author.

The shaft end of the leadscrew-side connector was designed with two flat faces instead of a purely cylindrical shape. When this shaft end is clamped in the coupling, the flat geometry provides a more defined torque-transmission interface than a circular shaft clamped only by friction. One of the coupling screws can be aligned parallel to one of the flat faces, which reduces the tendency of the interface to slip under load. This design improves the transmission capacity of the connector-to-leadscrew interface without changing the modular structure of the actuation chain. Figure 30 shows the exploded assembly view of the spur gear, leadscrew-side connector, coupling, and leadscrew.

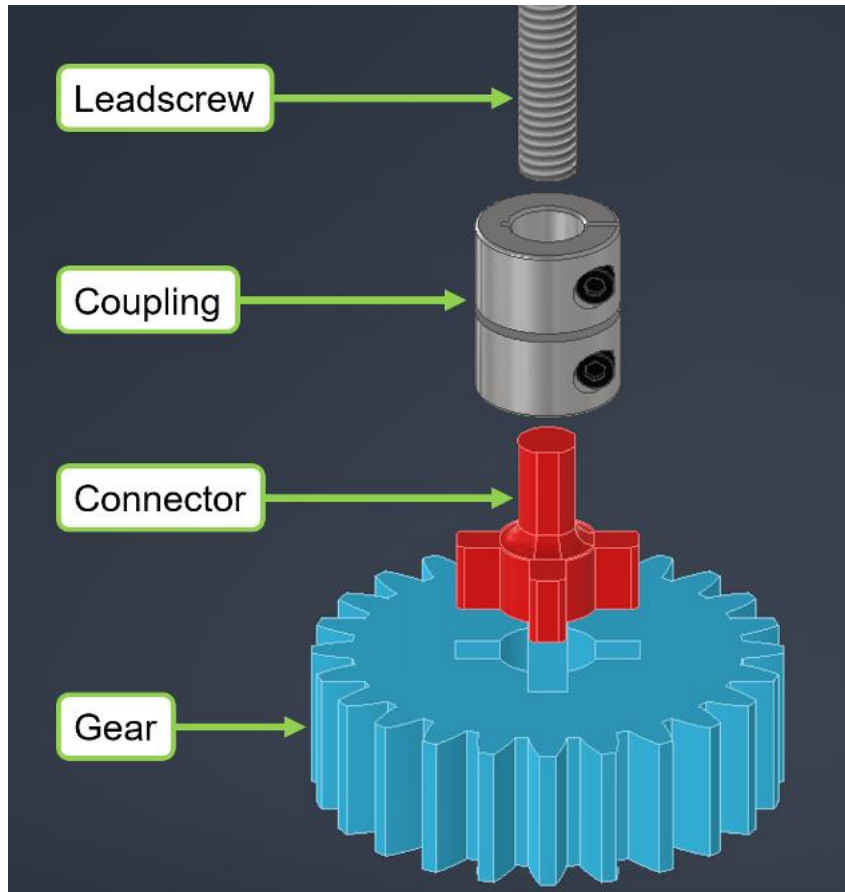


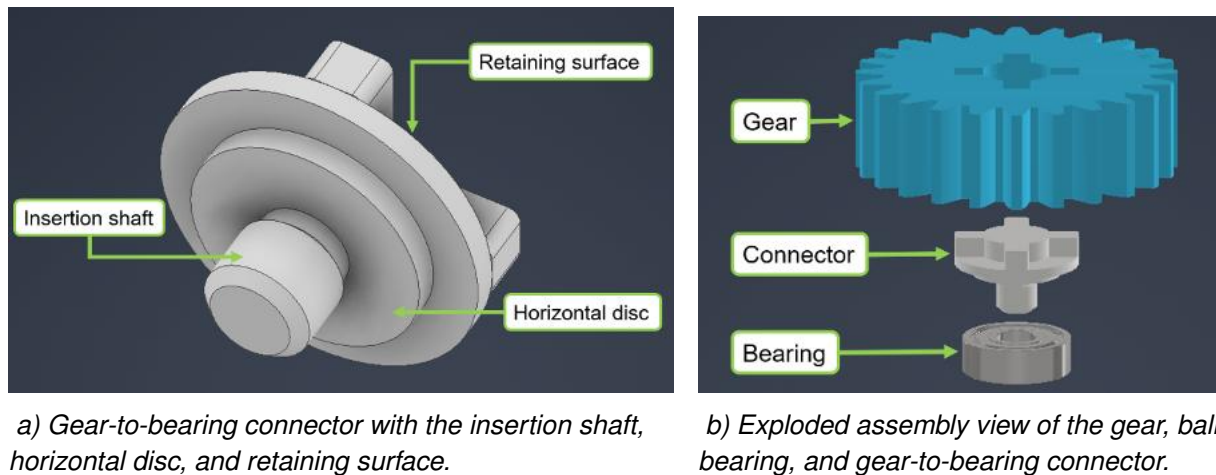
Figure 30: Exploded assembly view of the spur gear, leadscrew-side connector, coupling, and leadscrew. The figure was created by the author.

4.3.5 Gear Rotation Stability and Bearing Support

The rotational stability of the spur gear stage was supported by a bearing-based gear mounting concept. To allow smooth rotation, one ball bearing was placed below each spur gear. This support was necessary because torque-transmission quality depends on both gear tooth geometry and the parallel alignment of the two gear rotation planes. If the two spur gears do not rotate at the same level, the gear mesh can become uneven, which may cause unstable torque transmission, increased local loading, or damage to the printed transmission components. The gear-to-bearing connector supports both smooth rotation and stable gear alignment.

Figure 31 presents the design and assembly concept of the gear-to-bearing connector. The connector consists of three functional regions: the insertion shaft, the horizontal disc, and the

retaining surface. The insertion shaft was designed with a slight interference fit for the ball bearing. Its front chamfer supports insertion into the bearing, while the reduced shaft diameter at the end helps reduce the risk that irregular edges from the FDM-printed part interfere with the assembly. The horizontal disc contacts the inner ring of the ball bearing and acts as an axial reference surface to improve rotational stability. The retaining surface defines the assembly position of the gear relative to the connector and helps maintain stable horizontal rotation while still allowing the connector and gear to remain separate replaceable parts.



a) Gear-to-bearing connector with the insertion shaft, horizontal disc, and retaining surface.

b) Exploded assembly view of the gear, ball bearing, and gear-to-bearing connector.

Figure 31: Design and assembly concept of the gear-to-bearing connector used to support smooth and stable spur gear rotation. The figure was created by the author.

Overall, the drive and actuation method converts the motor command into controlled linear motion of the plunger contact interface. The NEMA 23 stepper motor, the TMC5160-TA-based custom driver board, the 1:1 spur gear stage, and the custom-designed connectors transmit motor torque to the leadscrew. The leadscrew and trapezoidal flange nut then convert the rotational input into linear displacement of the plunger contact interface, which applies compressive force to the syringe plunger during extrusion. After extrusion, the same actuation chain can move the plunger contact interface away from the syringe plunger without actively retracting the syringe plunger itself. The detailed timing and logic of this release motion are addressed in the control and compensation method.

4.4 Control and Compensation Method

The control and compensation method converts process parameters into executable actuator commands. Whereas the mechanical subsystem defines the structural interfaces and the drive and actuation subsystem defines the physical transmission chain, this section focuses on the logic used to determine the commanded extrusion motion. The method includes the distance-based target definition, compensation scaling, viscosity-constrained ramp planning, TMC5160 command generation, and the post-extrusion sequence.

The control sequence starts after the syringe and cannula have been manually prepared by the operator. The system does not perform automatic priming or syringe filling; it controls the extrusion step of the prepared syringe system. During operation, the input parameters

are converted into an ideal target distance, corrected by calibration-based compensation, and translated into TMC5160 ramp parameters. After the extrusion movement has reached its target, the plunger contact interface is retracted to release the mechanical contact condition, followed by a defined waiting time before the process is completed. The actual calibrated parameter values and resulting performance are evaluated in Chapter 5.

4.4.1 Control Scope and Process Sequence

The control scope was limited to the automated extrusion sequence after manual preparation of the syringe and cannula. Needle pre-extrusion, removal of trapped air, and the definition of the initial syringe state are performed manually by the operator and are therefore not treated as automated control functions in this thesis. The implemented control method does not perform syringe filling or priming. Instead, it controls the extrusion motion of an already prepared syringe system.

The process sequence was structured to separate parameter calculation, motion execution, release, and waiting time. After the operator-defined and system-defined input parameters have been provided, the target volume is converted into a target distance. This target distance is then modified by calibration-based compensation and converted into a TMC5160-executable ramp command. The extrusion movement is executed until the driver-side target distance is reached. After this movement has been completed, the plunger contact interface is retracted to release the mechanical contact condition at the syringe plunger. The waiting time starts after this release step and is used to account for delayed material outflow from the cannula. The control sequence itself is implemented as a state-based workflow to avoid conflicting motion commands. Extrusion, stopping, service motion, and fault handling are separated into defined motion states. This prevents a manual or service-related movement from being started while a precision-critical extrusion movement is still active.

The process-level sequence is summarized in Listing 4.1. The pseudocode is not intended to represent the complete implementation of the embedded software. Instead, it shows the methodological order in which the control and compensation steps are executed after manual preparation of the syringe and cannula.

```
1 Manual preparation by operator
2
3 Read input parameters
4 Calculate ideal target distance
5 Apply calibration-based compensation
6 Generate TMC5160 ramp command
7
8 Start extrusion movement
9 while target distance is not reached:
10 monitor driver-side actual distance
11 monitor process status
12
13 Stop extrusion movement
14 Retract plunger contact interface
15 Start waiting time
16
17 Finish process
```

Listing 4.1: Process-level pseudocode of the control and compensation sequence.

4.4.2 Input Parameters, Target Distance, and Compensation Scaling

The control calculation starts from a combination of operator-defined process parameters and predefined system parameters. The operator-defined parameters include the target fluid viscosity and the extrusion quantity. Additional system parameters, such as the syringe-specific stroke per milliliter, cannula geometry, and calibrated motion-scaling factors, are used internally to convert the requested process condition into a distance-based actuator command. This separation is important because the user interface provides the input channel, while the control and compensation method defines how these inputs are interpreted by the system.

The target extrusion quantity is converted into a linear target distance before it is sent to the motion execution layer. Since different syringes can require different plunger travel for the same nominal volume, the conversion is based on the calibrated stroke per milliliter of the syringe. The ideal target distance is therefore calculated from the requested quantity and the syringe-specific stroke parameter. In method terms, the target distance can be expressed as

$$x_{\text{target,ideal}} = \frac{V_{\text{target}}}{1000} \cdot s_{\text{ml}} \quad (4.1)$$

where $x_{\text{target,ideal}}$ is the ideal target distance in millimeters, V_{target} is the requested volume in microliters, and s_{ml} is the calibrated plunger travel per milliliter in mm/mL . This formulation treats the actuator command as a distance-controlled extrusion task rather than as a direct volumetric command.

The compensation strategy was introduced to address repeatable deviations observed under fixed process conditions. A calibration-based compensation approach is only meaningful if the dominant deviation is reproducible; otherwise, a corrected command scale would not reliably improve the delivered volume. The compensation approach was applied only under process conditions for which repeated trials indicated reproducible behavior. The formal repeatability and accuracy results are reported in Chapter 5. This observation suggested that the main limitation was not uncontrolled random scatter, but a systematic deviation between the ideal target distance and the delivered volume.

The compensation strategy modifies the effective relationship between commanded motion and delivered volume. The developed prototype is not a perfectly rigid system, and high-viscosity extrusion can introduce elastic deformation of the syringe, plunger, printed structure, and contact interfaces. Instead of attempting to describe all deformation sources analytically, the compensation method treats the system as a calibrated black-box relationship between command scaling and delivered volume. The compensation does not physically eliminate deformation. It adjusts the command scale so that the resulting delivered volume approaches the target volume under the selected process condition.

The compensation factor is determined from the measured volume deviation obtained with a standard parameter set. In the implemented control system, the relevant command-to-motion relationship is represented by the effective motion scale,

$$S_{\text{eff}} = M_{\text{step}} \cdot P_{\text{scale}} \quad (4.2)$$

where M_{step} denotes the value of `MICROSTEPS_PER_REAL_POSITION_UNIT`, and P_{scale} denotes the value of `POSITION_SCALE_FACTOR`. This effective scale defines the relation-

ship between the distance-based command and the internal motion units used by the controller.

For a selected viscosity condition, the system is first tested using the standard effective motion scale. The mean delivered volume measured under this condition is then compared with the target volume. The viscosity-specific compensation factor is defined as

$$C_\eta = \frac{V_{\text{target}}}{\bar{V}_{\eta,\text{std}}} \quad (4.3)$$

where C_η is the compensation factor for the selected viscosity condition, V_{target} is the target extrusion volume, and $\bar{V}_{\eta,\text{std}}$ is the mean delivered volume measured with the standard parameter set at the same viscosity. The effective motion scale is then updated according to

$$S_{\text{eff},\eta} = C_\eta \cdot S_{\text{eff},\text{std}} \quad (4.4)$$

where $S_{\text{eff},\text{std}}$ is the standard effective motion scale and $S_{\text{eff},\eta}$ is the compensated effective motion scale for the selected viscosity. This ratio-based update rule increases the effective motion scale when the measured delivered volume is lower than the target volume and decreases it when the measured delivered volume is higher than the target volume. The compensation is therefore not an analytical deformation model, but an experimentally determined correction of the command-to-volume relationship.

To extend the compensation strategy beyond individually calibrated viscosity values, the calibrated effective motion scales are organized as a viscosity-dependent interpolation map. For each calibrated viscosity condition, $S_{\text{eff},\eta}$ is calculated using the same definition in Equation 4.2. The resulting calibration points are then used to estimate the effective motion scale for viscosity values that were not directly calibrated. Because the investigated viscosity range spans several orders of magnitude, the interpolation is performed in the logarithmic viscosity domain:

$$S_{\text{eff}}(\eta) = I \left[\log_{10} \left(\frac{\eta}{1 \text{ mPa} \cdot \text{s}} \right) \right] \quad (4.5)$$

where η is the dynamic viscosity in $\text{mPa} \cdot \text{s}$, and I denotes the interpolation function constructed from the calibrated viscosity points. In this thesis, this interpolation-based compensation map is treated as a parameter-estimation method, not as a physical deformation model. Predictions inside the measured viscosity range are treated as interpolation, whereas viscosity values outside this range require additional experimental confirmation. The calibrated viscosity points, interpolation comparison, predicted parameter values, and validation results are reported in Chapter 5.

Listing 4.2 summarizes the role of the target-distance and compensation calculation. The pseudocode is not intended to reproduce the embedded implementation line by line. It is used to show the methodological sequence by which the input parameters, standard-parameter test result, and compensation factor are converted into an updated distance-based command before ramp planning and motor execution.

```

1 Load calibrated viscosity points
2
3 For each calibrated viscosity:
```

```
4 effective_motion_scale =  
5 microsteps_per_real_position_unit  
6 * position_scale_factor  
7  
8 Convert viscosity values  
9 to log10(viscosity)  
10  
11 Build interpolation model  
12 using calibrated effective motion scales  
13  
14 Read target viscosity  
15 Convert target viscosity  
16 to log10(target viscosity)  
17  
18 predicted_effective_motion_scale =  
19 interpolate effective motion scale  
20 at target viscosity  
21  
22 Decompose predicted effective motion scale  
23 into microsteps_per_real_position_unit and position_scale_factor  
24  
25 Use predicted scaling parameters  
26 for corrected command generation
```

Listing 4.2: Pseudocode for viscosity-dependent compensation interpolation.

The calibrated scaling values are treated as method parameters rather than as results in this chapter. Their numerical values, viscosity-dependent adjustment, and influence on the achieved filling accuracy are evaluated in Chapter 5.

4.4.3 Viscosity-Constrained Ramp Planning and TMC-Executable Command Generation

The ramp planning step converts the corrected distance-based command into a motion command that can be executed by the TMC5160-TA driver. After the target volume has been converted into a compensated target distance, the controller must determine a feasible velocity and acceleration profile for the selected viscosity condition. This step links the process-side constraints, such as viscosity and cannula geometry, with the actuator-side constraints of the stepper motor, leadscrew transmission, and driver ramp generator.

The viscosity-dependent constraint is treated as a limit on the allowable flow rate and plunger velocity. For a selected fluid viscosity and cannula configuration, the flow constraint model estimates the maximum admissible flow condition from the hydraulic load and actuator-related limits. The resulting allowable flow rate is converted into an allowable plunger velocity. This prevents the controller from selecting a ramp that would be faster than the process can reasonably support. The velocity limit used for ramp generation is selected from the most restrictive available condition, including the flow-based velocity limit, the peak velocity suggested by the ideal reference curve, and the user- or system-defined maximum linear speed.

The ideal 7-segment reference curve is used as a planning reference rather than as the directly executed motor trajectory. It provides a fluid-safe reference for estimating motion duration, peak velocity, and the qualitative shape of a smooth motion profile. However, the implemented

prototype communicates with the TMC5160-TA driver through its internal ramp generator. The jerk-limited reference curve is therefore not executed directly. Instead, its peak and timing information are reduced to a set of TMC-executable ramp parameters.

The TMC-executable ramp planner converts the constrained reference information into a trapezoidal or triangular executable approximation. The selected maximum velocity is first limited by the viscosity-dependent process constraint and then scaled by a safety factor. The acceleration and deceleration are derived from the selected velocity and an ideal-duration-based timing reference. A viscosity-dependent acceleration scale is used to reduce ramp aggressiveness for higher-viscosity conditions without creating separate control paths for different fluids. If the target distance is long enough, the command contains acceleration, constant-velocity, and deceleration phases. If the target distance is too short to reach the selected maximum velocity, the peak velocity is reduced and a triangular profile is generated.

The TMC5160-TA internal ramp generator is controlled by eight motion parameters that define the executable velocity profile. The parameter *maxVelocity* defines the maximum or plateau velocity of the motion. The parameters *startVelocity* and *stopVelocity* define the non-zero entry and exit velocities of the ramp. The parameter *firstVelocity* defines the transition velocity between the gentle and the main acceleration or deceleration zones. The parameter *firstAcceleration* governs the initial acceleration from *startVelocity* to *firstVelocity*, whereas *maxAcceleration* governs the main acceleration from *firstVelocity* to *maxVelocity*. On the deceleration side, *maxDeceleration* governs the main deceleration from *maxVelocity* back to *firstVelocity*, and *firstDeceleration* governs the final deceleration from *firstVelocity* to *stopVelocity*.

This two-stage acceleration and deceleration structure allows the motion command to introduce and remove load more gradually than a single abrupt ramp. This is relevant for high-viscosity extrusion because it reduces sudden loading of the syringe, plunger, and actuator-side structure while still allowing a higher peak velocity when the process condition permits. Figure 32 illustrates the conceptual role of these parameters in the executable ramp profile. The figure is used to explain the control meaning of the ramp-generator parameters and does not represent a directly measured velocity trace.

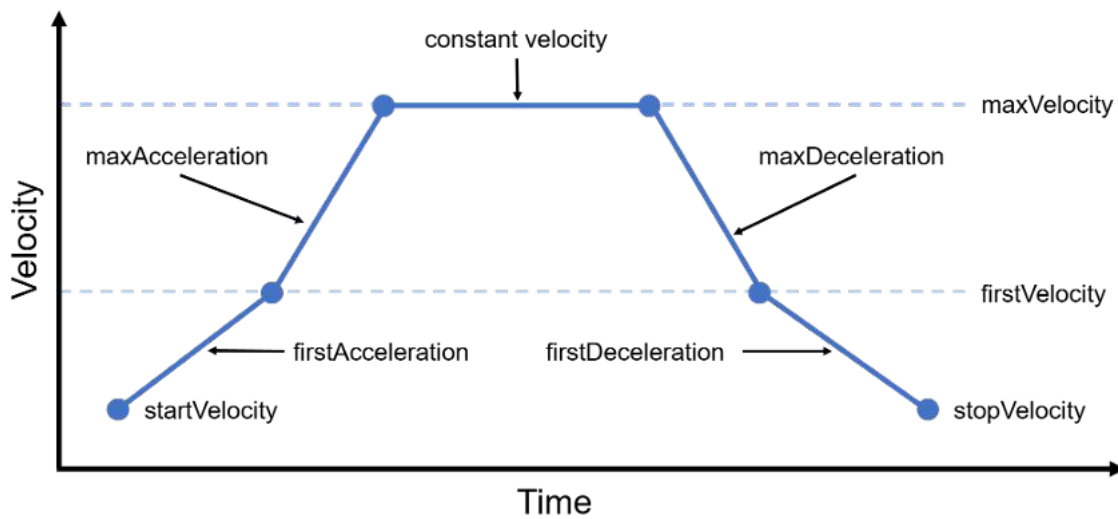


Figure 32: Conceptual velocity–time profile illustrating the role of the TMC5160 ramp-generator parameters. The profile shows the relationship between startVelocity, firstVelocity, maxVelocity, stopVelocity, firstAcceleration, maxAcceleration, maxDeceleration, and firstDeceleration. The figure was created by the author.

The resulting command is stored as a TMC-executable ramp command. This command contains the target quantity, target distance, maximum velocity, maximum acceleration, start velocity, stop velocity, first velocity, first acceleration, maximum deceleration, first deceleration, expected duration, direction, and validity information. The ramp planner acts as an intermediate layer between the ideal process-oriented motion reference and the low-level driver command. The command also includes a validity status, which confirms that essential ramp parameters, including the target quantity, stroke parameter, target distance, velocity, acceleration, deceleration, and expected duration, are valid positive values before the command is sent to the motor control layer. If this check fails, the command is reset and rejected.

Listing 4.3 summarizes the role of the viscosity-constrained ramp planning step. The pseudocode is used to show how the corrected target distance, viscosity-dependent constraints, and ramp-generator parameters are converted into a TMC-executable ramp command.

```

1 Read corrected target distance
2 Read selected viscosity
3 Load cannula and syringe parameters
4 Load motor and driver constraints
5
6 Evaluate flow constraint model
7 Convert allowable flow rate to allowable plunger velocity
8
9 Build ideal 7-segment reference curve as ideal reference prediction
10
11 velocity_limit = min(flow-based velocity, ideal reference peak
    velocity, maximum allowed linear speed)
12
13 Apply velocity safety scale
14 Select viscosity-dependent acceleration scale
15
16 Generate executable ramp approximation

```

```

17 if target distance is sufficient:
18     use trapezoidal ramp profile
19 else:
20     reduce peak velocity
21     use triangular ramp profile
22
23 Define TMC ramp-generator parameters:
24     maxVelocity,
25     maxAcceleration,
26     startVelocity,
27     stopVelocity,
28     firstVelocity,
29     firstAcceleration,
30     maxDeceleration,
31     firstDeceleration
32
33 Create TMC ramp command with:
34     target distance,
35     ramp-generator parameters,
36     expected duration,
37     direction,
38     validity flag
39
40 Pass TMC ramp command to motor control layer

```

Listing 4.3: Pseudocode for viscosity-constrained ramp planning and TMC-executable command generation.

4.4.4 Motion Mode Management, Passive Release, and Waiting Time

Motion mode management was used to prevent conflicting commands during the extrusion sequence. In the implemented system, the relevant operating situations were the active extrusion motion, controlled stopping of an extrusion motion, manual adjustment by disabling the motor output, and fault or rejected-command conditions. During active extrusion, the motion was treated as precision-critical because unintended interruption or superimposed movement could change the final actuator position and therefore affect the delivered volume. For this reason, the control logic only permits a controlled stopping request during active extrusion. Manual adjustment is not treated as an automated motion mode in this thesis; instead, the motor output can be disabled before extrusion so that the operator can adjust the mechanical state and define the initial condition of the syringe system.

After the driver-side target distance has been reached, the process is not considered fully stabilized immediately. High-viscosity extrusion can leave residual pressure in the syringe–cannula system and residual elastic deformation in the syringe, plunger, printed structure, and contact interfaces. To reduce this residual loading condition, the controller performs a plunger contact interface retraction after the extrusion movement. This step is referred to as post-extrusion passive release. It does not actively retract the syringe plunger and does not create suction at the cannula outlet. Instead, it moves the plunger contact interface away from the syringe plunger so that the residual contact load can be released more rapidly. This interpretation is consistent with the mechanical design of the plunger contact interface, which applies compressive force during extrusion without rigidly locking the syringe plunger to the actuator.

The waiting time is defined as the post-release observation interval required for the cannula outlet to reach a stable condition after commanded extrusion has ended. In the implemented prototype, this waiting time was not an automated software timer. It was measured manually by the operator as part of the experimental procedure. The interval starts after the plunger contact interface retraction has been completed, not immediately after the extrusion target distance is reached. This definition separates the commanded actuator motion from the pressure-delay behavior of the high-viscosity fluid. The passive release step is intended to shorten the stabilization time required after extrusion by reducing the residual pressure-related delay. The quantitative reduction in waiting time and its dependence on viscosity and cannula configuration are evaluated in Chapter 5.

Listing 4.4 summarizes the control role of motion mode management, passive release, and the manually observed waiting interval. The pseudocode is not intended to reproduce the complete embedded implementation. It is used to show the methodological order of the protected extrusion sequence and the post-extrusion release procedure.

```

1 Prepare syringe system manually
2 Disable motor output if manual adjustment is required
3 Set initial mechanical state
4
5 Request extrusion command
6 if extrusion command is allowed:
7     enter extrusion active state
8 else:
9     reject command
10
11 Execute TMC ramp command
12
13 while target distance is not reached:
14     monitor driver-side actual distance
15     monitor process status
16
17 if controlled stop is requested:
18     enter extrusion stopping state
19     ramp velocity to zero
20
21 After target-distance completion:
22     retract plunger contact interface
23     release residual contact load
24
25 Operator observes cannula outlet
26 Measure post-release waiting interval manually
27
28 Finish process
29 Return to idle state

```

Listing 4.4: Pseudocode for motion-mode guarded extrusion, passive release, and manual waiting-time observation.

4.4.5 Monitoring Signals and Control Boundary

The monitoring signals supervise the execution of the distance-commanded extrusion process. During operation, driver-side motion information and electrical status signals, including actual

position, motor current, bus voltage, driver temperature, and process status, were displayed to the operator. These signals allowed the operator to observe whether the actuator-side movement followed the commanded process state and to identify abnormal operating conditions during laboratory use. In this thesis, the monitoring layer is treated as a supervision and visualization function rather than an independent feedback controller.

The control boundary of the implemented prototype is defined as distance-command control with driver-state supervision. The system does not include a pressure sensor or flow sensor for closed-loop control during extrusion, and it does not update the delivered volume automatically based on outlet flow or measured fluid mass. Volumetric correction is instead performed before motion execution through calibration-based compensation and viscosity-dependent parameter estimation. This distinction is important because actuator-side motion completion does not necessarily imply immediate fluid stabilization at the cannula outlet. The final delivered volume, the effectiveness of the compensation strategy, and the influence of post-extrusion stabilization are therefore evaluated experimentally in Chapter 5.

4.5 User Interface and Guidance Design Method

4.5.1 Translation of Usability Heuristics into Interface Requirements

Building on the usability framework introduced in Chapter 3, the user interface was developed to translate Jakob Nielsen's 10 usability heuristics [27] into concrete interface requirements for the syringe pump platform. The focus was not to repeat the general need for user guidance, but to define how the heuristic principles were applied in the implemented laboratory interface. The interface design was treated as part of the experimental method because unclear interaction, missing process feedback, or unnecessarily complex screen structures could introduce avoidable variation during repeated filling trials.

All ten heuristics were considered during the interface design. In the context of the syringe pump platform, they were translated into requirements for system-state visibility, process-oriented terminology, consistent workflow states, error-preventive interaction, reduced memory load, user control, efficient operation for repeated use, access to advanced functions, minimalist presentation for routine operation, error recovery, and tab-specific user guidance. This translation allowed the interface to support both routine filling tasks and more detailed development or diagnostic tasks without overloading the standard workflow.

4.5.2 Interface Functions and Workflow Guidance

The interface provides the input channel for the process parameters required by the control and compensation method. In the basic operating workflow, the operator enters the target fluid viscosity and target extrusion quantity on the initial setting screen. These values are passed to the control subsystem, where target-distance calculation, compensation scaling, viscosity-dependent ramp planning, and TMC-executable command generation are performed. The system progress screen then presents the current operating state and the information required for recognition rather than recall, such as calculated process information and relevant status values. The interface itself is not described as a compensation or motion-planning module. Its

role is to make the required inputs explicit and to present the resulting process information in a form that can be checked by the operator before and during extrusion.

The interface was also structured to support different levels of use. For routine repeated extrusion, the basic workflow keeps the required interaction limited to the essential inputs and process-status display. This supports aesthetic and minimalist design by avoiding unnecessary information during standard operation. For more detailed configuration or diagnostic work, advanced functions are placed in the developer information area and its related subpages. These functions include manual motor-control settings, custom parameter adjustment, and access to current–time and volume–time curves. This separation supports flexibility and efficiency of use because routine users can operate the pump with minimal interaction, while experienced users can still access advanced information and settings when required.

Guided operation was implemented to support the manual and semi-automated parts of the laboratory workflow. Before extrusion, the operator prepares the syringe and cannula manually and may disable the motor output when mechanical adjustment is required. During extrusion, start and stop interactions are separated into clear states so that controlled stopping can be distinguished from normal process completion. After commanded extrusion and passive release, the interface supports the transition to the post-extrusion observation stage, in which the waiting interval is measured manually as part of the experimental procedure. In addition, each tab provides a corresponding user manual so that the operator can access tab-specific guidance without relying only on external documentation.

4.5.3 Usability Evaluation Based on Nielsen's Heuristics

The interface was evaluated using a heuristic review based on all ten usability heuristics introduced in Chapter 3. This evaluation was appropriate for the prototype stage because the objective was to identify whether the interface supported the intended laboratory workflow and avoided obvious usability problems. The review focused on the interface functions that directly affect preparation, parameter entry, extrusion execution, stopping, manual adjustment, advanced parameter access, process supervision, and post-extrusion handling.

The evaluation criteria were derived from the way each heuristic was translated into an implemented interface function. The review examined whether the interface made the process state visible, whether the terminology matched the experimental workflow, whether input and operation errors were reduced by the screen structure, and whether the operator could recover from unintended actions through controlled stopping or manual resetting of the mechanical state. It also assessed whether routine operation remained simple, whether advanced functions were accessible without cluttering the standard workflow, whether relevant calculated values were displayed directly, and whether tab-specific user manuals provided sufficient operational guidance.

Table 3 summarizes how the ten usability heuristics were translated into interface design and evaluation criteria for the developed syringe pump platform.

Table 3: Translation of Nielsen's 10 usability heuristics into interface design and evaluation criteria.

Usability heuristic	Application in the syringe pump interface
Visibility of system status	Display of the current process state, system progress, calculated process information, and relevant driver-side status values during operation.
Match between system and real world	Use of process-related terminology such as viscosity, extrusion quantity, process status, motor control, and volume–time behavior.
User control and freedom	Support for controlled stopping, motor-disable-based manual adjustment, and separation between routine operation and advanced configuration.
Consistency and standards	Consistent workflow states and tab structures for setting, extrusion progress, stopping, completion, developer information, and tab-specific guidance.
Error prevention	Separation of parameter input, start, stop, finish, and advanced setting functions to reduce unintended operation during precision-critical extrusion.
Recognition rather than recall	Display of calculated process information, progress information, and relevant status values so that the operator does not need to remember intermediate values.
Flexibility and efficiency of use	Separation between a basic mode for routine viscosity-and-quantity-based extrusion and advanced developer pages for manual motor control, customized parameters, and process curves.
Aesthetic and minimalist design	Restriction of the standard workflow to essential input and progress information, while keeping advanced curves and detailed parameter settings outside the routine operation screen.
Help users recognize, diagnose, and recover from errors	Use of clear stop, rejected-command, completion, and status states to support recognition of abnormal or unintended operation and controlled recovery.
Help and documentation	Provision of a user manual within each tab, allowing the operator to access tab-specific instructions directly during laboratory use.

The heuristic-based evaluation is not presented as clinical usability validation. Instead, it assesses whether the prototype interface is sufficiently clear, consistent, efficient, and supportive for controlled laboratory filling trials. The results of this evaluation are considered together with the experimental validation in Chapter 5.

4.6 Experimental Validation Method

4.6.1 Validation Structure Based on DQ, IQ, OQ, and PQ

The experimental validation method was structured using the logic of Design Qualification, Installation Qualification, Operational Qualification, and Performance Qualification. In this thesis, these terms are used as a qualification-inspired framework for laboratory validation of the developed syringe pump platform. They are not used to claim formal regulatory qualification or production release of a medical device. This structure separates verification of the design concept, verification of the physical setup, evaluation of operating functions, and final assessment of filling performance under representative laboratory conditions.

Design Qualification (DQ) was used to verify that the planned system design addressed the requirements derived in Chapter 3. This phase corresponds to a design review. It evaluates whether the mechanical subsystem, drive and actuation subsystem, control and compensation method, user interface, and validation procedure were defined in a way that could support viscosity-controlled syringe-based filling. The DQ step focuses on the consistency between requirements and design decisions rather than on measured filling performance.

Installation Qualification (IQ) was used to verify that the developed prototype and its components were assembled and prepared correctly before experimental operation. This phase corresponds to a setup checklist. It includes confirmation of the syringe fixation, standardized Luer Lock cannula connection, plunger contact interface, leadscrew and guide structure, motor and driver connection, electrical supply, user interface availability, and measurement equipment. The IQ step ensures that the system is in a defined and repeatable starting condition before operational testing begins.

Operational Qualification (OQ) was used to evaluate whether the implemented functions operated as intended within the specified laboratory operating range. This phase corresponds to measurements and experiments. It includes testing of parameter input, distance-based target generation, compensation scaling, viscosity-dependent ramp planning, TMC-executable command generation, extrusion execution, controlled stopping, plunger contact interface retraction, and monitoring display. The OQ step also includes repeated extrusion trials to evaluate volumetric deviation and repeatability under fixed process conditions.

Performance Qualification (PQ) was used to evaluate whether the complete platform could perform the intended filling task under representative laboratory conditions. In this thesis, PQ was implemented by producing and measuring a small number of implant-representative samples. This phase assesses the combined effect of mechanical positioning, actuation, compensation, user guidance, passive release, and post-extrusion observation on the final filling outcome. The PQ step therefore evaluates the integrated platform performance rather than an isolated subsystem function.

Table 4 summarizes how the qualification-inspired validation structure was translated into the experimental validation method used in this thesis.

Table 4: Qualification-inspired validation structure used for the experimental validation method.

Phase	Methodological role	Application in this thesis
DQ	Design review	Verification that the planned mechanical, actuation, control, user interface, and validation concepts address the requirements derived in Chapter 3.
IQ	Setup checklist	Verification that the prototype, syringe fixation, cannula connection, actuator, driver electronics, interface, and measurement equipment are correctly assembled and ready for testing.
OQ	Measurements and experiments	Verification that the implemented functions operate as intended across the specified laboratory operating range, including extrusion control, compensation, ramp generation, monitoring, and repeated filling trials.
PQ	Representative performance test	Validation that the integrated platform can fill implant-representative samples under representative laboratory conditions.

4.6.2 Design and Installation Qualification

The DQ step was performed by checking the consistency between the requirements and the implemented subsystem concepts. The mechanical design was reviewed with respect to syringe positioning, axial force transmission, plunger contact, actuator support, and linear guidance. The drive and actuation design was reviewed with respect to torque transmission, rotary-to-linear motion conversion, and driver integration. The control and compensation method was reviewed with respect to target-distance calculation, compensation scaling, viscosity-constrained ramp planning, and post-extrusion passive release. The user interface was reviewed with respect to parameter input, process visibility, error prevention, workflow guidance, and heuristic-based usability criteria.

The IQ step was performed before experimental measurements to confirm that the assembled prototype was ready for operation. The syringe had to be seated correctly in the snap-fit fixation, the cannula had to be attached through the standardized Luer Lock connection, and the plunger contact interface had to be aligned with the syringe plunger. The leadscrew, trapezoidal flange nut, stainless steel guide rod, and sliding bearing were checked for unobstructed motion. The stepper motor, TMC5160-TA driver board, embedded controller, display interface, power supply, and monitoring signals were also checked before running extrusion trials. This checklist-based procedure reduced the risk that setup errors would be interpreted as control or compensation errors during the experiments.

4.6.3 Operational and Performance Qualification

The OQ step was performed through controlled extrusion experiments. For each test condition, the operator prepared the syringe and cannula, entered the target viscosity and extrusion quantity, and executed the generated extrusion command. After extrusion, passive release, and post-extrusion stabilization, the delivered amount was measured and compared with the target volume. Repeated trials were used to evaluate whether the system behavior was reproducible under fixed process conditions. This distinction between repeatability and absolute deviation was important because the compensation strategy is only meaningful when the dominant error is systematic and reproducible.

The delivered volume was determined from gravimetric measurements and converted to volume using the supplier-specified specific gravity of the tested liquid. Two weighing approaches were used depending on the viscosity-dependent handling behavior of the liquid. This distinction was introduced to reduce systematic measurement errors caused by preparation loss, incomplete collection, stringing, or residual material at the cannula outlet.

For low-viscosity liquids, the delivered mass was measured using a laboratory glass Petri dish as the receiving container. The empty Petri dish was weighed before the test, and the total mass of the Petri dish with the delivered liquid was measured after extrusion and post-extrusion stabilization. This approach was selected because low-viscosity liquids can be unintentionally lost during syringe filling, air removal, or initial-state preparation. If the syringe mass difference were used for these liquids, preparation-related losses could be incorrectly included in the extrusion result.

For high-viscosity liquids, the delivered mass was determined from the mass difference of the syringe before and after extrusion. This approach was selected because high-viscosity liquids show stronger cohesion and are less likely to flow cleanly into a receiving dish, especially when a small-diameter cannula is used. Measuring the receiving dish could therefore be affected by incomplete transfer, stringing, or residual material at the cannula outlet. Under these conditions, the syringe mass difference provides a more practical estimate of the mass displaced from the syringe during extrusion.

The measured mass difference was converted into delivered volume according to

$$V_{\text{delivered}} = \frac{\Delta m}{\rho}, \quad (4.6)$$

where $V_{\text{delivered}}$ is the delivered volume, Δm is the measured mass difference, and ρ is the density derived from the supplier-specified specific gravity. For the Petri-dish-based method, Δm is the difference between the mass of the filled Petri dish and the empty Petri dish. For the syringe-based method, Δm is the difference between the syringe mass before and after extrusion. The specific gravity was not measured independently in this thesis, but was taken from the official supplier documentation of the tested liquid. The density used for the volume conversion was calculated as

$$\rho = SG_{\text{sup}} \cdot \rho_{\text{water}}, \quad (4.7)$$

where SG_{sup} is the supplier-specified specific gravity and ρ_{water} is the reference density of water used for the conversion. The volumetric deviation was then calculated from the difference between the delivered volume and the target volume:

$$e_V = \frac{V_{\text{delivered}} - V_{\text{target}}}{V_{\text{target}}} \cdot 100 \quad (4.8)$$

The operational experiments also evaluated the influence of viscosity-dependent control parameters. The effective motion scale and compensation parameters were determined from calibration tests and, where applicable, estimated through the viscosity-dependent interpolation method. The resulting parameters were then used to generate TMC-executable ramp commands for the corresponding process conditions. The OQ measurements assessed whether the control method could translate the input parameters into executable and repeatable extrusion behavior across the investigated viscosity range.

The PQ step was performed after the subsystem functions and operational behavior had been verified. A small number of implant-representative samples were filled using the integrated platform. These trials were used to assess whether the complete system could support the intended filling task when mechanical positioning, actuation, compensation, user guidance, passive release, and post-extrusion observation were applied together. The same gravimetric measurement principle was used to determine the delivered amount, with the weighing approach selected according to the viscosity-dependent handling behavior of the tested liquid. The final evaluation of filling accuracy, repeatability, waiting-time reduction, and sample filling performance is presented in Chapter 5.

4.7 Chapter Summary

This chapter described how the platform requirements were translated into mechanical, actuation, control, user-interface, and validation methods for the syringe pump platform. The mechanical and drive sections defined the physical force path from the motor to the syringe plunger, while the control section described how process inputs are converted into compensated actuator commands. The user-interface and validation sections described how the process is operated, monitored, and evaluated under laboratory conditions.

5 Results and Evaluation

5.1 Overview of the Qualification Results

This chapter presents the qualification results of the viscosity-controlled syringe pump platform and evaluates whether the developed system meets the methodological requirements defined in Chapter 4. It also examines whether the platform can support controlled high-viscosity syringe-based extrusion under defined laboratory conditions. The results are not organized as a description of individual components, but as an evaluation of the complete platform from design qualification to operational extrusion behavior and integrated filling performance.

The validation results are structured according to the DQ/IQ/OQ/PQ-inspired validation framework introduced in Chapter 4. The design qualification and installation qualification results summarize whether the system design addressed the derived requirements and whether the assembled prototype was ready for controlled testing. The operational validation results evaluate repeated extrusion behavior, volumetric deviation, compensation performance, viscosity-dependent parameter estimation, motion execution, and process monitoring. The user interface is evaluated separately using the interface requirements derived from Jakob Nielsen's 10 usability heuristics [27], because its function is to support operator guidance, process visibility, and error-preventive interaction rather than to directly determine volumetric accuracy. The integrated filling performance results then assess post-extrusion passive release, waiting-time behavior, and implant-representative filling under representative laboratory conditions.

The platform is evaluated at three levels: subsystem readiness, controlled extrusion behavior, and integrated filling performance. Detailed numerical results are reported in the corresponding sections, while the interpretation of remaining limitations and their implications for further development is addressed in the discussion chapter.

5.2 Design Qualification and Installation Qualification Results

5.2.1 Design Qualification Results

The design qualification results summarize whether the developed platform addressed the requirements derived in Chapter 3. Within the validation structure, DQ evaluated the design basis before filling accuracy was tested experimentally. The DQ was performed by comparing the requirement groups with the corresponding design solutions developed in Chapter 4.

Table 5 summarizes whether the consolidated requirement areas from Chapter 3 were addressed by the design and method decisions described in Chapter 4. The table links each requirement area to the corresponding design or method response and documents the design basis for the subsequent qualification steps.

Table 5: Design qualification results based on the consolidated requirement areas derived in Chapter 3.

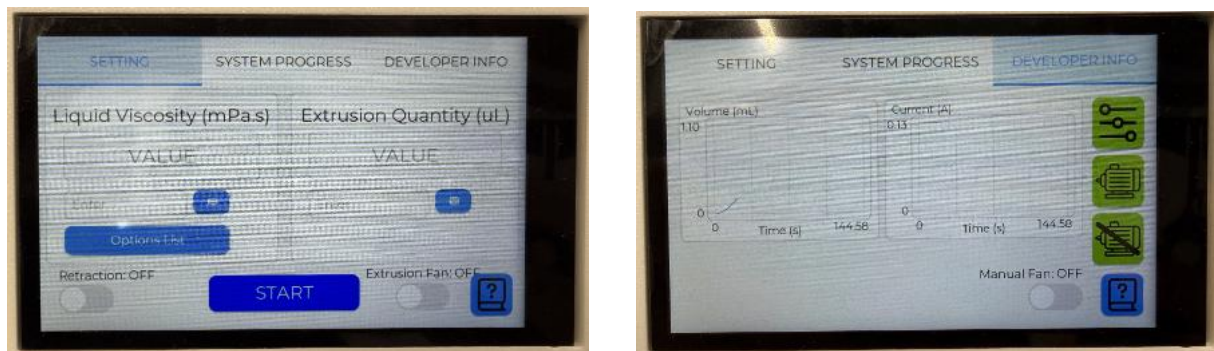
Requirement area from Chapter 3	Design or method response	Outcome
Syringe compatibility and positioning	Syringe-specific fixation and standardized Luer Lock connection.	Addressed
Mechanical stability and force transmission	Defined force path, plunger contact interface, actuator support, and linear guidance.	Addressed
Viscosity-dependent motion control	Conversion of target volume and viscosity into executable motion parameters.	Addressed
Compensation of repeatable deviations	Calibration-based compensation and viscosity-dependent parameter interpolation.	Addressed
Process transparency and user guidance	Process-state visualization, guided workflow, and Nielsen-heuristic-based interface design.	Addressed
Validation-oriented operation	DQ/IQ/OQ/PQ-inspired validation structure and gravimetric measurement method.	Addressed

The DQ outcome showed that all consolidated requirement areas were covered by corresponding design or method decisions. This result does not yet demonstrate final filling accuracy. Instead, it confirms that the platform was sufficiently defined to proceed with installation verification, operational validation, and integrated filling performance evaluation.

5.2.2 User Interface Design Qualification Based on Nielsen's Usability Heuristics

The user interface was included in the design qualification because process transparency and user guidance were defined as requirements in Chapter 3. This qualification step was based on Jakob Nielsen's 10 usability heuristics [27] and was performed as a qualitative prototype-level review. The objective was not to conduct a quantitative usability study. Instead, the review checked whether the implemented interface supported process visibility, consistent operation, error-preventive interaction, routine use, access to advanced functions, and tab-specific guidance during laboratory filling trials.

Representative interface screens documented the evidence for the heuristic-based design qualification. The screenshots document the implemented interface structure rather than a simulated mock-up. They show how routine parameter input, advanced configuration, process monitoring, protected operation, invalid-parameter handling, and embedded documentation were implemented in the prototype interface.

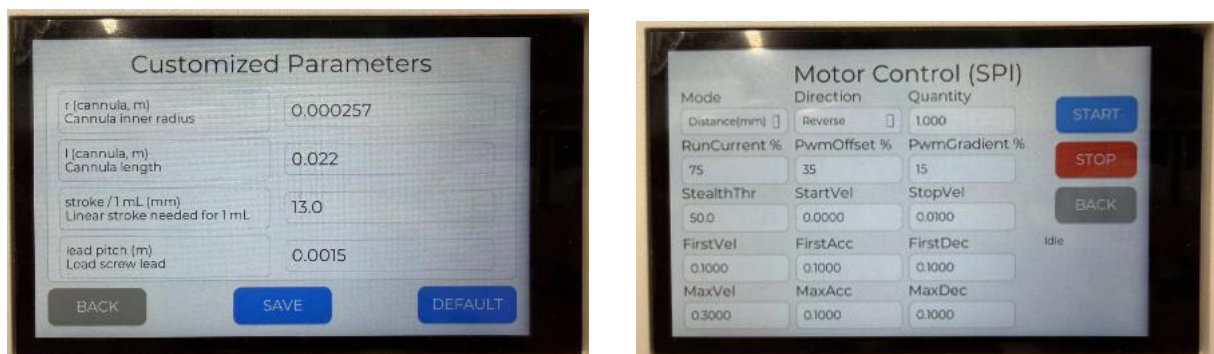


a) SETTING tab for routine operation.

b) DEVELOPER INFO tab for advanced access.

Figure 33: Routine input and advanced interface structure. The figure was created by the author.

Figure 33 documents the separation between routine operation and advanced access. In the SETTING tab, the routine workflow is limited to the required process inputs, namely liquid viscosity and target extrusion quantity. This supports efficient repeated operation and avoids unnecessary information in the standard workflow. The DEVELOPER INFO tab provides access to advanced functions, including volume–time and current–time curves, motor-disable control, manual fan control, customized parameter settings, and motor-control functions. This separation supports flexibility and efficiency of use while maintaining an aesthetic and minimalist routine operation screen.

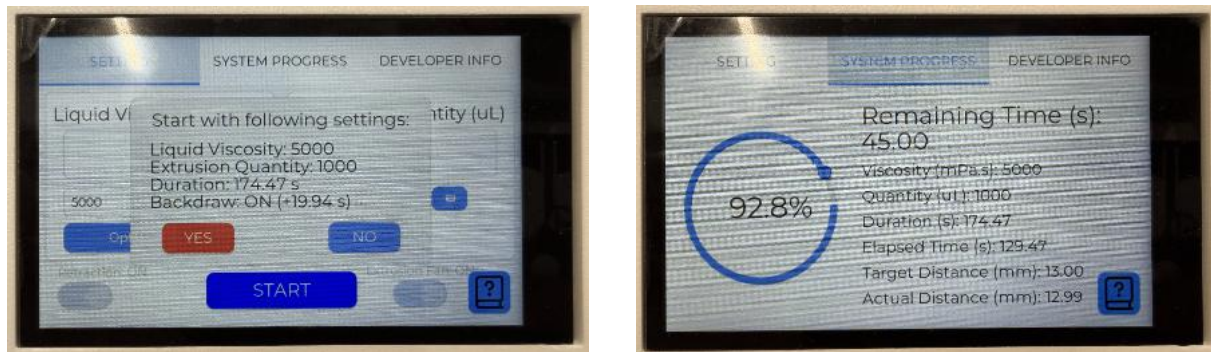


a) Customizer parameter tab.

b) Motor control tab.

Figure 34: Advanced parameter customization and motor-control access. The figure was created by the author.

Figure 34 documents the advanced configuration functions. The customizer parameter tab allows system-dependent parameters, such as cannula inner radius, maximum linear motion speed, leadscrew lead, and plunger tip radius, to be adjusted when the hardware configuration changes. The motor control tab provides access to detailed TMC-related motor-control parameters, including direction, quantity, run current, stealth threshold, maximum velocity, maximum acceleration, and maximum deceleration. These functions are not part of the standard filling workflow, but they support manual tuning, development testing, and prototype adjustment. Their placement in advanced screens supports user control and freedom without overloading routine operation.

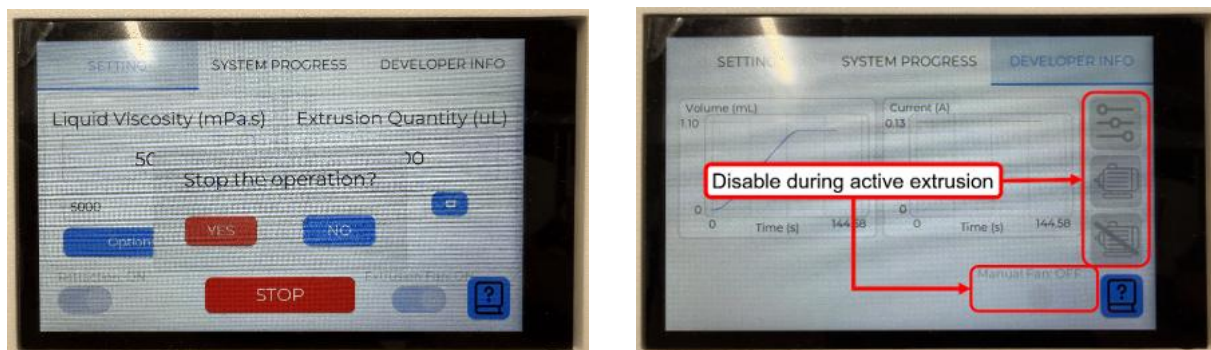


a) Pre-extrusion confirmation dialog.

b) System progress tab during extrusion.

Figure 35: Pre-extrusion confirmation and process-state monitoring. The figure was created by the author.

Figure 35 shows how the interface supports parameter confirmation and process monitoring. Before extrusion is started, the confirmation dialog displays the selected viscosity, target quantity, calculated duration, and retraction setting. This allows the operator to check the intended process condition before motion execution. During extrusion, the system progress tab displays relevant process information, including remaining time, viscosity, quantity, duration, elapsed time, target distance, actual distance, and progress percentage. This supports visibility of system status and recognition rather than recall, because the operator does not need to remember the entered parameters or estimate the remaining process time manually.

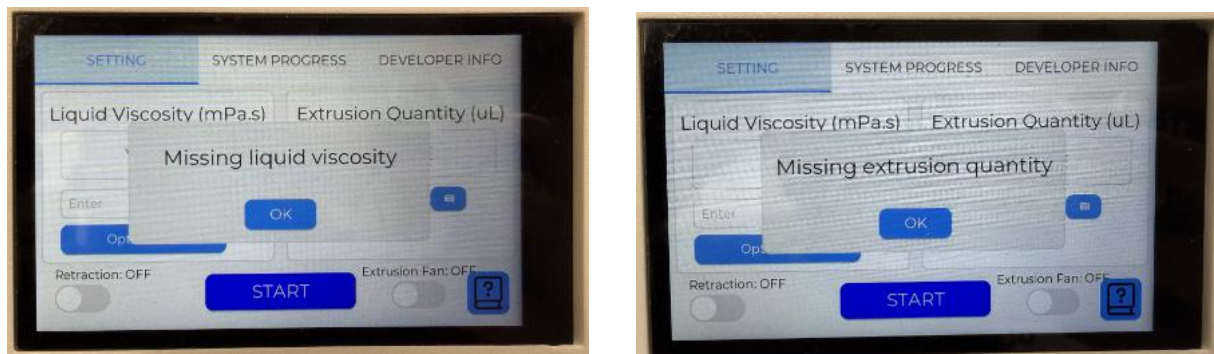


a) STOP confirmation dialog.

b) Disabled controls during active extrusion.

Figure 36: Protected interaction during active extrusion. The figure was created by the author.

Figure 36 shows the error-preventive interaction during active extrusion. If the operator requests stopping during extrusion, a confirmation dialog is displayed before the stop command is executed. This reduces the risk of unintended interruption. During active extrusion, advanced controls that could interfere with the ongoing process, such as customized parameter access and motor-control access, are disabled. This design preserves user control through a protected stopping path while preventing unintended changes to precision-critical process settings during motion execution.



a) Missing liquid viscosity warning.

b) Missing extrusion quantity warning.

Figure 37: Invalid parameter warnings for required process inputs. The figure was created by the author.

Figure 37 shows the invalid-parameter warning behavior. If the liquid viscosity is missing, the interface displays the warning message "Missing liquid viscosity". If the target extrusion quantity is missing, the interface displays the warning message "Missing extrusion quantity". These warnings prevent incomplete process inputs from being passed to the extrusion workflow and directly identify the missing parameter. This supports error prevention and helps the operator recognize, diagnose, and correct input-related errors before extrusion is started.



a) User manual entry in the SETTING tab.

b) Tab-specific user manual content.

Figure 38: Embedded help and tab-specific user documentation. The figure was created by the author.

Figure 38 shows the embedded help function. The user manual entry is available directly within the interface and uses a recognizable icon structure. The corresponding user manual page provides operation-related information for the selected tab. This supports help and documentation by allowing the operator to access context-specific instructions without relying only on external documentation.

Table 6 reports the heuristic-based design qualification outcome. Each usability heuristic was linked to one or more implemented interface functions shown in the representative screenshots. The table therefore serves as a structured summary of the qualitative review rather than as a separate quantitative usability assessment.

Table 6: User interface design qualification based on Jakob Nielsen's 10 usability heuristics.

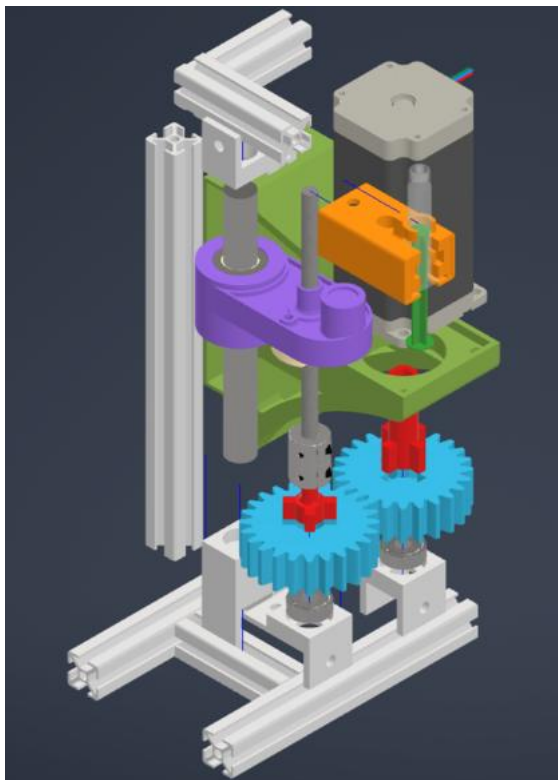
Usability heuristic	Interface response	Outcome
Visibility of system status	System progress, process state, calculated values, remaining time, target distance, and actual distance are displayed during operation.	Addressed
Match between system and real world	Interface terminology follows the filling workflow, including viscosity, quantity, duration, retraction, motor control, and volume–time behavior.	Addressed
User control and freedom	Start confirmation, protected stopping, motor-disable-based manual adjustment, and advanced configuration access are separated into defined interaction paths.	Addressed
Consistency and standards	Setting, progress, developer information, motor-control, parameter-customization, and user manual screens follow a consistent tab-based interface structure.	Addressed
Error prevention	Required input checks, start confirmation, stop confirmation, and disabled controls during active extrusion reduce unintended operation.	Addressed
Recognition rather than recall	Selected process parameters, calculated values, progress information, and warning messages are shown directly on the interface.	Addressed
Flexibility and efficiency of use	Routine users can operate the pump through the basic viscosity-and-quantity workflow, while experienced users can access developer functions when required.	Addressed
Aesthetic and minimalist design	Routine screens are limited to essential input and progress information, while detailed curves and customized settings remain in the developer area.	Addressed
Help users recognize, diagnose, and recover from errors	Invalid-parameter warnings identify missing viscosity or quantity before extrusion is started, and status-related dialogs support recognition of unintended operation.	Addressed
Help and documentation	Each tab provides access to a corresponding user manual for context-specific operating guidance.	Addressed

The heuristic-based review confirmed that the prototype interface addressed all ten usability principles. This result shows that the graphical user interface was sufficiently structured to support controlled laboratory operation, reduce avoidable operator-dependent variation, and make the process state and required user actions visible during repeated filling trials.

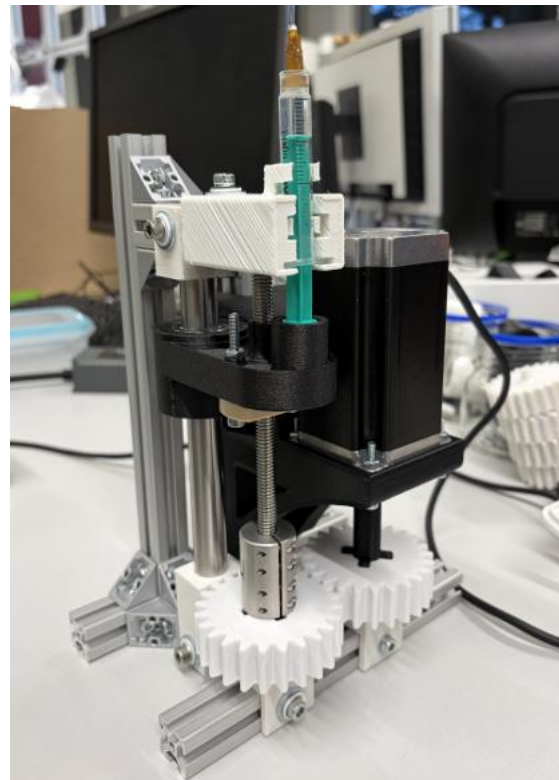
5.2.3 Installation Qualification Results

The installation qualification evaluated whether the assembled prototype was ready for operational qualification testing. This step corresponds to the IQ part of the qualification-inspired validation structure. Before the extrusion experiments were started, the mechanical assembly, drive chain, electronics, user interface, and measurement setup were checked using a predefined setup checklist. The purpose was to ensure that subsequent OQ results would not be dominated by avoidable installation or preparation errors.

Figure 39 documents the installation-related elements checked during IQ. The CAD-based view was used as an installation reference to locate the main mechanical and actuation components included in the checklist. The physical prototype image records the assembled test-ready state before OQ testing. In addition, the KERN PCD balance was documented separately as the gravimetric measurement device prepared for the subsequent extrusion trials. The figure is therefore not used to repeat the design explanation from Chapter 4, but to connect the installation checklist with the actual prototype configuration used for the experiments.



a) CAD-based installation reference.



b) Assembled test-ready prototype.

Figure 39: Installation qualification overview showing the CAD-based installation reference and the assembled test-ready prototype. The figure was created by the author.

The installation qualification focused on elements that directly affect extrusion repeatability and measurement reliability. These include the syringe seating, cannula connection, plunger contact alignment, linear motion path, drive and actuation chain, driver electronics, monitoring display, and gravimetric measurement setup. The CAD-based reference supports the identification of these elements, while the prototype image documents that the system was assembled in the configuration used for the following tests.

Figure 40 documents the KERN PCD balance prepared for the subsequent gravimetric evaluation. In the installation qualification, the balance is documented as part of the measurement setup readiness. The figure does not present measurement results, but confirms the availability of the gravimetric measurement device before operational qualification testing.



Figure 40: KERN PCD balance used as the gravimetric measurement device for subsequent extrusion trials. The figure was created by the author.

Table 7 summarizes the installation qualification outcome. The checklist was used to confirm that the prototype was in a defined and repeatable starting condition before operational qualification testing.

Table 7: Installation qualification checklist before operational qualification testing.

Installation item	Verification criterion	Outcome
Syringe fixation	Syringe seated in the fixation without visible misalignment.	Verified
Cannula connection	Cannula attached through the standardized Luer Lock connection.	Verified
Plunger contact interface	Contact surface aligned with the syringe plunger before extrusion.	Verified
Linear motion path	Leadscrew, trapezoidal flange nut, stainless steel guide rod, and sliding bearing move without obstruction.	Verified
Drive and actuation chain	Motor, gear stage, connectors, coupling, and leadscrew assembled in the intended transmission path.	Verified
Driver electronics and controller	TMC5160-TA driver board, embedded controller, power supply, and display interface available for operation.	Verified
Monitoring display	Process state and relevant driver-side status information visible before testing.	Verified
Measurement setup	KERN PCD balance, receiving container or syringe-weighing setup, and supplier-specific-gravity information prepared for the tested liquid.	Verified

The IQ outcome showed that the prototype could be brought into a defined test-ready state before OQ testing. This result is important because the following extrusion measurements depend on repeatable syringe positioning, a stable actuation path, a functioning control interface, and a consistent gravimetric measurement procedure. The installation qualification therefore served as the transition from design implementation to operational qualification.

5.3 Operational Qualification Results

5.3.1 Motion Execution and Process Monitoring

The first OQ step evaluated whether the implemented motion-control and monitoring functions operated as intended during extrusion. This check was performed before the quantitative volume measurements were interpreted. Its purpose was to confirm that the prototype could generate an executable motion command, execute the commanded extrusion movement, display the relevant process information, complete the post-extrusion release sequence, and document relevant operational constraints under the defined laboratory operating conditions.

Figure 41 documents the representative operational setup used during this OQ step. The figure documents the assembled syringe pump platform together with the GUI monitoring interface used during motion execution. Since the extrusion movement cannot be represented dynamically in a static image, the figure is used to document the operational configuration and the availability of process monitoring during the test.

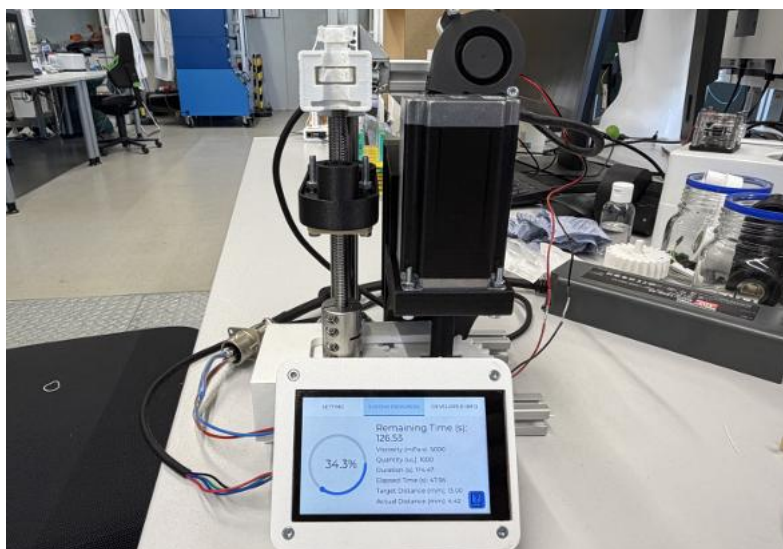


Figure 41: Syringe pump prototype and GUI monitoring interface during operational qualification testing. The figure was created by the author.

The operational check also included the use of customized parameters and manual adjustment functions before extrusion. Hardware-dependent parameters, such as cannula geometry, maximum linear motion speed, leadscrew lead, and plunger-related settings, were treated as configuration inputs for command generation. Manual adjustment was limited to the preparation phase before extrusion, for example by using the motor-disable function to define the initial mechanical state. During active extrusion, functions that could interfere with the ongoing motion were protected or disabled, so that the extrusion sequence remained controlled.

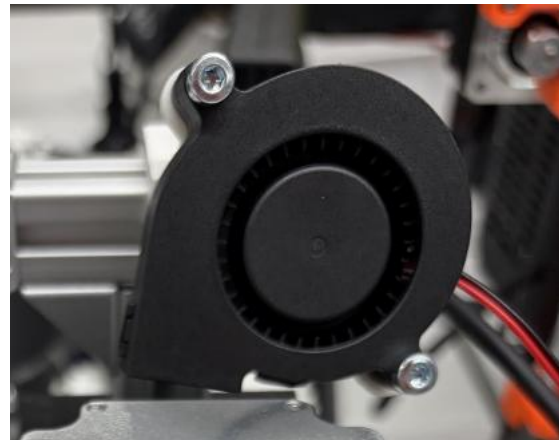
During long operating periods, motor heating may become an operational constraint. The motor surface temperature is relevant not only for component lifetime, but also for handling and user safety. Motor temperature was therefore measured under extended operating conditions to assess whether additional cooling was required. The measurement was performed with a FLIR E60 thermal camera. To improve the consistency of the thermal image measurement, the black industrial tape was attached to a defined position on the motor housing and used as a fixed measurement target. Before measurement, sufficient time was allowed for the tape and the motor housing surface to approach thermal equilibrium. This reduced the influence of the initial temperature difference after the tape had been applied. The surface emissivity of the thermal camera was set to 0.95, corresponding to the high-emissivity surface of the black industrial tape.

Under these measurement conditions, the highest measured surface temperature at the measurement target was 81.7 °C, as shown in Fig. 42 a). This value was still below the estimated maximum housing temperature of 103 °C, which was derived from the motor temperature-rise limit and an ambient laboratory temperature of approximately 23 °C. However, the measured surface temperature was already high enough to be considered relevant for handling and user

safety. For this reason, the 5 V 5015 radial fan shown in Fig. 42 b) was added as an optional local cooling measure.



a) Motor housing temperature after long-duration operation.



b) 5 V 5015 radial fan used for motor cooling.

Figure 42: Motor thermal observation and selected radial fan for optional cooling. The figure was created by the author.

The radial fan was selected instead of an axial fan because it directs airflow more locally toward the motor housing. This reduces unnecessary airflow toward the syringe side of the setup and therefore limits potential disturbance of the filling condition. The fan was mounted on the rear frame with a defined distance and angle relative to the motor housing. It is driven by the 5 V output of the custom driver board and can be activated manually from the DEVELOPER INFO tab or selected before extrusion when cooling is required.



Figure 43: Installed 5 V radial fan mounted on the rear frame and angled toward the motor housing for localized cooling. The figure was created by the author.

Table 8 reports the operational qualification results for motion execution and process monitoring. The table focuses on whether the motion command, manual adjustment, customized

parameter use, monitoring functions, active-operation protection, post-extrusion release, and motor thermal management were available or documented during OQ testing.

Table 8: Operational qualification results for motion execution and process monitoring.

Operational item		Qualification evidence	Result
Customized setup	parameter	Hardware-dependent parameters were entered before operation and used as input for command generation.	Verified
Manual adjustment before extrusion		Motor-disable-based manual adjustment was available before extrusion to define the initial mechanical state.	Verified
Command generation		The selected viscosity, target quantity, and customized parameters were converted into a valid TMC-executable motion command.	Verified
Motion execution		The extrusion motion was executed until the commanded target distance was reached.	Verified
Process monitoring		The interface displayed process state, elapsed time, remaining time, progress percentage, target distance, and actual distance during operation.	Verified
Active-operation protection		Advanced controls that could interfere with extrusion were disabled during active operation, while protected stopping remained available.	Verified
Post-extrusion release	passive re-	The plunger contact interface retraction was executed after target-distance completion.	Verified
Motor thermal management		Motor heating during long-duration operation was documented with thermal imaging, and a 5 V 5015 radial fan was integrated as an optional localized cooling measure.	Documented

The OQ check verified that the prototype could generate, execute, and monitor the extrusion command under the tested laboratory conditions. It also identified motor heating as an operational constraint and addressed it with an optional cooling measure. These results provided the functional basis for the subsequent repeatability, baseline deviation, and compensation tests.

5.3.2 Repeatability and Baseline Deviation

After motion execution and process monitoring had been verified, repeated baseline extrusion trials evaluated the repeatability of the syringe pump platform before compensation. The objective of this step was not to demonstrate final filling accuracy, but to determine whether the uncompensated extrusion behavior was sufficiently repeatable under defined process conditions. This distinction is important because calibration-based compensation is only meaningful

when the observed deviation is systematic and reproducible rather than dominated by random trial-to-trial scatter.

The baseline tests were performed under controlled experimental conditions to reduce avoidable variation between trials. Four viscosity conditions were investigated: 1 mPa·s water, 5000 mPa·s oil, 30000 mPa·s oil, and 60000 mPa·s oil. The 5000, 30000, and 60000 mPa·s oils were Wacker silicone oils, for which the supplier-specified specific gravity at 25 °C was taken as 0.97. For all tests, the same B. Braun Injekt Luer Lock Solo 2 mL disposable syringe and B. Braun 0.80 × 22 mm cannula type were used, the target extrusion volume was set to 1 mL, and the initial syringe barrel scale was kept between 1.3 and 1.7 mL. Each syringe was used only once. The same uncompensated effective motion scale of 36634.5 and the same motor run current of 75% were used for all baseline trials. The tests were performed in a temperature-controlled laboratory environment at approximately 23 °C. Before each measurement, the KERN PCD balance was tared. Water was measured using a glass Petri dish as the receiving container, whereas the higher-viscosity oils were evaluated using the syringe mass difference before and after extrusion. After each oil-based extrusion, the plunger contact interface was retracted by 3 mm and the delivered amount was evaluated after a standardized post-release waiting interval of 3 min. For the water-based tests, the pressure-delay effect was negligible under the tested conditions, and the delivered amount was evaluated after visible completion of the extrusion process. The detailed influence of post-extrusion passive release on waiting-time behavior is evaluated later in the performance qualification results.

Table 9 reports the baseline repeatability results before compensation. For the water-based tests, the initial and measured masses refer to the empty and filled Petri dish, respectively. For the oil-based tests, they refer to the syringe mass before and after extrusion. The measured mass difference was first converted into delivered volume and then compared with the 1 mL target volume to calculate the volumetric deviation. A negative deviation indicates that the delivered volume was lower than the target volume, whereas a positive deviation indicates that the delivered volume was higher than the target volume. Mass values are reported to three decimal places according to the balance readability, while volumetric deviations are rounded to two decimal places for tabular consistency.

Table 9: Baseline repeatability results before compensation.

Trial	Initial mass [g]	Measured mass [g]	Volumetric deviation [%]
Viscosity: 1 mPa·s water			
1	48.505	49.507	0.20
2	48.505	49.507	0.20
3	48.505	49.508	0.30
4	48.504	49.502	-0.20
5	48.505	49.506	0.10
Mean absolute deviation			0.20
Maximum intra-condition difference			0.50
Viscosity: 5000 mPa·s			
1	4.238	3.330	-6.39
2	4.215	3.309	-6.60
3	4.483	3.573	-6.19
4	4.481	3.570	-6.08
5	4.327	3.424	-6.91
Mean absolute deviation			6.43
Maximum intra-condition difference			0.83
Viscosity: 30000 mPa·s			
1	4.334	3.561	-20.31
2	4.359	3.591	-20.82
3	4.288	3.512	-20.00
4	4.275	3.501	-20.21
5	4.276	3.509	-20.93
Mean absolute deviation			20.45
Maximum intra-condition difference			0.93
Viscosity: 60000 mPa·s			
1	4.422	3.750	-30.72
2	4.372	3.696	-30.31
3	4.374	3.699	-30.41
4	4.409	3.731	-30.1
5	4.216	3.537	-30
Mean absolute deviation			30.31
Maximum intra-condition difference			0.72

The baseline trials showed repeatable behavior within each viscosity condition. Although the mean absolute deviation differed between viscosity levels, the maximum intra-condition difference between the five repeated trials remained below 1 percentage point for each tested viscosity. This indicates that the uncompensated deviation was not dominated by uncontrolled random variation. Instead, the results suggest a reproducible baseline offset under the defined process conditions. The tested range from nominal 1 to 60000 mPa·s was therefore considered suitable for subsequent compensation evaluation.

The repeatability result provides the basis for the compensation step. Since the trial-to-trial variation within each viscosity condition remained below the defined tolerance, the observed baseline deviation can be treated as a systematic and correctable process deviation. The following section evaluates whether calibration-based compensation and viscosity-dependent parameter estimation can reduce this systematic deviation between commanded displacement volume and delivered volume.

5.3.3 Compensation Parameter Selection and Viscosity-Dependent Interpolation

The baseline repeatability results established the need for viscosity-specific correction. This subsection focuses on how corrected effective motion scale values were selected for the calibrated viscosity conditions and how these values were used to estimate a compensation setting for an intermediate viscosity. The effective motion scale was selected because it directly changes the relationship between commanded actuator displacement and delivered volume.

Four viscosity conditions were used as calibration points for the compensation parameter selection: 1 mPa·s water, 5000 mPa·s oil, 30000 mPa·s oil, and 60000 mPa·s oil. For each condition, the effective motion scale was adjusted experimentally until the delivered volume was within the defined tolerance range. These calibrated values were not treated as final performance validation results. Instead, they served as anchor points for building a viscosity-dependent compensation relationship.

The calibration procedure was based on the baseline repeatability results in Table 9. For each viscosity condition, the mean error from the baseline test was first used together with the effective motion scale update relation introduced in Section 4.4.2, Equation 4.4. This provided an initial estimate for the corrected effective motion scale. This update was only used as a first approximation. In practice, the extrusion error was not expected to follow an ideal linear relation with the effective motion scale. The delivered volume can also be affected by viscosity-dependent pressure build-up, syringe compliance, plunger friction, and delayed material release. For this reason, the initially estimated value did not always meet the target accuracy directly. Further adjustment was required until the delivered volume fell within the defined tolerance range.

The 5000 mPa·s oil condition can be used as an example. In the baseline test, this condition showed a mean error of -6.43%, which means that the delivered volume was lower than the target value. The effective motion scale had to be increased. Using Equation 4.4 with $S_{\text{eff, std}} = 36634.50$ gives

$$(1 + 0.0643) \cdot 36634.50 = 38990.98$$

The initial estimate for $S_{\text{eff},\eta}$ was 38990.98. A subsequent extrusion test with this value showed that the delivered volume was then higher than the target value. The new measured error was used for another scaling step, followed by another experimental check. This iterative adjustment was continued until an effective motion scale was found that satisfied the target accuracy for the corresponding viscosity condition.

Table 10 reports the calibrated extrusion results used for parameter selection. The table follows the same structure as the baseline repeatability table, but each viscosity block additionally reports the selected effective motion scale used for that calibrated condition. For the water condition, the baseline setting was retained because the delivered volume was already within the defined tolerance range. Since the previous baseline evaluation had already confirmed repeatable behavior within each viscosity condition, the calibrated table uses the maximum absolute deviation instead of the maximum intra-condition difference. This metric was selected to evaluate how far the calibrated result deviated from the ideal zero-error condition.

Table 10: Calibrated extrusion results used for compensation parameter selection.

Trial	Initial mass [g]	Measured mass [g]	Volumetric deviation [%]
Viscosity: 1 mPa·s water Selected effective motion scale: 36634.50			
1	48.505	49.507	0.20
2	48.505	49.507	0.20
3	48.505	49.508	0.30
4	48.504	49.502	-0.20
5	48.505	49.506	0.10
Mean absolute deviation			0.2
Maximum absolute deviation			0.30
Viscosity: 5000 mPa·s Selected effective motion scale: 38531.0			
1	4.481	3.508	0.31
2	4.475	3.511	-0.62
3	4.428	3.462	-0.41
4	4.377	3.403	0.41
5	4.445	3.470	0.52
Mean absolute deviation			0.45
Maximum absolute deviation			0.62
Viscosity: 30000 mPa·s Selected effective motion scale: 40598.0			
1	4.280	3.310	0.00
2	4.285	3.318	-0.31
3	4.289	3.317	0.21
4	4.235	3.268	-0.31
5	4.308	3.339	-0.10
Mean absolute deviation			0.19
Maximum absolute deviation			0.31
Viscosity: 60000 mPa·s Selected effective motion scale: 41605.0			
1	4.675	3.710	-0.52
2	4.556	3.583	0.31
3	4.626	3.659	-0.31
4	4.664	3.690	0.41
5	4.696	3.730	-0.41
Mean absolute deviation			0.36
Maximum absolute deviation			0.52

The calibrated effective motion scale increased with viscosity, but the relationship was not linear over the investigated range. The calibrated values were therefore used as discrete anchor points for the viscosity-dependent interpolation map defined in Section 4.4.2, Equation 4.5. A monotonic interpolation was selected because it preserves the increasing trend of the calibrated data and avoids artificial oscillations between neighboring calibration points. In this context, the calibrated trials defined effective motion scale values that were repeatable and sufficiently close to the target volume, while the interpolation map was used to estimate the corresponding parameter for an intermediate viscosity condition.

For the interpolation step, the four calibrated compensation points were used as anchor points. The viscosity values were first transformed into the logarithmic viscosity domain before the interpolation map was evaluated. The calibrated input data were

$$\eta = [1, 5000, 30000, 60000] \text{ mPa} \cdot \text{s}$$

and the corresponding selected effective motion scale values were

$$S_{\text{eff}} = [36634.50, 38531.0, 40598.0, 41605.0]$$

For the validation condition, the target viscosity was

$$\eta_{\text{target}} = 40000 \text{ mPa} \cdot \text{s}$$

which corresponds to a logarithmic viscosity value of

$$\log_{10}(40000) = 4.60206$$

The monotonic interpolation map defined in Section 4.4.2 was then evaluated at this logarithmic viscosity value. This resulted in an interpolated effective motion scale of

$$S_{\text{eff}}(40000) = 41001.71$$

Therefore, the value of 41001.71 was obtained from the monotonic interpolation routine at the logarithmic target viscosity. It was not introduced as an additional manually tuned calibration value. The calculation sequence used to obtain this value is summarized in Listing 5.1.

```

1 Input calibration points:
2 viscosity_mpas = [1, 5000, 30000, 60000]
3
4 effective_motion_scale = [36634.50, 38531.0, 40598.0, 41605.0]
5
6 Transform viscosity axis:
7 log_viscosity = log10(viscosity_mpas)
8
9 Target condition:
10 target_viscosity = 40000 mPa*s
11 target_log_viscosity = log10(40000)
12 target_log_viscosity = 4.60206
13
14 Interpolation:
15 Build the monotonic interpolation map

```

```

16 from log_viscosity and effective_motion_scale
17 using the interpolation routine defined in Section 4.4.2
18
19 Evaluate this interpolation map at target_log_viscosity
20
21 Output:
22 interpolated_effective_motion_scale = 41001.71

```

Listing 5.1: Interpolation calculation for the 40000 mPa·s validation condition.

Figure 44 presents the monotonic interpolation based on the calibrated compensation points. The viscosity axis is displayed on a logarithmic scale to represent the wide investigated viscosity range. The interpolated point for 40000 mPa·s corresponds to the calculation sequence summarized in Listing 5.1.

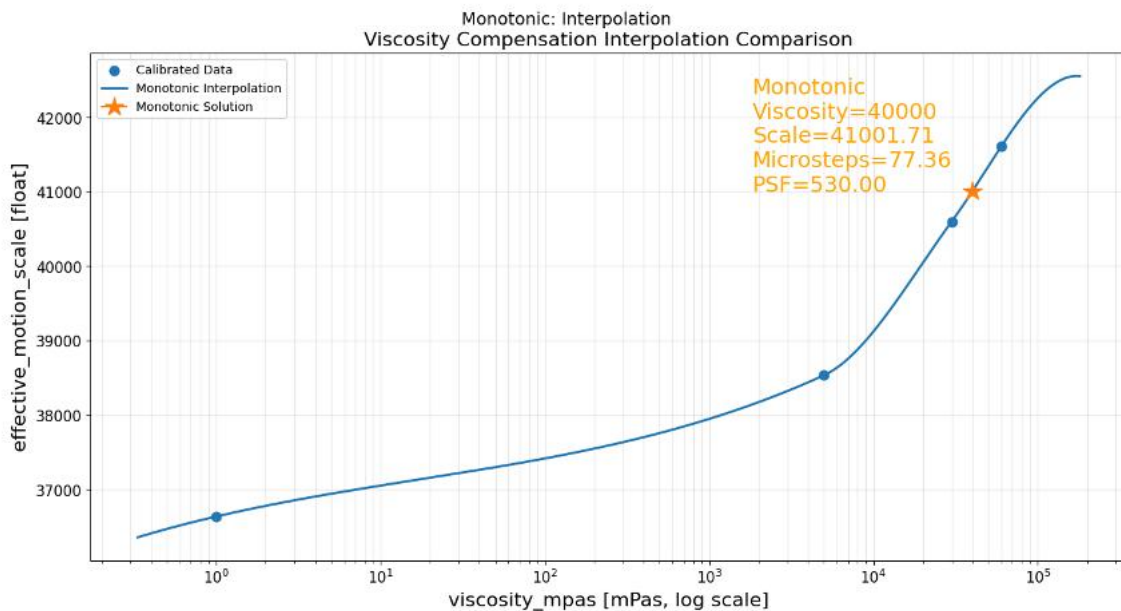


Figure 44: Monotonic interpolation of the viscosity-dependent effective motion scale based on the calibrated compensation points. The interpolation follows the compensation map defined in Section 4.4.2, Equation 4.5. For the nominal 40000 mPa·s validation liquid, the interpolation predicted an effective motion scale of 41001.71. The figure was created by the author.

The interpolated effective motion scale was then decomposed into the two motor-control sub-elements introduced in Section 4.4.2: the microsteps per real position unit and the position scale factor. The relationship between these two sub-elements and the effective motion scale follows Equation 4.2. This conversion was required because the effective motion scale is not implemented as a single firmware parameter. In the SPI-based TMC51X0 implementation, the position scale factor also converts physical distance, velocity, and acceleration commands into internal controller units before they are passed to the TMC converter. This scaling preserves low-speed and low-acceleration commands that could otherwise be lost during integer-based parameter conversion.

The experimentally observed executable parameter range was used to define the practical decomposition limits. The default microsteps per real position unit were kept at 70.0 when

possible, while the practical upper limit of the position scale factor was set to 530.0. During parameter tuning, settings that required a position scale factor above this limit no longer produced reliably executable motor commands. Therefore, if the interpolated effective motion scale would require a position scale factor above 530.0, the position scale factor was fixed at this practical upper limit and the remaining compensation was assigned to the microsteps per real position unit.

For the nominal 40000 mPa·s validation liquid, the interpolated effective motion scale of 41001.71 could not be applied by keeping the microsteps per real position unit at the default value of 70.0, because this would require a position scale factor of 585.74. This value exceeds the practical upper limit of 530.0. The position scale factor was therefore fixed at 530.0, and the remaining compensation was assigned to the microsteps per real position unit. The resulting parameter decomposition is summarized in Listing 5.2.

```

1  Input:
2  target_viscosity = 40000 mPa*s
3  interpolated_effective_motion_scale = 41001.71
4
5  Practical parameter limits:
6  default_microsteps_per_real_position_unit = 70.0
7  max_position_scale_factor = 530.0
8
9  Initial decomposition check:
10 required_position_scale_factor =
11 interpolated_effective_motion_scale
12 / default_microsteps_per_real_position_unit
13
14 Computed value:
15 required_position_scale_factor = 585.74
16
17 Decision:
18 if required_position_scale_factor <= max_position_scale_factor:
19     position_scale_factor =
20         required_position_scale_factor
21
22     microsteps_per_real_position_unit =
23         default_microsteps_per_real_position_unit
24
25 else:
26     position_scale_factor =
27         max_position_scale_factor
28
29     microsteps_per_real_position_unit =
30         interpolated_effective_motion_scale
31         / max_position_scale_factor
32
33 Final motor-control parameter set:
34 effective_motion_scale = 41001.71
35 position_scale_factor = 530.0
36 microsteps_per_real_position_unit = 77.36

```

Listing 5.2: Parameter decomposition for the 40000 mPa·s validation condition.

The nominal 40000 mPa·s validation liquid was prepared by mixing the 30000 and 60000 mPa·s Wacker silicone oils. Although these two oil grades were included as calibrated

anchor points, the mixed 40000 mPa·s condition itself was not used as a calibration point. It was reserved as an intermediate validation condition for the subsequent performance qualification. The interpolated motor-control parameter set was used without additional manual retuning. This separation between calibrated anchor points and the interpolated validation condition allows the following section to evaluate whether the viscosity-dependent compensation strategy can be applied to an uncalibrated viscosity within the tested range.

5.4 Performance Qualification Results

5.4.1 Compensated Extrusion Performance

The performance qualification evaluated the final extrusion performance of the viscosity-dependent compensation strategy under a representative high-viscosity condition. In contrast to the operational qualification, which established the calibration points and interpolation relationship, this section evaluates the compensated extrusion result obtained from an interpolated parameter. The purpose was to determine whether the selected compensation strategy could be applied directly to an uncalibrated viscosity condition within the investigated range.

The mixed 40000 mPa·s condition was used as the performance validation condition. This viscosity condition was not included as one of the calibrated anchor points in the previous OQ subsection. Instead, the effective motion scale was predicted by monotonic interpolation from the calibrated values at 1, 5000, 30000, and 60000 mPa·s. The interpolated effective motion scale was 41001.71, corresponding to the motor-control parameter set summarized in Listing 5.2. This interpolated parameter set was used for the compensated extrusion trials without additional manual retuning. As in the other oil-based compensated trials, the plunger contact interface was retracted after extrusion and the delivered amount was measured after a 3 min post-release waiting interval.

Table 11 reports the compensated extrusion performance at 40000 mPa·s. The same gravimetric reporting convention as in the previous OQ measurements was used. Since this was an oil-based test, the initial and measured masses refer to the syringe mass before and after extrusion.

Table 11: Compensated extrusion performance at 40000 mPa·s using the interpolated effective motion scale.

Trial	Initial mass [g]	Measured mass [g]	Volumetric deviation [%]
Viscosity: 40000 mPa·s			
Interpolated effective motion scale: 41001.71			
1	4.334	3.358	0.62
2	4.294	3.331	-0.72
3	4.328	3.366	0.82
4	4.268	3.307	-0.93
5	4.297	3.322	0.52
Mean absolute deviation			0.72
Maximum absolute deviation			0.93

For the tested 40000 mPa·s condition, the interpolated effective motion scale was sufficient to keep all five repeated extrusion trials within the defined 1% tolerance without additional retuning. The maximum absolute deviation was 0.93%, indicating that the interpolated parameter remained inside the required accuracy range for this intermediate viscosity condition.

The 40000 mPa·s validation result supports the feasibility of the viscosity-dependent compensation strategy within the investigated viscosity range. The result does not imply universal validity for all fluids or geometries, but it shows that the calibrated anchor points and monotonic interpolation were sufficient to select a working compensation parameter for an intermediate high-viscosity condition under the defined laboratory setup.

5.4.2 Post-Extrusion Passive Release and Waiting-Time Behavior

The final part of the performance qualification evaluated how post-extrusion passive release influenced pressure-delay behavior. The previous sections focused on motion execution, compensation parameter selection, and compensated extrusion accuracy. This subsection addresses a different limitation of high-viscosity extrusion: the delayed material response after the commanded actuator motion has ended. The purpose was to determine whether waiting alone was sufficient for high-viscosity fluids, or whether the plunger contact interface had to be retracted to release the residual contact load after extrusion.

A reference test was performed without post-extrusion passive release at 30000 mPa·s. In this test, the extrusion motion was completed, the plunger contact interface was not retracted, and the system was left to stabilize for 30 min before the final mass was evaluated. This test condition was used to assess whether a long waiting period alone could compensate for the residual pressure and delayed outflow behavior in the syringe–cannula system.

Table 12 reports the extrusion results obtained at 30000 mPa·s without post-extrusion passive release. The table follows the same reporting structure as the previous gravimetric result tables, but the condition explicitly documents that no plunger contact interface retraction was applied after extrusion.

Table 12: Extrusion results at 30000 mPa·s without post-extrusion passive release and with 30 min waiting time.

Trial	Initial mass [g]	Measured mass [g]	Volumetric deviation [%]
Viscosity: 30000 mPa·s			
Post-extrusion passive release: not applied; waiting time: 30 min			
1	4.372	3.469	-6.91
2	4.444	3.588	-11.75
3	4.300	3.389	-6.08
4	4.340	3.461	-9.38
Mean absolute deviation			8.53
Maximum intra-condition difference			5.70

The no-release test indicated that a long waiting time alone was not sufficient to obtain repeatable high-viscosity extrusion results within the 1% tolerance. Although the system was left to stabilize for 30 min, the maximum intra-condition difference remained above 1%. This behavior indicates that, for the tested 30000 mPa·s condition, the residual pressure and contact-load state after extrusion could not be controlled reliably by waiting alone.

The influence of post-extrusion passive release depended strongly on viscosity. For water and similarly low-viscosity fluids, the pressure-delay effect was negligible under the tested conditions. The observed deviation after extrusion was below 0.1%, so the effect of residual pressure relaxation could be neglected for this low-viscosity case. In contrast, the 30000 mPa·s test showed that the lack of passive release prevented the system from achieving repeatable results within the defined tolerance. For high-viscosity fluids, post-extrusion passive release was therefore required with the existing hardware configuration to obtain a stable post-extrusion condition.

The comparison with the standard oil-based test protocol showed that post-extrusion passive release substantially reduced the waiting interval required for practical evaluation. In all oil-based baseline, calibrated, and compensated trials with passive release, the delivered amount was evaluated after a standardized 3 min post-release waiting interval. By contrast, the no-release reference test at 30000 mPa·s was evaluated only after a 30 min waiting period and still did not achieve repeatable results within the defined tolerance. This comparison indicates that waiting alone was not sufficient for the tested high-viscosity condition, whereas passive release allowed the process to be evaluated after a much shorter and standardized post-release interval. This reduction is important because it shortens the total extrusion operation elapsed time without reducing the commanded extrusion speed. The passive release step provides a practical way to manage pressure-delay behavior in high-viscosity extrusion while maintaining the existing motion-control strategy.

5.5 Chapter Summary

This chapter presented the qualification results for the viscosity-controlled syringe pump platform. The design qualification showed that the consolidated requirement areas derived in Chapter 3 were addressed by corresponding design and method decisions. At the prototype level, the user interface design qualification confirmed that the interface supported process visibility, guided operation, protected interaction, advanced parameter access, and tab-specific help according to Nielsen's usability heuristics. The installation qualification further confirmed that the assembled prototype, drive chain, control interface, and gravimetric measurement setup could be brought into a defined test-ready state before operational testing.

The operational qualification confirmed that the implemented motion-control and monitoring functions operated as intended under the tested laboratory conditions. The system could generate TMC-executable motion commands, execute extrusion movements, display process information, protect active operation from interfering inputs, and perform post-extrusion plunger contact interface retraction. The baseline extrusion trials showed repeatable behavior within each viscosity condition from 1 to 60000 mPa·s, while the mean absolute deviation increased with viscosity under the uncompensated setting. This repeatable baseline behavior provided the basis for selecting viscosity-specific effective motion scale values. The calibrated effective motion scale values were then used as anchor points for monotonic interpolation, which predicted an effective motion scale of 41001.71 for the intermediate 40000 mPa·s validation condition.

The performance qualification evaluated the compensated extrusion result and the effect of post-extrusion passive release. At 40000 mPa·s, the interpolated effective motion scale was used without additional manual retuning, and the compensated extrusion trials remained within the defined 1% volumetric tolerance. The 40000 mPa·s result supports the feasibility of the viscosity-dependent compensation strategy within the investigated viscosity range and under the defined laboratory setup. The waiting-time evaluation further showed that, for high-viscosity extrusion, waiting alone was not sufficient to obtain repeatable results within the defined tolerance. With post-extrusion passive release, oil-based extrusion trials could be evaluated after a standardized 3 min post-release waiting interval, whereas the no-release reference test at 30000 mPa·s still showed insufficient repeatability after 30 min. The waiting-time comparison indicates that passive release reduced the practical stabilization burden of the high-viscosity extrusion process under the tested laboratory conditions. Taken together, the results support the intended process chain. The prototype could be installed in a repeatable test-ready state, execute distance-based extrusion commands, generate repeatable baseline deviations, correct these deviations through viscosity-dependent compensation, and reduce post-extrusion stabilization time by releasing the plunger contact interface after extrusion. The remaining limitations and the implications for further development are discussed in the following chapter.

6 Discussion

6.1 Interpretation of the Results

The developed syringe pump cannot be interpreted adequately as a stand-alone actuator. The qualification-inspired evaluation showed that the final extrusion behavior depended on the interaction between commanded motor displacement, syringe positioning, force transmission, viscosity-dependent command generation, user-guided operation, gravimetric evaluation, and post-extrusion handling. The DQ and IQ results are treated as the boundary conditions under which the later extrusion results should be interpreted. They indicate that the prototype, measurement setup, and operating workflow were sufficiently defined before quantitative performance was evaluated.

The OQ results mainly support the operational readiness of the prototype. The system was able to generate TMC-executable motion commands, execute the commanded extrusion movement, display relevant process information, prevent interfering user inputs during active operation, and perform post-extrusion plunger contact interface retraction. The observation of motor heating during longer operation also showed that operational constraints outside volumetric accuracy must be considered during repeated laboratory use. The integrated radial fan should be understood as an operational support measure, not as a direct contributor to filling accuracy.

The baseline and compensation results provide the main explanation for why a calibration-based strategy was meaningful. Under the uncompensated setting, the mean volumetric deviation changed with viscosity, while the repeated trials within each viscosity condition remained sufficiently consistent. This pattern indicates that the error was not dominated by uncontrolled random scatter, but by a repeatable deviation between commanded actuator displacement and delivered volume. For the tested configuration, the effective motion scale was a suitable correction parameter. The 40000 mPa·s validation condition was particularly relevant because it tested the interpolation map instead of another manually tuned calibration point. The interpolated effective motion scale of 41001.71 was applied without additional manual retuning, and the compensated trials remained within the defined 1% volumetric tolerance.

The post-extrusion passive release results indicate that displacement compensation alone is not sufficient to describe the complete high-viscosity filling process. For low-viscosity fluids such as water, the pressure-delay effect was small enough to be neglected under the tested conditions. For the tested high-viscosity condition, however, waiting alone did not provide repeatable results within the defined tolerance. The comparison between the no-release reference test and the standard oil-based test protocol indicates that post-extrusion passive release reduced the practical stabilization burden of the process. Without passive release, the 30000 mPa·s reference test still showed insufficient repeatability after 30 min. With passive release, the oil-based trials could be evaluated after a standardized 3 min post-release waiting interval under the tested conditions. This comparison indicates that the post-motion pressure state of the syringe–cannula system must be managed rather than treated as a purely time-dependent waiting problem.

6.2 Limitations of the Developed Platform

The calibrated compensation parameters define the most important boundary of the present results. The effective motion scale values established in this work are valid for the tested combination of syringe, cannula, fluid family, temperature range, mechanical setup, and outlet condition. They should not be interpreted as universal values for syringe-based high-viscosity filling processes in general. The results show that the selected configuration can achieve the required accuracy over the investigated viscosity range from 1 to 60000 mPa·s, but this does not mean that the same parameters remain valid when the flow path or boundary conditions are changed.

Additional exploratory testing with a smaller cannula suggested that the compensation strategy depends on cannula geometry. With the smaller cannula, the system was still able to maintain a maximum intra-condition difference below 1 percentage point, suggesting that the extrusion process remained repeatable within a fixed condition. However, the maximum absolute deviation could no longer be kept below 1%. This distinction is important because the system may remain repeatable after the cannula is changed, while the calibrated effective motion scale may no longer be accurate for the new hydraulic resistance. A different cannula requires additional calibration before the same 1% volumetric accuracy can be expected.

A further limitation is that the present validation was performed with the cannula outlet exposed to atmospheric pressure. If the outlet is connected to a closed container or a confined filling volume, the terminal pressure may be higher than assumed in the current tests. This would increase the effective back pressure acting against extrusion and may change the pressure build-up, syringe deformation, delayed outflow, and compensation behavior. The current system also does not directly measure pressure or flow rate. Applications with a closed outlet condition would therefore require additional adjustment or calibration before the same volumetric accuracy can be expected.

An additional operational limitation was observed when the commanded motion speed exceeded a certain practical operating range. Under these conditions, the motor produced unexpected vibration and acoustic noise. This effect may be related to resonance-like effects in the motor and drive system, although it was not investigated systematically within the scope of this thesis. The corresponding motion-speed limitation is not a fixed physical value, because the effective motion scale changes the relation between the commanded motion parameter and the actual motor operating condition. The speed range at which vibration occurs may therefore shift when a different effective motion scale is used. In an additional 1 mL water-based observation under conditions where vibration and acoustic noise occurred, the mean absolute volumetric error increased from approximately 0.2% to 0.6%, and the maximum absolute deviation increased from approximately 0.5% to 0.7%. Although the result still remained within the defined 1% tolerance for the 1 mPa·s condition, the increase in deviation indicates that vibration can influence extrusion accuracy. A stronger influence may be expected for higher-viscosity fluids because higher hydraulic resistance increases the sensitivity of the system to motion instability, pressure build-up, and mechanical transmission behavior. It is also unclear whether long-term operation under these vibration conditions could affect motor durability or the stability of the mechanical transmission. Future tests should evaluate the usable motion-speed range for each relevant effective motion scale before this operating condition is used as a long-term process setting.

The long-term reliability of the printed mechanical components must also be considered. Most

structural and interface components of the prototype were manufactured using FDM 3D printing. No failure of these parts was observed during the reported experiments, and the printed structure was sufficient for prototype-level laboratory validation. However, repeated operation may lead to fatigue, wear, local deformation, or changes in the contact and alignment conditions of printed parts. If such effects occur, affected components should be replaced before they introduce unintended deviations into the extrusion results. The current results support the short-term functional suitability of the printed prototype, but they do not provide a complete long-term durability assessment.

The experimental validation is also limited by the number of calibration and validation points and by the assumptions used in the volume calculation. The viscosity-dependent interpolation was based on calibrated anchor points at 1, 5000, 30000, and 60000 mPa·s, while the intermediate validation was performed at 40000 mPa·s. This result supports the feasibility of the method within the investigated range, but it does not justify extrapolation beyond this range. In addition, the delivered volume was calculated from measured mass differences and supplier-specific gravity values, which were not independently measured for each test condition. These limitations define the accuracy boundary of the present validation procedure and motivate the future development directions summarized in Chapter 7.

7 Summary and Future Prospects

7.1 Summary of the Thesis

This thesis developed and validated a viscosity-controlled syringe pump platform for controlled syringe–cannula extrusion of high-viscosity oils under laboratory conditions. The work was motivated by the need to prepare reproducible filling trials for the central component of a urological implant, while keeping the technical focus on the syringe–cannula extrusion process. The investigated process could not be reduced to motor displacement alone, because the delivered volume was affected by fluid viscosity, cannula resistance, syringe mechanics, plunger friction, structural compliance, and delayed material release. For this reason, the pump was treated as an integrated process platform combining mechanical support, actuator control, user guidance, compensation, and validation-oriented operation.

The platform was structured as a set of interacting subsystems rather than a single motorized syringe holder. The mechanical design defined the syringe fixation, standardized cannula connection, plunger contact interface, actuator support, and linear guidance required for repeatable extrusion. The drive and actuation design converted motor rotation into controlled plunger displacement through the transmission and leadscrew system. In the control subsystem, target quantity and viscosity were translated into TMC-executable motion parameters, while the effective motion scale was used as the main compensation parameter. The user interface and validation procedure completed the process platform by supporting parameter input, process visibility, protected operation, gravimetric measurement, and the DQ/IQ/OQ/PQ-inspired evaluation structure.

The qualification results established the operational basis for evaluating the extrusion experiments. DQ linked the requirement areas derived in the earlier chapters to the corresponding design and method decisions, while IQ verified that the assembled prototype, drive chain, electronics, user interface, and gravimetric measurement setup were ready for testing. OQ demonstrated that the system could generate TMC-executable motion commands, execute the commanded extrusion movement, display process information, protect active operation from interfering inputs, perform post-extrusion plunger contact interface retraction, and document operational constraints such as motor heating during longer operation.

The extrusion trials demonstrated why viscosity-dependent compensation was necessary for the investigated high-viscosity filling process. Under the uncompensated setting, the mean volumetric deviation changed with viscosity, whereas the repeated trials within each viscosity condition remained sufficiently consistent. This repeatable deviation provided the basis for calibration-based correction. The effective motion scale was selected as the main compensation parameter, and calibrated values at 1, 5000, 30000, and 60000 mPa·s were used as anchor points for monotonic interpolation. The intermediate 40000 mPa·s condition then tested the interpolation map instead of another manually tuned calibration point. Using the interpolated effective motion scale of 41001.71, the compensated trials remained within the defined 1% volumetric tolerance under the tested laboratory conditions.

The passive-release tests added a second important finding: displacement compensation alone was not sufficient to manage the post-extrusion behavior of high-viscosity oils. For low-viscosity fluids such as water, the pressure-delay effect was negligible under the tested conditions. For the tested high-viscosity condition, however, waiting alone did not provide repeatable

results within the defined tolerance. With post-extrusion passive release, the oil-based trials could be evaluated after a standardized 3 min post-release waiting interval, whereas the no-release reference test at 30000 mPa·s still showed insufficient repeatability after 30 min. Overall, this thesis demonstrates that controlled high-viscosity syringe-based extrusion requires repeatable mechanical positioning, viscosity-dependent compensation, and management of the post-extrusion pressure-delay state within the syringe–cannula system.

7.2 Future Prospects

One immediate development step is the integration of the current laboratory prototype into a more compact device. In the present setup, the pump structure, power supply, controller, display, and driver electronics are still partly separated or arranged as a laboratory prototype. A future version should integrate these elements into a single compact unit while maintaining accessibility for parameter adjustment, maintenance, and experimental operation. Such integration would improve handling, reduce setup variability, and provide a more consistent basis for repeated filling trials.

Mechanical robustness should also be improved, especially for components that carry load or define alignment during high-viscosity extrusion. The current prototype demonstrated functional suitability for laboratory validation, but many components were manufactured by 3D printing. Future versions could replace selected load-bearing or alignment-critical components with stiffer and more durable materials. This would reduce the risk that structural deformation, local wear, or alignment changes influence the delivered volume during high-viscosity extrusion.

Beyond short-term validation, repeated extrusion tests over extended operating periods are required. The present work demonstrated the short-term feasibility of the developed platform under defined laboratory conditions, but it did not provide a complete lifetime or durability assessment. Future testing should examine whether repeated syringe exchange, repeated plunger loading, motor heating, vibration, fan operation, and printed component wear affect extrusion repeatability over time. This would help define maintenance intervals and replacement criteria for components that may degrade during repeated use.

The drive train also requires optimization at very low extrusion speeds. During the present development, unexpected motor vibration and acoustic noise were observed at very low motion speeds, which may be related to resonance-like effects. One possible improvement is to adjust the gear ratio so that the motor can operate in a more suitable speed range while the output motion at the plunger remains slow enough for high-viscosity extrusion. This could improve low-speed motion quality and reduce the risk of vibration-related disturbances or long-term motor loading.

A further step toward a more state-aware control concept is the integration of additional sensing functions. A pressure sensor could provide direct information about pressure build-up in the syringe–cannula system and allow the control strategy to consider both commanded displacement and the actual hydraulic state during extrusion. This could support more precise process control, especially when the outlet condition, cannula geometry, or fluid viscosity changes. A temperature sensor could also be integrated near the motor or motor housing to support automated thermal management. Instead of requiring the operator to decide before extrusion

whether the fan should be activated, the system could switch on the cooling fan automatically when a defined motor-temperature threshold is exceeded. Such sensing functions would move the platform from purely parameter-based operation toward a more state-aware control concept.

The stall-detection capability of the TMC driver could also be used to reduce manual preparation effort. This function could support sensorless initialization routines, such as sensorless homing or plunger flange detection, without adding additional mechanical limit switches. In a future version, these routines could be integrated into the controller so that the initial preparation stage would no longer depend only on manual adjustment. For example, the system could automatically move to a defined reference position, detect the plunger flange contact condition, and prepare the actuator position before extrusion. This would reduce operator-dependent setup variation and support a semi-automated or fully automated preparation workflow.

Finally, the compensation strategy should be evaluated using a broader experimental database. The present validation was limited to a defined syringe, cannula, fluid family, and viscosity range. Future studies should include additional viscosity levels, different cannula geometries, different syringe sizes, and different outlet boundary conditions. Such data would help determine whether the effective motion scale approach should remain a configuration-specific calibration method or could be developed into a more general process model for viscosity-dependent syringe–cannula extrusion.

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